THREE ESSAYS ON THE ECONOMETRIC ANALYSIS OF EMERGING ISSUES IN ELECTRICITY MARKETS

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Abstract of the Dissertation

Three Essays on the Econometric Analysis

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This dissertation comprises three essays that examine emerging issues in electricity markets using econometric methods. The first essay estimates a dynamic model of generator behavior to study how transmission expansion affects investment in electricity markets. The second essay investigates learning-by-doing and industrial policy in the Chinese solar panel industry. The third essay develops an econometric model to measure the impacts of artificial intelligence on United States power grids.

To my family.

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Chapter 1

Investment and the Transfer of Power: Dynamic Effects of Transmission in Electricity Markets

1.1 Introduction

Climate change remains the most important environmental issue facing the world at present. In order to prevent catastrophic impacts of climate change, economies need to decrease greenhouse gas emissions. Electricity was associated with the highest share of carbon emissions in the United States, 31 percent in 2022 according to the Energy Information Administration, and has continuously provided the largest source of greenhouse gas emissions for decades (U.S. Energy Information Administration, 2022). Renewable energy provides the clearest pathway to a reduction in electricity-related greenhouse gas emissions. Currently, a major project is underway throughout the developed world and the United States in particular to replace fossil fuel generators with renewable generation.

However, two persistent problems plague the renewable energy transition. First, the unequal dispersion of potential for renewable resources makes renewable generation infeasible in vast swaths of the United States. The climate of states like New York and New Hampshire presents significantly less potential for renewable generation than that of states like Texas and Nevada. Texas and Nevada have abundant sun and wind, which have allowed renewable energy to flourish. The area required by solar and wind also means that the locations with the highest propensity for renewable generation often are further from cities. Figure 1.10 illustrates this geographic heterogeneity, showing combined technical potential for wind and solar generation across U.S. counties. Second, renewable resources tend to be intermittent. The sun shines only for so many hours a day, and when the wind blows demonstrates marked variability. As electricity must always balance, intermittency can cause instability in electric grids. If too much electricity enters the system at one time, it can cause an overload and lead to system outages. Storage has been proposed as a potential solution for the intermittency problem but remains costly and limited. Storage also does not provide a solution in areas with limited ability to take advantage of any renewable energy.

Increasing transmission, the ability to send electricity between zones, provides another potential solution to the problem of intermittency and dispersion. Time-zone and geographic differences ensure that the peak hours for sunlight in Kansas coincide well with the peak demand hours in Indiana, and increasing long-distance transmission increases market interconnectedness, which decreases risks of instability and renewable energy curtailment, the forced shutdown of renewable resources due to over-supply. While long-distance transmission has previously proved largely infeasible both technically and economically, advancement in technologies such as High-Voltage Direct Current (HVDC) and High-Voltage Alternating Current (HVAC) provide a method through which to increase the range of transmission and potentially facilitate greater connection of renewables to the grid. Figure 1.11 shows the current state of HVDC infrastructure in the United States, with existing lines shown as solid and proposed expansions shown as dashed. Similarly, the transition from fossil fuel resources that could be easily transported to renewable resources increases economic viability of transmission expansion. Despite its potential merits as a technique to increase adoption rates for renewable energies, transmission expansion has largely escaped examination by economists.

How does long-run investment in transmission impact capacity investment in renewable resources? At face value, the direction of the welfare effects of transmission expansion may seem trivially positive as is the case in most problems of friction reduction; however, because of the substantial cost of increasing transmission capacity combined with the dynamic effects of firm investment, the sign of welfare change remains ambiguous without further study. Increasing transmission tends to lead to equilibration of prices across markets and thus may lead lower prices to cause exit in some markets and higher prices to cause entrance in others. Exiting firms may cause blackout and decreased reliability, which can cause significant losses in consumer welfare due to the high inelasticity of the demand curve for electricity. Beyond first order welfare and reliability effects, emissions produced must be examined.

Two main counterfactual experiments are performed to examine policy impacts. First, the impacts of the increased subsidization of investment in long-range transmission infrastructure are examined through the lens of the Bipartisan Infrastructure Law. This enters the model through changes in the adjacency matrix representing the ability to send electricity between locations. Second, the impacts of subsidization in renewable energy technology through the Inflation Reduction Act are examined. By performing the experiments separately and together, it can be determined if subsidization of transmission infrastructure is an alternative to subsidization of renewable generation or simultaneous subsidization policies complement each other.

Recent work within the industrial organization literature has taken a profound interest in the determinants of long-run investment in the electricity market as it relates to energy transition. Butters et al. (2021) examine the issue of battery investment to reduce intermittency as it relates to the California Independent System Operator (CAISO) market while Elliott (2022) uses an oligopolistic model to consider the potential impacts of carbon taxes and capacity payments on the resource mix in Western Australia. While both consider the dynamics of the electricity market, both also assume that prices are constant throughout the entire region and cannot be used to analyze locations with more than one settlement point. More recent work by Gowrisankaran et al. (2024) connects changes in the price differential between natural gas and coal to widespread changes in capacity investment away from coal to natural gas. Arkolakis and Walsh (2023) integrates the impacts of transmission into a macro model and considers how the implementation of proposed projects might impact further adoption of renewable energy. Though Arkolakis and Walsh's work considers a similar question, the question is approached from a macro perspective and so, the dynamics of individual firms are neglected providing an important source of future research the current work addresses.

Gonzales et al. (2022) considers the implications of interconnection between the Northern and Southern Chilean electricity markets for long-run entry and exit. My work extends their work by generalizing the model to deal with larger and more complex grids. My work also goes further by utilizing variation in spatial transmission capacity to examine transmission's impact without significant temporal variation in transmission infrastructure.

The inclusion of generator investment based on price differentials between locations marks a major contribution to the literature. Real-world limitations on the movement of electricity make ignoring transmission perilous, and the need for investment in renewables makes a valuation of transmission expansion including its effects on renewables investment increasingly pertinent. Consideration of how interconnected electricity markets function together even on short time scales also remains conspicuously absent from the industrial organization literature around electricity providing a gap for research.

I examine the behavior of generators providing electricity in the wholesale market by building a theoretical model of the short-run electricity market and connecting the model to a dynamic game in the long-run electricity market where generators enter and exit. The model builds on prior work in electricity by adding constrained trade between regions in electricity. I estimate the short-run model using maximum likelihood, and I estimate the long-run model using a fixed-point generalized method of moments methodology using the strategy for estimation capacity investment in dynamic games found in Gowrisankaran and Schmidt-Dengler (2024). I run the counterfactuals by adjusting the available set of long-range transmission lines in the United States and the fixed costs and entrance costs of the generators in line with the legislation considered.

The paper is organized as follows. Section 1.2 reviews the prior literature related to transmission as well as investment in the electricity market. Section 1.3 describes the structure of the market analyzed and presents descriptive statistics about the market. Section 1.4 presents a structural model, while Section 1.5 describes the estimation strategy using the structural model. Section 1.6 provides a description on results and model-fit. Section 1.7 details counterfactuals. Section 1.8 concludes the paper and provides further research directions.

1.2 Literature Review

The literature related to the question of the impact of congestion on investment by generators in the electricity market falls in three distinct categories: discussions of the problems related to a changing resource mix in the electricity market, examinations of investment decisions by firms in the electricity market, and considerations of the impacts of congestion on firm behavior.

A rich body of literature exists related to the current energy transition the United States and the world is undergoing as it relates to electricity markets and firm behavior in these markets. Gowrisankaran et al. (2016) examines intermittency of renewable resources and how the intermittency impacts the electricity market. Gowrisankaran et al. (2016) finds that intermittency accounts for a substantial portion of social costs related to renewable energy and that intermittency presents a major barrier to a grid powered mainly be renewable energy. More recently, Gowrisankaran et al. (2024) examines the rationale for utility's shift away from coal and towards natural gas. The paper utilizes an estimation technique popularized by Gowrisankaran and Schmidt-Dengler (2024) to understand how the increasing price differential between coal and natural gas led to a shift in the resource mix and substantial changes in capacity investment. My estimation strategy borrows substantially from this strain, especially in the creation of an estimation strategy but utilizes a dynamic game to consider the actions of multiple generation companies together, takes into account the shift towards renewables, and integrates transmission into the model.

Related to the body of work on a changing resource mix comes a substantial group of papers that ask the question of how firms choose to invest in and enter electricity markets. Butters et al. (2021) examines the impact of California's battery capacity mandate on battery adoption and electricity markets. The paper builds a model of the operations of battery storage facilities in the day ahead and real time markets and relates the operations model to the long-run market for storage capacity. The paper concludes with a series of implications relating to policy and also the impacts of differing assumptions about depreciation and capital changes over time. The paper finds that the mandate increases social welfare slightly after considering capital costs even before taking into account the impacts on reliability.

Elliott (2022) builds on Butters et al. (2021) by building a model of investment in generators in the Australian electricity market. Elliot's model divides generation companies into strategic firms whose entry and exit impacts price and fringe firms and then uses this division to model a dynamic game in which firms choose to add or subtract generation from their portfolio in the long-run. Elliot utilizes the model to run a series of counterfactuals with the primary focus on changes to carbon taxes and the capacity market. Both models provide a useful framework to consider the impact of policy on long-run firm behavior in the electricity market.

Holland et al. (2022) provides the first model from the literature to consider multiple wholesale markets simultaneously. Holland uses EIA data on generation mix and load to model the impacts of several market policies including increasing transmission on renewables adoption and social welfare in the electricity market. While Holland et al. (2022) proves ambitious, the model considers regional markets as homogeneous and makes the strong assumption that increasing transmission simply means all generators have access to all load. The current work builds on this modelling by considering a more realistic model of transmission and congestion in an electricity market with equilibria determined at the node level.

As the current work examines the impact of transmission on investment, I focus heavily on the impacts of transmission on markets and the firms that operate in them. Arkolakis and Walsh (2023) builds a macro-style model of the process of growing the economy while also converting to clean energy. Within the model, transmission is considered as well as the impacts of transmission expansion; however, very little consideration is given as to how the energy markets function and the endogenous process of market investment by resources.

Gonzales et al. (2022) performs the most similar research to my work. The paper also has a model of long-run dynamics within the Chile electricity market and estimates the impacts of a grid expansion connecting the Northern and Southern regions of Chile on entry and exit of renewables and fossil fuels. The short-run model used to relate transmission expansion to prices is very simplistic and does not allow for settlement at multiple points simultaneously. It also requires generation data, which are not available for the United States. As the American grid is substantially larger and more complex, the expansion of Gonzales et al. (2022) I perform by adding these features presents a valuable contribution.

Finally, Gonzales et al. (2022) does not seriously consider the interdependence of generators in its long-run model. My model utilizes a dynamic game for the long-run stage, which allows for a more accurate model of transmission's impact on capacity investment. This is because within a transmission framework, long-run capacity choices are fundamentally interdependent. When considering transmission, the decision to invest in solar in Nebraska impacts the choices of Indiana coal plant owners, making this consideration valuable.

The current work builds heavily on prior work conducted by Butters et al. (2021), Elliott (2022), Gonzales et al. (2022), Gowrisankaran et al. (2024) related to investments in electricity markets to understand the determinants of investments in increased capacity at the grid level.

My work expands on prior work by incorporating congestion data from multiple markets simultaneously to examine how increased transmission impacts prices in the real time market and linking this to the dynamics of generation investment in the long-run market. The work advances the field by combining the considerations made by Gonzales et al. (2022) for transmission in short-run electricity markets with a capacity adoption model with a dynamic game similar to the long-run model found in Elliott (2022). In estimation, the work takes its inspiration from recent advances in the estimation of capacity games from Gowrisankaran and Schmidt-Dengler (2024) and Pakes and Ericson (1998). Combined, these frameworks allow for a more realistic analysis of transmission's impacts.

1.3 Setting and Data

1.3.1 Setting

In most of the United States, the electricity market is divided between the wholesale and retail markets. In wholesale markets, generators sell electricity to utilities who then sell the electricity to end-users in the retail market. Wholesale market prices are generated through the locational marginal price (LMP) mechanism. Generators submit bids of price-quantity pairs that determine how much an individual generator is willing to supply at a series of set prices. The equilibrium price is thus the price at which the last economically dispatched generator bids for the final unit of electricity. This price is paid to all generators that are dispatched at a particular market location.

I focus on the wholesale market for electricity in the Eastern Interconnection and ERCOT. I choose to focus on the regions with wholesale markets because wholesale markets have readily available price data and because vertically integrated retail markets function differently. The focus on the Eastern Interconnection comes from the fact that the majority of US ISOs are in the Eastern Interconnection. The addition of ERCOT provides the ability to perform counterfactuals examining expanded interconnection between ERCOT and the Eastern grid.

1.3.2 Dataset Creation

Data for the short-term model came from EIA form 923, EIA form 860, EIA form 930 data, EPA's Power Markets Data, the individual ISOs, S&P capital IQ Global, Homeland Infrastructure Foundation Level Database (HIFLD), and Meteostat. The model was estimated on data from 2018 to 2023 at the zonal level. The zonal level was chosen to increase the proportion of binding transmission constraints in the data and to increase granularity of the dataset in terms of both generators and load sources.

From each individual ISO, I took price and load data at the zonal level. I then took hourly fuel mix data at the zonal to estimate generation by generator by hour. For fossil fuel generators covered by CAMPD, the generation by hour was available. However, for other generators, the generation needed to be estimated. In order to estimate generation by hour by generator, EIA Forms 923 and 860 were combined with ISO data on fuel mix. Percentage of generation by the generator was found by taking the percentage of total generation for the zone by month by generation owner from Form 923. The value was then multiplied by the generation for the fuel type for the hour. If generation by the generator was greater than generation capacity, the generation was reduced to the generation capacity, and generation was redistributed to the remaining generators.

In order to estimate transmission capacity and line losses between zones, the HIFLD data on the electricity grid was utilized. The data allowed me to build a full dataset of transmission capacity and line losses in 2022. A line was considered to connect two zones if the line started at a substation in one zone and ended at a substation in another. Using this approach for connection estimation has the downside of neglecting flows that went between multiple substations and zones. However, this works well with the assumption of the grid as a bipartite network from the model. The HIFLD data contained information on type of line, voltage, length, and location. This was used to estimate flow limits and line losses based on engineering assumptions according to the technique discussed in Appendix 1.13 following Arkolakis and Walsh (2023). To get historical changes in transmission capacity and line losses, I solved for transmission capacity additions and retirements based on S&P Capital IQ's transmission projects dataset. These were added to the transmission capacity and losses from the HIFLD data to get the transmission capacity from 2001 to 2024.

With the approximate generation data for generators that did not report to the EPA and the exact costs for generators that did report to the EPA, the exports from each zone to each other zone were estimated. To estimate exports, I first added net exports to zones outside of the area of interest. Net exports to outside areas by hour were taken from EIA form 930, which provides data on flows between balancing authorities at an hourly level. Where a balancing authority outside of the dataset connected to multiple zones in the dataset, net exports are added to load proportionately to load in each of the connected regions.

Given data on load and approximate generation, an optimal dispatch problem was solved for each hour based on the generation by generator, transmission capacity and line losses. Generation was dispatched from zones with low LMP to zones with high LMP in order of marginal cost based on heat rate and fuel cost until all load was satisfied and all generation was distributed. Marginal cost was estimated using fuel cost data from EIA Form 860 and heat rate data from EPA CAMPD data. For resources without marginal cost data from EIA, an order of dispatch was provided by resource type with solar, hydro, geothermal, and wind dispatched first with ties distributed based on capacity size. Nuclear was then distributed followed by fossil fuel resources for which marginal cost data at the monthly level was available. Firms with lower marginal costs per unit of generation were dispatched to regions earlier. I use these export data to estimate hourly marginal cost curves in the short-term model.

The characteristics of the zones impacting marginal cost of the generators were taken from EIA form 860, Meteostat, and S&P Capital IQ Global. EIA provided information about the generation companies in terms of ownership and characteristics of generators. Meteostat provided data on weather hourly in the zones. In order to get an average weather for the zones, a centroid was taken from each zone. Weather data thus represent the data at the weather station closest to the centroid of the zone.

Data for the long-term model came from EIA Form 860, EIA's Annual Energy Outlook (AEO), NREL's Standard Scenarios, and S&P Capital IQ Global. The long-term model was estimated on historical data going back to 2001 with forward-looking generators basing capacity decisions on revenues estimated until 2050 after which the fuel mix and load profile are assumed constant and an infinite sum is taken. The year 2050 was chosen because it was the last year of projections from AEO and because it is a date at which most pledges for energy transition are based.

EIA form 860 provided data on capacity of generators between 2001 and 2024. These data were aggregated by owner, zone, and resource type to get capacity for long-term model estimation. For resources of types not modelled such as nuclear and hydroelectric, the capacity was taken from the aggregate capacity by zone. Meteostat and Prism provided historical long-term weather data. S&P Capital IQ Global and EIA provided data on historical natural gas prices and coal prices.

Most data from the projections for years in the future come from EIA's Annual Energy Outlook (AEO) for 2023 and NREL's Standard Scenarios with Cambium. The AEO has data on projected load by region, projected natural gas and coal price, and projected capacity and generation for nuclear and other generation types. In order to get projections for long-term changes in transmission capacity for the long-term model, NREL's Standard Scenarios were utilized. The Standard Scenarios also provided long-run data on transmission and distribution losses, load growth, and future capacity for fuels not in the model.

The current study will consider data only from Independent System Operators in the Eastern Interconnection and the Electric Reliability Council of Texas (ERCOT). For a map of the zones of interest, refer to Figure 1.5. The process of constructing the zone map is described in Appendix 1.11.2.This includes the Southwest Power Pool (SPP) which governs parts of the Midwest from Oklahoma to Wyoming. The Eastern Interconnection also includes the Midcontinent Independent System Operator (MISO) which governs the Midwest east of Mississippi to Nebraska as well as Pennsylvania-New Jersey-Maryland Interconnection (PJM), New York Independent System Operator (NYISO), and ISO New England, which governs all states on the Eastern Seaboard North of New York. Outside of the Eastern Interconnection, data are also provisioned from the ERCOT, which includes much of Texas.

1.3.3 Descriptive Statistics

Table 1.1 shows how prices and loads differ between the different Independent system operators. There is significant variation with ISO-NE having the highest mean price and SPP having the lowest. Load also varies by ISO with PJM, the largest ISO also having the highest load. Figures 1.6 and 1.8 show how load varies across the sample by zone. Load differs both within a time dimension and across the spatial dimension with peaks in load profiles in differing by time zone and weather conditions and larger more densely populated zones having more load. Table 1.2 provides descriptive statistics on the estimated features of the hourly dispatch model including hourly exports by generator and hourly generation by generator type. Table 1.3 provides descriptive statistics for the short run characteristics and the variables that feed into the short-run model.

Figure 1.7 shows how estimated capacity factor differs by ISO. Capacity factor is the ratio between total generation and total capacity and indicates the efficiency of a particular generator or generators in a particular location.

1.4 Model

1.4.1 Preliminaries

The modelling framework borrows substantially from Butters et al. (2021), Elliott (2022), and Gonzales et al. (2022). The model consists of a static short-run model and a dynamic long-run model. The primary source of variation arises from the introduction of location-specific prices and spatial heterogeneity in generators, as well as the introduction of a dynamic game into a model with transmission constraints.

While previous models typically assumed either a single price for all locations or prices varying only by broad regions, this model allows all markets to clear interdependently. Although allowing many settlement locations increases the size of the state space—making a short-run dynamic model intractable for considering complex phenomena like curtailment—this tradeoff is necessary to capture granular transmission effects. Because decreasing curtailment is a benefit of transmission expansion, excluding it likely leads only to an conservative estimate of the benefits of increasing transmission.

Within the short-run, a combined Independent System Operator (ISO) solves an optimal dispatch problem, choosing how much electricity to offer and at what nodes to minimize costs. In the long-run, a capacity game occurs where generator owners choose optimal capacity based on profit expectations derived from the short-run model's prices. While the short-run model functions primarily as an engineering problem solved by a social planner and estimated using econometrics, the long-run model is solved by competitive equilibrium.

1.4.2 Network Topology and Aggregation

The network is modelled as a bipartite graph consisting of demand nodes indexed by i and generation nodes indexed by j. Electricity flows from generation nodes j to demand nodes i. Figure 1.12 illustrates the mechanics of this network. The adjacency matrix A represents the transmission capacity, where A_{ij} is the maximum electricity that can flow between node j and node i. Line losses, governed by the particular technologies used at locations, are represented by the matrix B, where B_{ij} represents the proportion of electricity lost during transmission from j to i. At each generation node j, there are various fuel resources indexed by k (e.g., wind, solar, coal, natural gas) with individual generators indexed by g.

Fuel resources are divided into endogenous and exogenous categories in the long-run investment model. Wind, solar, coal, and natural gas are modeled endogenously. The "other major fuel sources" category—primarily hydroelectricity and nuclear—is taken as exogenous to reduce the state space. This is consistent with historical data, where year-to-year variation is much higher in fossil and renewable sources than in hydro or nuclear, which are more heavily influenced by government investment and regulatory constraints than by short-run market forces.

1.4.3 Short-run Operations Model

Under certain conditions, the generator's profit maximization problem is identical to the social planner's cost minimization problem. Following Gilbert et al. (2004) and Cramton (2017), the conditions for equivalency include: no market power among generators, no gamesmanship or non-competitive bidding, increasing and continuous marginal cost curves without jumps, and linear constraints.

The assumption of increasing marginal cost curves is consistent with both economic theory and bid offer data provided by generators. While gamesmanship is strictly policed by the Federal Energy Regulatory Commission, the assumption of no market power is the strongest. As Gilbert et al. (2004) notes, transmission constraints can engender market power; however, due to competitive pressures and bidding regulations, sustained market power in the electricity market proves elusive. Given the value of this equivalency for solvability, I proceed by solving the problem as a social planner's problem.

The ISO solves a static problem each hour t, choosing the quantity q_{ijkt} to send to each individual demand node i from each resource k at supply node j. The ISO minimizes the total cost of generation subject to system constraints, which is equivalent to maximizing the negative of the cost function:

$$\pi_t = \max_{q_{ijkgt} i,j,k,g,t \in I,J,K,G,T} \left[-\sum_{I=1}^{I} \sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{g=1}^{G} c_{jkgt}(x_{jkgt}, q_{ijkgt}, \varepsilon_{ijkgt}) \right]. \tag{1.1}$$

The cost of dispatching a generator, c_{jkgt}, depends on resource characteristics x_{jkgt}, the quantity dispatched, and an error term ε_{ijkgt} representing incorrect specification of the cost function visible to the ISO but not the econometrician.

The optimization is subject to three primary linear constraints:

1. Resource Constraint (Kirchhoff's Law): The system must balance such that total load equals total generation minus line losses. Demand is assumed perfectly inelastic.

$$\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{g=1}^{G}(1-B_{ij})q_{ijkgt} = L_{it} \quad \forall i. \tag{1.2}$$

2. Transmission Constraint: The ISO cannot transmit more electricity between two locations than the line capacity permits.

$$\sum_{k=1}^{K}\sum_{g=1}^{G} q_{igjkt} \leq A_{ij} \quad \forall i,j,t. \tag{1.3}$$

3. Capacity Constraint: A resource cannot be dispatched beyond its maximum capacity O_{jt}^{MAX} at any time t. This capacity varies over time due to weather conditions (affecting solar/wind) and seasonal factors (affecting thermal efficiency).

$$0 \leq \sum_{i} q_{igjkt} \leq O_{jkgt}^{MAX} \quad \forall j,k,g,t. \tag{1.4}$$

Dispatch and Pricing

If no constraints bind, resources set marginal cost equal to price and produce where the price is highest. However, when constraints bind, prices diverge. If the lowest-cost resource faces a transmission constraint, it dispatches up to A_{ij} in the highest-price location, then moves to the second-highest, repeating until either the maximum generation O_{jt}^{MAX} is reached or marginal revenue equals marginal cost.

The Locational Marginal Price (LMP) at any node is equal to the marginal cost of the last economically dispatched resource serving that node. When transmission constraints bind, cheap power from low-cost zones cannot fully reach high-demand areas, causing LMPs to separate spatially.

I rely on the following assumptions for the solution:

1. Blackouts are not allowed; there is sufficient electricity to satisfy all load at every node at any time.

2. No resource is ever transmission-constrained at *all* locations simultaneously.

3. Within each generation node j, at least one resource is not at maximum capacity at any given time.

1.4.4 Econometric Specification

The solution method for the operations model (detailed in Appendix 1.9) is reminiscent of trade models. We can define a latent variable $\tilde{V}$ representing the conditions for the marginal resource. The error for a constrained generator equals the difference between its marginal cost and that of the marginal generator, adjusted for the shadow costs of transmission and capacity constraints.

The structural equation is given by:

$$\pi_t = \max_{\{q_{ijkgt}\}_{i,j,k,g,t \in I,J,K,G,T}} [-\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{g=1}^{G} c_{jkgt}(x_{jkgt}, q_{ijkgt}, \varepsilon_{ijkgt})] \quad s.t. \tag{1.5}$$

$$\underbrace{\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{g=1}^{G}(1-B_{ij})q_{ijkgt} = L_{it}}_{\text{Resource constraint}} \quad \forall i$$

$$\underbrace{\sum_{k=1}^{k}\sum_{g=1}^{G} q_{ijkgt} \leq A_{ij}}_{\text{Transmission constraint}} \quad \forall i,j,t$$

$$\underbrace{0 < \sum_{i} q_{ijkgt} \leq O_{jkgt}^{MAX}}_{\text{Capacity constraint}} \quad \forall j,k,g,t.$$

To identify the error terms, consider generator $jg'k$. For any nodes ijk' with transmission constraints but without capacity constraints:

$$\tilde{\varepsilon}_{igj'kt} = \varepsilon_{igjkt} - \varepsilon_{i'g'jkt}. \tag{1.6}$$

For nodes $i'jk'$ that are neither transmission nor capacity constrained:

$$\tilde{\varepsilon}_{i'jk't} = \varepsilon_{i'jk't}. \tag{1.7}$$

By assumption, each j node has one resource not capacity constrained, and each resource has at least one i node with which it is not transmission constrained. Plugging Equation 1.7 into 1.6 allows us to solve for each ε_{ijkt}

Define P_t as a multivariate normal probability density function with mean zero and variance Ω_1 from which the errors ε_{ijkt} are drawn. This yields the likelihood function for the set of resources excluding the marginal resource:

$$L_1 = \prod_{t=1}^{t=T} P_t(\tilde{V}_{111t}, ..., \tilde{V}_{IJKt}). \tag{1.8}$$

By maximizing this likelihood function, I estimate the parameters ν_k and β to solve the short-run model.

Resource Dynamics in the Short-Run

In this framework, renewable resources (wind, solar) have marginal costs averaging to zero, reflecting that their fuel inputs are free. Conversely, fossil fuel resources have consistently positive marginal costs and control the marginal generation, choosing when to burn fuel.

In the long run, these differences drive market entry and exit. Fossil fuel resources face increasing fixed costs due to regulation and maintenance, while renewables experience decreasing entry costs due to technological improvement. As transmission expands, prices equalize spatially: prices paid to fossil resources generally decrease, while those paid to renewables increase. These short-run profit signals drive the long-run transition, leading to increased renewable capacity and fossil fuel exit.

1.4.5 Long-run Model

The short-run and long-run models interact through prices and capacity. Figure **??** illustrates this relationship: the ISO solves the optimal dispatch problem hourly, which generates prices and costs that feed into generators' expected profits. These expected profits then drive long-run capacity investment decisions, which in turn affect future dispatch outcomes.

In the short run, expanding transmission capacity leads to price convergence between regions through spatial arbitrage. Figure 1.13 illustrates this mechanism: low-cost regions (net exporters) see prices rise as they gain access to distant markets, while high-cost regions (net importers) experience price declines as cheaper power flows in. The welfare effects include producer surplus gains in exporting regions and consumer surplus gains in importing regions.

These short-run price changes create dynamic investment incentives. Figure 1.14 shows how expanded market access shifts the marginal benefit curve for capacity investment outward in resource-rich regions, leading to increased equilibrium capacity. Higher prices in previously constrained exporting regions incentivize entry by renewable generators who can now sell to distant markets.

The long-run model characterizes the decision of firms to invest in or reduce capacity. Firms invest in capacity based on potential profits and retire capacity based on the expected losses. The value of a set amount of capacity given state variables in the current period $\upsilon_{gt}(\Theta)$ is equal to the max with respect to the current period's chosen investment ΔO_g of profits from all generators in the current period $\Pi_{jg}(\Theta)$ plus the unobservable error of the chosen amount of investment today plus the discounted expected value of the generator in the next period. Profits $\Pi_{jg}(\Theta)$ are the sum for a particular node of the aggregate demand times the average price minus the aggregate cost times the aggregate demand minus fixed costs times total capacity minus entrance cost times positive investment. The problem is shown in Equation 1.9.

The problem is not stationary as both entry costs and fixed costs are integrated AR(1) processes. Our primary parameters to estimate are ψ_{ik}, the per-period average increase or decrease in fixed and entry costs, ρ_{ik}, the persistence of fixed and entry costs, and $\sigma^2_{\zeta_i}$, the standard deviation of the error for fixed and entry costs. In theory, over time, fixed costs will increase for coal and natural gas leading to retirement of fossil fuel plants while entry costs will decrease for solar and wind plants leading to investments in renewable generation. Over time, the model simulates a renewable energy transition. The short-run transmission network impacts the long-run choices of investment and retirement through changes in revenue and costs, which impact the profits and change the desirability of owning and investing in capacity.

$$\upsilon_{gt}(\Theta) = \max_{\Delta O_g} \sum_j \Pi_{jg}(\Theta) + \varepsilon_{jg}(\Delta O_{jg}) + \beta \mathbb{E} \upsilon_{jgt+1}(\Theta') \quad s.t. \tag{1.9}$$

$$\Pi_{jg} = \textstyle\sum_k [\underbrace{D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) P_{jkg}}_{\text{Revenue}}$$

$$\underbrace{- C_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) - F_k O_{jgk} - E_k max(\Delta O_{jgk}, 0)]}_{\text{Costs}}$$

$$\underbrace{O'_{jgk} = O_{jgk} + \Delta O_{jgk}}_{\text{Evolution of capacity}}$$

$$F'_k = \psi_{1k} + \rho_{1k} F_k + \zeta'_{k1}, \zeta_1 \sim N(0, \sigma^2_{\zeta_1})$$

$$E'_k = \psi_{2k} + \rho_{2k} E_k + \zeta'_{k2}, \zeta_2 \sim N(0, \sigma^2_{\zeta_2})$$

1.5 Estimation Strategy

1.5.1 Identification

The parameter set β is identified based on variations in choices to dispatch resources of the same fuel type with different characteristics. The parameter set ν_k is identified by variations chosen in quantities dispatched by resource type after controlling for within resource heterogeneity. The variance of any particular error is not identified separately from the parameters, but the covariance of the residuals identifies the correlation between errors at a particular timestep.

The ability to estimate the impact of changing transmission capacity through altering transmission capacity matrix A and line loss matrix B comes from the model's estimation of the parameters above, β and ν_k. Using these parameters, the system of resources for dispatch can be solved as a linear program giving the shadow prices related to transmission capacity. Using the fully solved model, counterfactuals can also be run by altering these matrices.

The eight primary parameters of interest in the long-run model are the per-period change in fixed cost by resource type, ψ_{1k}, the per-period change in entrance cost by resource type, ψ_{2k}, the auto-regressive coefficient in the law of motion for fixed costs ρ_{1k}, the auto-regressive coefficient in the law of motion for entry cost ρ_{2k}, the initial fixed cost, F_{0k}, the initial entrance cost, E_{0k}, and the variance of the error terms, $\sigma^2_{\zeta_{1k}}$ and $\sigma^2_{\zeta_{2k}}$. It should be noted that each of these parameters are estimated for coal, natural gas, solar, and wind, so in total 32 parameters are estimated or 8 parameters per resource. The total sample size for the long-run is equal to the total number of included generation companies times the total number of resources those companies hold times the number of zones the generators participate in times the total number of years in the long-run sample. This is well above the minimum size for statistical power.

The initial fixed cost by resource type, F_{0k}, is identified by the capacity retirement choice in the starting period by generator type after accounting for expected demand, costs, and prices. This occurs when ΔO_{jk} is negative. The autoregressive coefficient in the entrance cost equation,ρ_{1k}, is identified by the level of correlation between negative capacity investment decisions in different periods after accounting for other parameters and factors. The per-period change in the fixed cost, ψ_{1k}, is identified by decisions to change capacity retirements over time after accounting for changes in prices, demand, and costs. The initial entrance cost by resource type, E_{0k}, is identified by the amount of increase in capacity in the starting period by resource type after accounting for fixed costs and changes in prices and expected demand. The autoregressive coefficient in the entrance cost equation, ρ_{2k}, is identified by the level of correlation between positive capacity investment decisions in different periods after accounting for other parameters and factors. The per-period change in entrance costs by resource type, ψ_{2k}, is identified by changes in entrance by resource type after accounting for other parameters and factors.

The variance of the error terms $\sigma^2_{\zeta_{1k}}$ and $\sigma^2_{\zeta_{2k}}$ are identified by variations in decisions by resources to decrease or increase capacity respectively for resources of the same type after accounting for differences in demand, prices, and costs.

1.5.2 Short-run Estimation

The data do not include information on where electricity generated by a particular resource in a particular location is transported. However, with the generation data and the the transportation data available, the exact locations electricity is sent to can be determined by assuming that energy goes to the location of highest price, so all low-cost energy is dispatched to the region with highest returns given losses and transmission constraints before high-cost energy is dispatched. This reasonably approximates the unobserved quantity flows by resource type.

Where data on flows and generation are available at a more granular level than the ISO such as at the zonal level, the zonal level is used as the area of observation, and demand nodes i and supply nodes j are for zone. This makes prices more consistent with the prices experienced by resources. Where these data are available only at the ISO level, these nodes are ISOs, and prices are aggregated to the ISO level by taking a weighted average based on load. Where available, these weighted averages are taken from official sources. When not available, I calculate them.

The data do not include information on losses for all transfers between all zones and ISOs. As such, losses are calculated outside the model using the loss data I do have and data on line lengths, composition, and flows to estimate the losses. This is appropriate because losses are not endogenous to the model but are purely a result of technical constraints related to lines.

The operations model will be solved using the simulated maximum likelihood method. Errors will be drawn from a multivariate normal distribution for each time, which will allow me to solve for the logarithm of the likelihood function from Equation 1.8;

$$log(L_1) = \sum_{t=1}^{T} log(P_t(\tilde{V}_{111t}, ..., \tilde{V}_{IJKt})). \tag{1.10}$$

Errors will be simulated, and parameters will be changed based on gradient descent until the maximum of the log-likelihood function is found.

1.5.3 Long-run Estimation

The long-run model estimation utilizes the dynamic game investment estimation methodology from Pakes and Ericson (1998) and utilizes the algorithm from Gowrisankaran and Schmidt-Dengler (2024) to find feasible choices to speed up algorithmic performance. The estimation strategy is thus an iterative fixed point algorithm, which solves for a value function within the iterated step to find equilibrium choice probabilities. The estimation utilizes the generalized method of moments (GMM) framework to find parameters.

Following the methodology from Gowrisankaran and Schmidt-Dengler (2024), I compute the equilibrium through a nested fixed point procedure where I iterate start with a belief of each generation company of the probability of rival actions, solve the value function iteratively for the belief, and iterate until all beliefs are correct. Diverging from the framework, I use an oblivious equilibrium concept, so the beliefs must only be correct with regards to the beliefs of generation companies about rival competitive capacity.

The state space is discretized as follows. Fixed cost is divided into 20 bins for from zero to the maximum revenue per megawatt of a resource type. Entry cost is divided into 20 bins from zero to ten times the maximum revenue per megawatt of a resource type. Demand state space is divided into 5 bines from .75 to 1.5 times previously seen max load. The own capacity state space is divided into 15 bins from 0 to 1.5 times max generation of a generation owner. The alternative capacity state space is divided into ten bins from 0 to 1.5 times the maximum load. The average price state space is divided into 10 bins from 0 to 1.1 times the maximum average price in the data. The possible investments are restricted to blank with 10 possible investment choices. Investment is restricted, so negative capacity is not permitted.

In order to estimate the model, the parameters from the short-term model are estimated via maximum likelihood. These parameters are used within the marginal cost function to solve the linear program to get quantities and prices in the short-run with expected changes to the grid in the future. A subset of dates are solved using the parameters for all elements of the state space. The results are then fed into the long-run model, which uses the short-run results for profits for each year.

The long-term model has 32 parameters to estimate including the integrated and auto-regressive terms in the fixed cost and entry cost, the initial fixed cost and initial entry cost, and the standard deviation for both errors. Each parameter needed to be estimated for all four resource types. In order to estimate the 32 parameters, at least 32 moments must be used. I follow Gowrisankaran et al. (2024) in choice of moments to get 36 total moments for an overidentified model. For natural gas, solar, and wind, the moments include an indicator for positive investment interacted with quantity built, total quantity squared, and an indicator for investments in capacity of large size, which is specific to resource based on the data. I interact these moments with total capacity and also include investment variance as a moment. I add one additional moment, which is the amount of capacity retired to help identify fixed cost parameters. For coal, I do the same but focus on moments related to retirements instead of investment with one investment-related moment.

1.6 Results

1.6.1 Short-run Model Estimates

Table 1.4 presents estimates from the short-run dispatch model for export sensitivity and base cost parameters across resource types. The intercept terms represent base marginal costs in \$/MWh, while the export coefficients capture how marginal costs vary with the quantity exported from each zone. Table 1.5 reports coefficients for natural gas generator characteristics. The natural gas spot price coefficient of 6.93 implies that a \$1/MMBtu increase in the Henry Hub spot price raises the marginal cost of natural gas generation by approximately \$6.93/MWh, consistent with average heat rates in the sample. The negative coefficient on heat content reflects that generators with higher fuel efficiency face lower marginal costs. Table 1.6 presents analogous results for coal resources, where the coal price coefficient of 8.68 reflects the higher heat rates typical of coal plants.

Figure 1.1 displays model fit for the short-run dispatch model. The model slightly underestimates average prices but accurately captures price dispersion across zones. The correlation between predicted and actual locational marginal prices is relatively strong, indicating that the model captures the primary drivers of spatial price variation.

1.6.2 Long-run Model Estimates

Table **??** reports parameter estimates for the long-run capacity investment model. The estimates for ψ_{2k}, which governs the drift in entry costs over time, are negative for coal (-0.106) and positive for solar (0.017) and wind (0.095), consistent with declining costs for renewable technologies relative to fossil fuels. The autoregressive coefficients ρ_{1k} and ρ_{2k} indicate moderate persistence in cost shocks, with wind showing the highest persistence

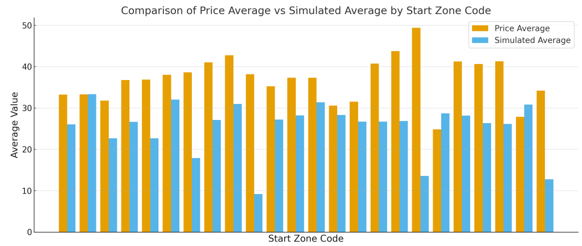

Figure 1.1: Model Fit for the Short-run.

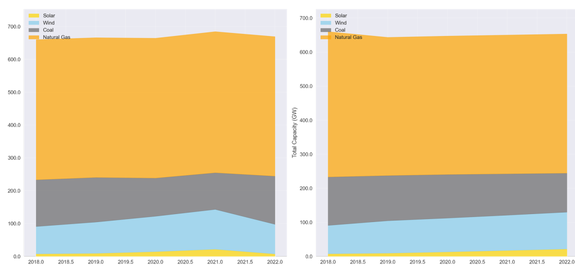

Figure 1.2: Model Fit for the Long-run.

in entry cost shocks ($\rho_{2k} = 0.223$).

Figure 1.2 compares simulated capacity paths against historical data from 2001–2024. The model matches aggregate capacity levels well but overpredicts solar and wind investment relative to historical patterns while generating relatively flat coal and natural gas capacity trajectories. This discrepancy likely reflects factors outside the model including state-level renewable portfolio standards, federal tax credits with complex phase-out schedules, and permitting delays that constrained renewable development during portions of the sample period.

1.6.3 Robustness Checks

I perform a suite of robustness checks detailed in Appendix Section 1.15 and summarized in Table 1.10. I show the sensitivity of my a relaxation of assumptions. In particular, I show the robustness of my estimates to the allowance of blackouts and market power, varying assumptions about demand and its structure, changes to the specific dataset structure, and changes to other specific model features. The evidence presented is relatively compelling that the results are robust under slight relaxation. Some changes like market power have more impact on the estimated impacts as the level of deviation from the baseline assumption rises.

1.7 Counterfactuals

I use the estimated model to evaluate three sets of policy counterfactuals: upgrades to existing transmission infrastructure, addition of new long-distance transmission lines, and active federal legislation affecting both transmission and generation investment. In all counterfactual exercises, I hold estimated parameters fixed and modify only the primitives of the model—entry costs, line capacities, and per-MWh revenue—during simulation. This ensures that behavioral responses (investment, retirement, and dispatch) emerge endogenously from the estimated model rather than from re-fitting. The full set of counterfactuals run is described in Table 1.7 and Figure 1.15.

1.7.1 Transmission Infrastructure Upgrades

The first set of counterfactuals examines the effects of upgrading existing transmission lines to High-Voltage Direct Current (HVDC) technology, which reduces line losses and increases transfer capacity. I simulate outcomes under full replacement of all inter-zonal lines as well as partial replacement of randomly selected lines at varying intensities.

Table **??** reports the price impacts of transmission upgrades. Full replacement of all lines with HVDC technology reduces average wholesale prices by 6% across the study region, generating welfare gains of approximately $4 billion per year. This figure represents roughly one-third of total congestion costs estimated by Grid Strategies for the region. Notably, most of these gains can be achieved with partial upgrades: replacing 50% of transmission lines still delivers price reductions exceeding 5%.

Figure 1.3 illustrates how price reductions scale with the percentage of lines upgraded. The relationship exhibits diminishing returns, with the marginal benefit of additional up-

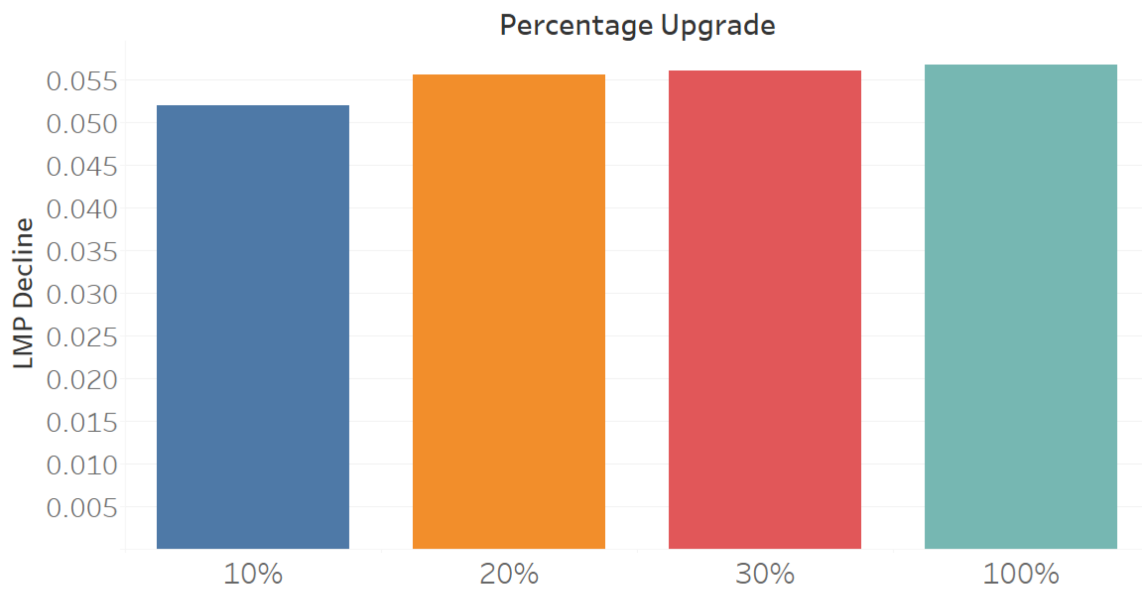

Figure 1.3: Price Decrease by Percentage Grid Upgrade

grades declining as congestion on the most constrained corridors is relieved.

The effects of transmission upgrades vary substantially by region. The Northeast, where congestion costs are highest and renewable resources are scarce, experiences the largest price declines. The Midwest—home to abundant wind resources that are often curtailed due to transmission constraints—sees modest price increases as expanded export capacity raises local prices toward the national average. This price convergence, while increasing costs for some Midwestern consumers, reflects more efficient allocation of generation resources and reduced curtailment of low-cost renewable energy.

1.7.2 New Long-Distance Transmission

The second set of counterfactuals examines the addition of specific long-distance transmission projects. I first consider the limiting case of removing all transmission constraints, then evaluate specific proposed lines.

Removal of All Transmission Constraints

Eliminating all transmission constraints reduces average prices in every ISO except SPP and MISO, where prices increase as these regions become net exporters. This counterfactual also dramatically accelerates the renewable energy transition in the long-run model, as generators in renewables-rich regions can access distant markets without congestion-related price discounts.

Grain Belt Express

The Grain Belt Express is a proposed 800-mile HVDC transmission line designed to carry 5,000 MW of wind power from western Kansas to Indiana and points east. In the long-run counterfactual, adding this line increases solar and wind adoption in the Eastern Interconnection by over 10% by 2050, with the largest increases concentrated in SPP and western MISO where the line originates.

ERCOT-Eastern Interconnection Connection

A major HVDC connection between ERCOT and the Eastern Interconnection would allow Texas's abundant wind and solar resources to serve load in the Southeast and beyond.

1.7.3 Federal Legislation

The final set of counterfactuals evaluates three pieces of federal legislation affecting electricity markets. These counterfactuals and their relationships are described in Table 1.8. I implement these policies as *policy wedges*—time-varying modifications to model primitives that activate at each policy's enactment date while holding all estimated parameters fixed. This approach isolates the causal effect of each policy from the underlying structural relationships estimated from pre-policy data. To ensure comparability across scenarios, I use identical random seeds and shock draws for all simulations, so that differences across scenarios reflect only the policy treatment and not sampling variation. I run four scenarios: (i) both policies as enacted, (ii) IRA only, (iii) BIL only, and (iv) neither policy (the baseline).

Big Wires Act

The BIG WIRES Act (S.2827, 118th Congress) would mandate that each FERC Order No. 1000 transmission planning region achieve a Minimum Interregional Transfer Capacity (MITC) equal to at least 30% of its coincident peak electrical demand, or an increase of 15% over current capacity, whichever is lower, by 2035 as decribed in Hickenlooper and Peters (2023). The legislation is motivated by the widening gap between planned interregional transmission builds and anticipated need documented in the DOE's U.S. Department of Energy (2023), driven by barriers including insufficient coordination across planning regions, cost allocation disputes, and local opposition Joskow (2021); Pfeifenberger and Staff (2021).

I follow the modeling approach of Botterud et al. (2024), who evaluate the BIG WIRES Act using the GenX capacity expansion model across 64 zones aggregated into 12 model regions that approximate the FERC Order No. 1000 planning regions. Their framework distinguishes interregional transmission (built between zones in different planning regions) from intraregional transmission (built within a region), and assumes that absent the Act, no new interregional transmission is built—reflecting the observed tendency of Balancing Authorities to prioritize investment within their own planning region. Under the Act, interregional capacity is expanded until each region satisfies the MITC requirement, with GenX determining the least-cost allocation of new capacity across corridors.

To implement this counterfactual in my model, I add inter-zonal transfer capacity consistent with the corridor-level builds reported in Botterud et al. (2024). I model the MITC requirement as time-varying capacity additions that phase in via a ramp function:

$$A_{ij,t}^{\text{MITC}} = A_{ij,t}^{\text{baseline}} + \Delta A_{ij}^{\text{MITC}} \cdot \rho(t; t_0^{\text{MITC}}, R) \tag{1.11}$$

where $\Delta A_{ij}^{\text{MITC}}$ is the additional transfer capacity on corridor (i,j) required to satisfy the MITC constraint, calibrated to be consistent with the corridor-level estimates in Botterud et al. (2024). The largest capacity additions are concentrated in the Eastern Interconnection: between the Midwest and Mid-Atlantic, Southeast and Florida, Mid-Atlantic and Carolinas, Midwest and Central, and Mid-Atlantic and Southeast, which together account for approximately 80% of total new interregional transmission builds under the Act.

Inflation Reduction Act

The Inflation Reduction Act (IRA) provides substantial tax credits for renewable energy investment, effectively reducing entry costs for solar and wind generators. I model the IRA through two channels that enter the simulation as policy wedges on the estimated model's primitives.

First, the IRA reduces entry costs for eligible renewable technologies. I implement this as a proportional reduction in the entry cost E_k for eligible technology k:

$$E_{k,t}^{\text{IRA}} = E_{k,t}^{\text{baseline}} \cdot (1 - \tau_k^{\text{IRA}} \cdot \rho(t; t_0, R)) \tag{1.12}$$

where $\tau_k^{\text{IRA}} \in [0,1)$ is the technology-specific cost reduction implied by the investment tax credit, t_0 is the IRA enactment date (August 2022, implemented as the first full period on or after enactment), and $\rho(t; t_0, R)$ is a ramp function that phases the credit in over R periods to avoid discontinuities in the investment decision. I set R to one year as a conservative default.

Second, the IRA production tax credit provides an ongoing per-MWh subsidy to eligible generators, which I model as a revenue shifter:

$$P_{k,t}^{\text{eff}} = P_t + s_k^{\text{IRA}} \cdot \rho(t; t_0, R) \tag{1.13}$$

where $s_k^{\text{IRA}} \geq 0$ is the production credit value in the same per-MWh units as the model's prices. This enters the generator's per-period profit function, raising effective revenue for eligible resources without altering the ISO's dispatch problem or the structure of the Bellman equation. Retirement decisions respond endogenously to the changed profit environment.

Eligibility is determined by a mapping from technology type to IRA-eligible status. In the baseline implementation, eligibility depends on technology k alone and does not vary by region, though the framework accommodates region-specific eligibility if needed.

Bipartisan Infrastructure Law

The Bipartisan Infrastructure Law (BIL) provides funding for transmission infrastructure including grants for interregional transmission projects. I model the BIL as a time-varying expansion of inter-zonal transfer capacities:

$$A_{ij,t}^{\text{BIL}} = A_{ij,t}^{\text{baseline}} + \Delta A_{ij}^{\text{BIL}} \cdot \rho(t; t_0^{\text{BIL}}, R) \tag{1.14}$$

where $\Delta A_{ij}^{\text{BIL}} \geq 0$ represents the additional transfer capacity on corridor (i, j) funded by the BIL, in the same units as the baseline line limits A_{ij}, and t_0^{BIL} is the BIL enactment date (November 2021). The adjusted capacities enter the ISO dispatch problem's constraint set, expanding the feasible region for inter-zonal power flows. The dispatch objective and optimality conditions are otherwise unchanged.

I consider two specifications for the BIL transmission expansion. The first is corridor-specific, assigning $\Delta A_{ij}^{\text{BIL}}$ values to individual lines consistent with announced project funding. The second is a system-wide specification in which each corridor receives a proportional expansion $\Delta A_{ij}^{\text{BIL}} = \alpha \cdot A_{ij,t}^{\text{baseline}}$, parameterized by a single scalar $\alpha > 0$.

Combined Policy Effects

Running these policies in combination reveals whether transmission investment and renewable subsidies act as substitutes or complements. Formally, I compare the welfare gain from the joint policy $\Delta W^{\text{IRA+BIL}} = W^{\text{as\_enacted}} - W^{\text{no\_ira\_no\_bil}}$ against the sum of individual effects $\Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}$, where $\Delta W^{\text{IRA}} = W^{\text{no\_bil}} - W^{\text{no\_ira\_no\_bil}}$ and $\Delta W^{\text{BIL}} = W^{\text{no\_ira}} - W^{\text{no\_ira\_no\_bil}}$. Complementarity implies $\Delta W^{\text{IRA+BIL}} > \Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}$.

The results indicate that these policies are largely complementary: renewable subsidies increase the supply of low-cost generation, while transmission investment ensures this generation can reach high-demand markets. The mechanism is intuitive—the IRA lowers entry costs, inducing additional renewable capacity in resource-rich regions, but the value of this capacity is limited by transmission constraints that prevent it from reaching high-price markets. The BIL relaxes these constraints, raising the effective price received by new entrants and amplifying the IRA's investment incentive. Implementing both policies together yields welfare gains exceeding the sum of individual effects, suggesting that a portfolio approach to energy transition policy dominates single-instrument strategies. I include estimates of the costs of the policies and show that all are welfare improving even after considering costs.

1.7.4 Welfare Decomposition

I run a welfare decomposition where I demonstrate how gains accrue to producers and consumers at the zonal level. While the total system-wide surplus is obviously positive for both producers and consumers, the distribution of the surplus is significantly varied. In some regions, mainly ERCOT and specific zones within PJM, welfare is uniformly improved. This occurs because the insurance effect dominates with producers exporting more electricity during low-cost hours and importing during high-cost hours. For regions with significant renewable energy and low prices before transmission expansion, producer welfare increases while consumer welfare decreases. The opposite occurs in regions with fewer renewable resources and higher initial prices. I quantify these effects and provide useful intuition for policy-makers as to what transfers could be made to make the policy Pareto-improving.

1.8 Conclusion

This paper develops and estimates a dynamic model of generator behavior in electricity markets with transmission constraints, linking short-run dispatch decisions to long-run capacity investment. The model captures how transmission infrastructure shapes price formation across interconnected markets and how these prices influence firms' incentives to invest in different generation technologies.

Three main findings emerge from the analysis. First, transmission constraints impose substantial costs on electricity consumers: upgrading existing infrastructure to modern HVDC standards would reduce wholesale prices by 6% and generate $4 billion in annual welfare gains. Second, the benefits of transmission investment are regionally heterogeneous, with the largest gains accruing to transmission-constrained regions in the Northeast. Third, in the long run, expanded transmission capacity accelerates the renewable energy transition by allowing generators in renewables-rich regions to access distant markets, increasing solar and wind adoption by over 10% by 2050 under scenarios including major projects like the Grain Belt Express.

The policy counterfactuals indicate that transmission investment and renewable energy subsidies act as complements rather than substitutes. Policies like the Inflation Reduction Act increase the supply of renewable generation, while transmission investments under the Bipartisan Infrastructure Law and Big Wires Act ensure this generation can reach consumers. Implementing both types of policies together yields benefits exceeding the sum of their individual effects.

Several limitations suggest directions for future research. The model operates at the zonal rather than nodal level, preventing analysis of specific substation-to-substation connections. This limitation makes the model better suited to evaluating large-scale interregional transmission than individual line additions. Additionally, the model treats demand as exogenous; long-run shifts in the geographic distribution of load in response to electricity price changes represent an important extension. Finally, the model does not incorporate storage, which may interact with transmission as alternative solutions to renewable intermittency.

Despite these limitations, the results provide evidence that transmission investment offers a cost-effective complement to direct renewable subsidies in accelerating the energy transition. As policymakers evaluate proposals to expand interregional transmission capacity, models that capture both the short-run operational benefits and long-run investment effects can inform the efficient allocation of infrastructure spending.

Tables

ISO	Mean Price	Mean Load	Median Price	Median Load
ISO-NE	50	13,623	45	13,308
NYISO	43	17,686	39	17,297
ERCOT	40	44,328	23	41,901
SPP	37	30,420	30	29,246
MISO	38	73,945	34	72,179
PJM	41	146,112	37	88,236

Table 1.1: Average and Median Price and Load by ISO

Variable	Mean	Standard Deviation	Median
Load	5,075	7,055	2,296
Price	37.8	79.5	27.6
SO_2 Emissions	526	417	470
Solar Generation	2	13	0.03
Wind Generation	44	61	20
Natural Gas Generation	29	66	33
Coal Generation	162	217	64
Exports	18.9	76.3	0.53

Table 1.2: Descriptive Statistics: Short-run

Variable	Mean	Standard Deviation	Median
Fuel Cost	603	2,093	324
Heat Content	3.6	6.0	1.03
Natural Gas Spot Price	3.8	2.3	2.7
Coal Spot Price	166	128	98
Temperature	13	12	15
Loss Percentage	0.05	0.06	0.05
Transmission Constraint	0.51	1.00	0.50
Capacity Constraint	0.06	66	33

Table 1.3: Descriptive Statistics: Short-run

Resource	Intercept	Signif.	Export Coeff.	Signif.
Electricity used for energy storage	30.97	***	0.02	***
Natural Gas	37.72	***	0.01	***
Nuclear (incl. Uranium, Plutonium, Thorium)	46.44	***	0.01	***
Coal	30.97	***	32.47	***
Distillate Fuel Oil (diesel, No. 1–4)	33.23	***	0.01	***
Landfill Gas	33.91	***	0.00	***
Water at hydro or hydrokinetic technologies	36.27	***	-0.02	***
Other Biomass Gas (digester, methane, etc.)	27.64	***	-0.11	***
Petroleum Coke	31.50	***	0.06	***
Wood/Wood Waste Solids	40.91	***	0.06	***
Blast Furnace Gas	30.34	***	0.00	***
Waste heat (no direct fuel source)	37.43	***	0.01	***
Black Liquor	30.97	***	0.01	***
Other Gas	41.99	***	-0.12	***
Other	30.97	***	0.10	***
Purchased Steam	30.97	***	-0.02	***
Agricultural By-Products	29.56	***	-0.06	***
Petroleum Products	34.94	***	0.00	***

Note: Export coefficients represent the marginal relationship between exports and output by energy resource. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The log-likelihood of the regression is 7.662301.

Table 1.4: Short-run Model Estimates by Resource Type

Variable	**Estimate**	**Significance**
Natural Gas Price	6.93	***
Dewpoint	2.46	***
Temperature	0.00	
Heat Content	-2.22	***
Generator Startup	-2.3	***
Generator Near Capacity	-2.3	***

Notes: Specification includes hourly and monthly dummies, zone-code dummies, and severe weather dummies.

Table 1.5: Estimates for Natural Gas Covariates

Variable	**Estimate**	**Significance**
Coal Price	8.68	***
Dewpoint	2.46	***
Temperature	4.00	***
Heat Rate	-1.30	***
Generator Startup	-2.3	***
Generator Near Capacity	-2.3	***

Notes: Specification includes hourly and monthly dummies, zone-code dummies, and severe weather dummies.

Table 1.6: Estimates for Coal Covariates

Table 1.7: Counterfactual Simulation Scenarios

		Modified Primitives		
Scenario	**Description**	Line Capacity A_{ij}	Entry Cost E_k	Revenue P_k^{eff}
Panel A: Transmission Infrastructure Upgrades				
Full HVDC	All inter-zonal lines replaced with HVDC	✓		
Partial HVDC	Random subset at intensity $\alpha \in (0,1)$	✓		
Panel B: New Long-Distance Transmission				
No constraints	All transmission limits removed	✓		
Grain Belt Express	5,000 MW HVDC, KS to IN	✓		
ERCOT–Eastern IC	HVDC link, ERCOT to South-east	✓		
Panel C: Federal Legislation				
BIG WIRES Act	MITC $\geq$ 30% peak demand by 2035	✓		
IRA $\times$ BIL Factorial Design				
Baseline	Neither IRA nor BIL			
IRA only	IRA enacted, no BIL		✓	✓
BIL only	BIL enacted, no IRA	✓		
As enacted	Both IRA and BIL	✓	✓	✓
Panel D: Cross-Cutting Analysis				
Welfare decomposition	Producer/consumer surplus by zone for each scenario	*Applied to all scenarios*		

Notes: Each scenario holds estimated structural parameters fixed and modifies only the indicated model primitives. A_{ij} denotes inter-zonal transfer capacity, E_k denotes technology-specific entry cost, and P_k^{eff} denotes effective per-MWh revenue inclusive of production tax credits. All policy wedges phase in via a ramp function $\rho(t;t_0,R)$ to avoid discontinuities. Panel C scenarios use identical random seeds and shock draws to isolate policy effects from sampling variation.

Table 1.8: Federal Legislation: Factorial Design and Complementarity Test

	Bipartisan Infrastructure Law	
Inflation Reduction Act	Not enacted	Enacted
Not enacted	W^{baseline}	$W^{\text{BIL only}}$
Enacted	$W^{\text{IRA only}}$	$W^{\text{as enacted}}$

Comparison	**Definition**
IRA marginal effect	$\Delta W^{\text{IRA}} = W^{\text{IRA only}} - W^{\text{baseline}}$
BIL marginal effect	$\Delta W^{\text{BIL}} = W^{\text{BIL only}} - W^{\text{baseline}}$
Joint effect	$\Delta W^{\text{joint}} = W^{\text{as enacted}} - W^{\text{baseline}}$
Complementarity	$\Delta W^{\text{joint}} - \left(\Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}\right) > 0$

Notes: W denotes total welfare (consumer plus producer surplus). The top panel displays the 2×2 factorial structure of the federal legislation counterfactuals. The bottom panel defines the comparisons used to test policy complementarity. Complementarity holds when the joint welfare gain exceeds the sum of individual gains, indicating that transmission investment (BIL) and renewable subsidies (IRA) are mutually reinforcing.

Figures

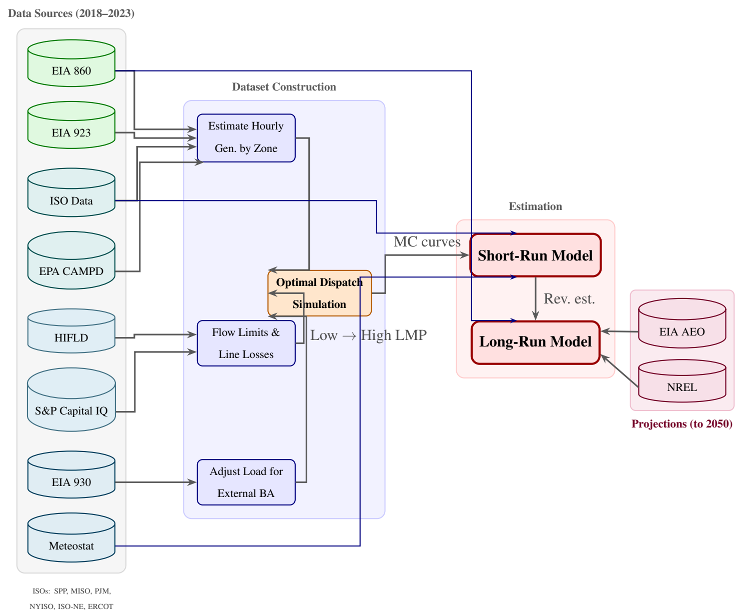

Figure 1.4: Dataset Construction Diagram.

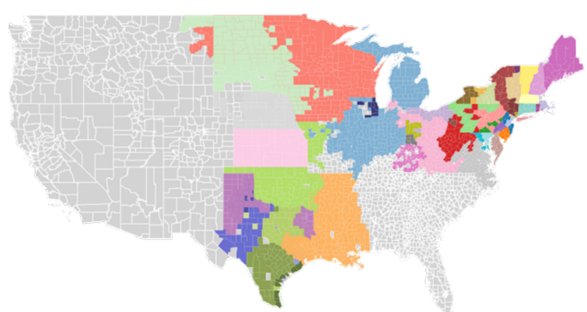

Figure 1.5: Mapped Zones from Market.

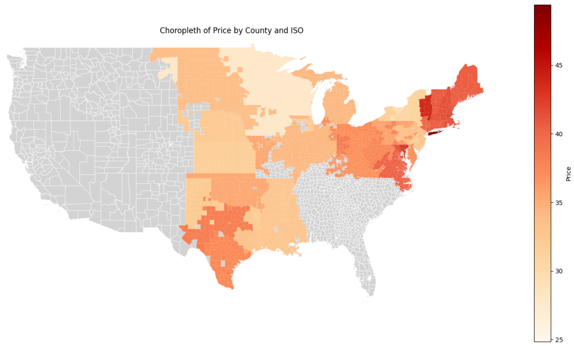

Figure 1.6: Locational Marginal Price by Zone in Data.

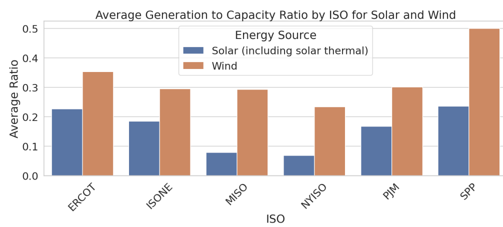

Figure 1.7: Estimated Capacity Factor for Solar and Wind by ISO.

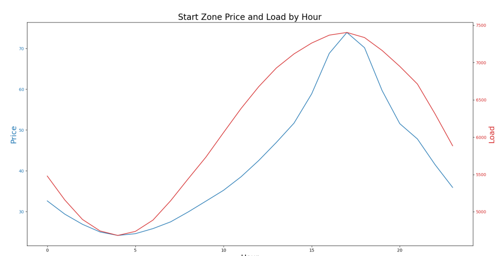

Figure 1.8: Price and Load by Hour for entire dataset.

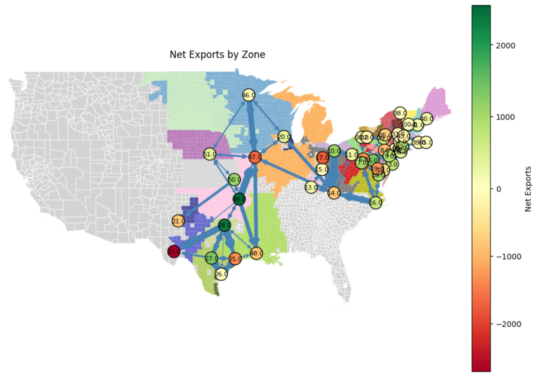

Figure 1.9: Map of Exports between Zones in the Model.

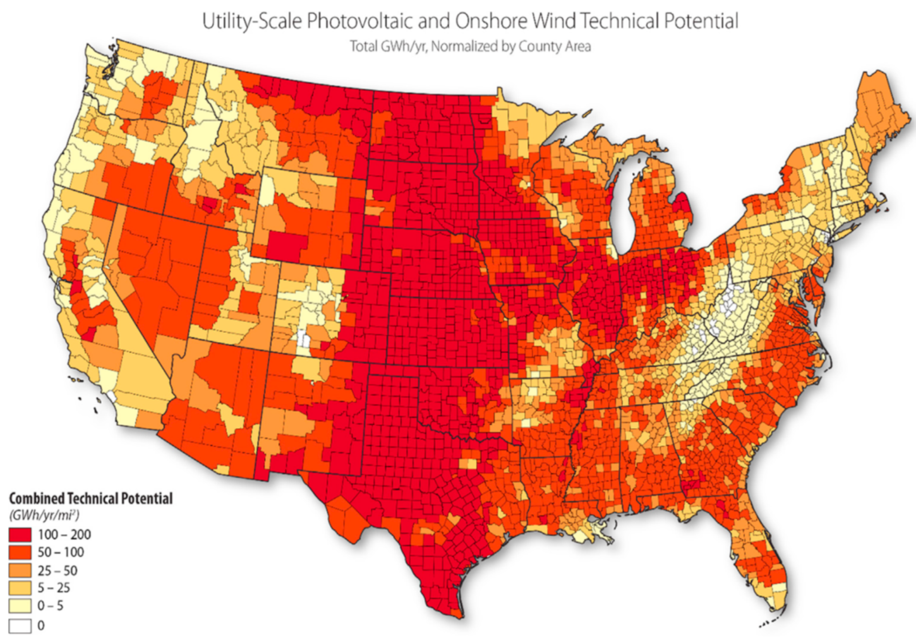

Figure 1.10: Combined Technical Potential for Wind and Solar Generation by County (GWh/yr/mi^2). Darker shading indicates higher combined renewable potential. Source: Department of Energy.

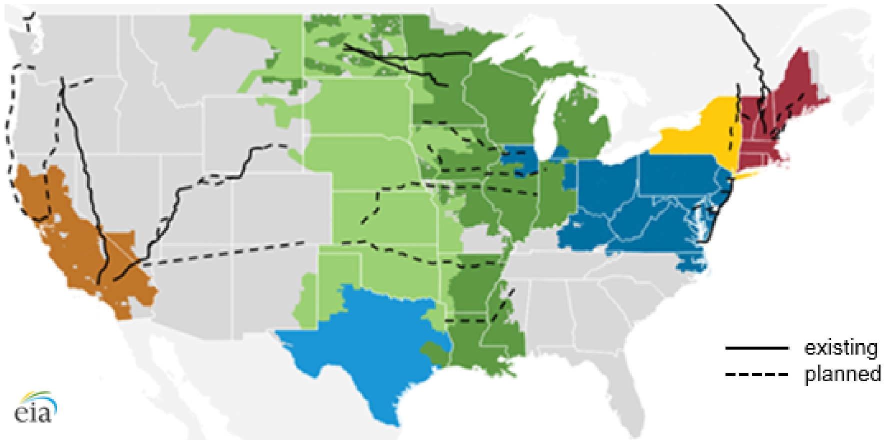

Figure 1.11: Existing and Planned High-Voltage Direct Current Transmission Lines. Solid lines indicate existing HVDC infrastructure; dashed lines indicate proposed expansions. Source: EIA.

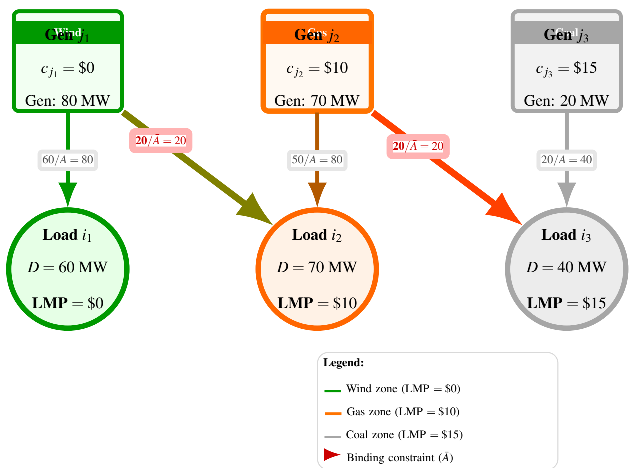

Figure 1.12: Network Topology Illustrating Congestion Pricing. Binding export constraints create price separation across zones. Each zone's LMP equals the local marginal cost when transmission constraints bind.

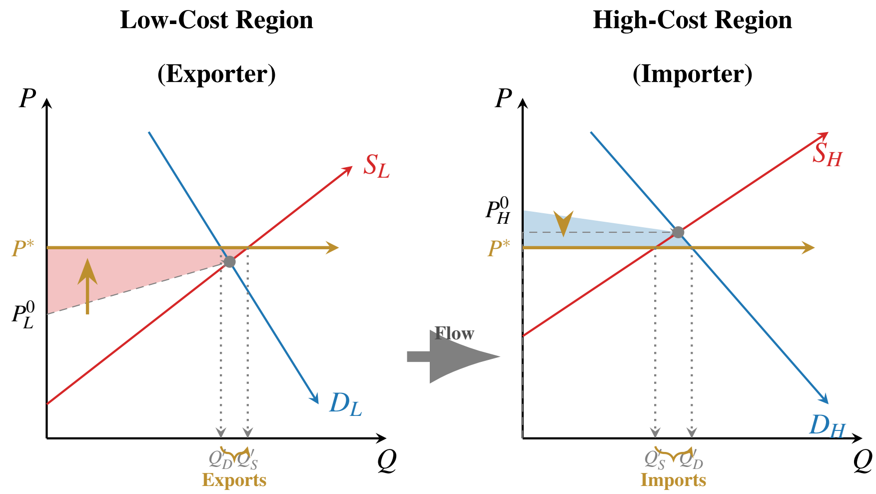

Figure 1.13: Static Effects of Transmission Expansion: Price Convergence and Spatial Arbitrage. In the low-cost (exporting) region, prices rise and producer surplus increases (red shaded area). In the high-cost (importing) region, prices fall and consumer surplus increases (blue shaded area).

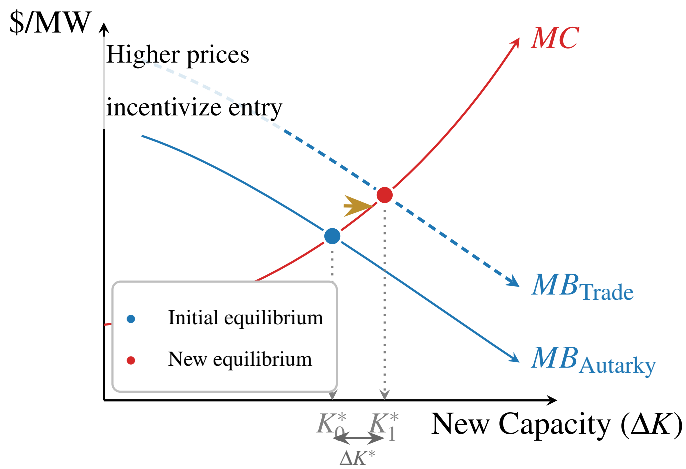

Figure 1.14: Dynamic Effects of Transmission Expansion: The Investment Signal. Expanded market access shifts the marginal benefit curve outward for generators in resource-rich regions, leading to increased equilibrium capacity investment.

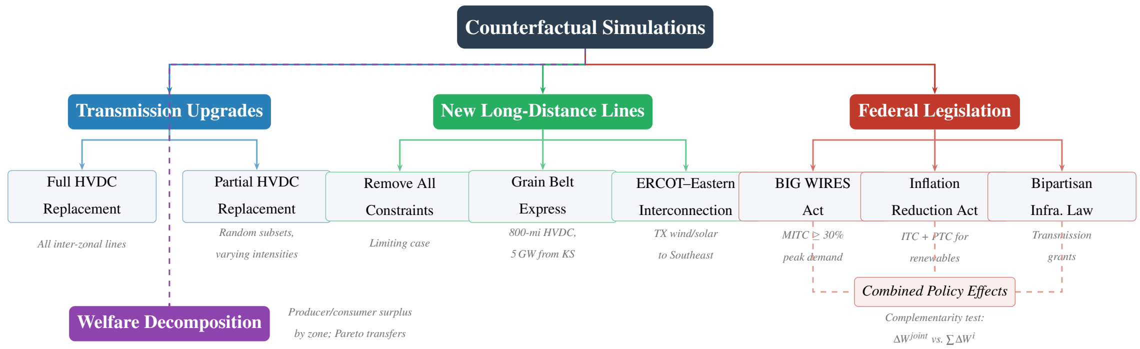

Figure 1.15: Visual description of counterfactuals run and how they relate.

1.9 Solving the Operations Model

This appendix derives the likelihood function used to estimate the short-run operations model. Section 1.9.1 establishes the equivalence between the generator profit maximization problem and a social planner's cost minimization problem, then formulates the ISO's dispatch problem. Section 1.9.2 derives the first-order conditions from the Lagrangian. Section 1.9.3 specifies the cost function and derives the likelihood function for estimation.

1.9.1 The ISO Dispatch Problem

Under standard conditions, the generator's profit maximization problem is equivalent to a social planner's cost minimization problem. Following Gilbert et al. (2004) and Cramton (2017), the conditions for this equivalence include: (i) no market power among generators, (ii) no gamesmanship or non-competitive bidding, (iii) increasing marginal cost curves, and (iv) linear constraints. The assumption of increasing marginal costs is consistent with both bid offer data and economic theory. Gamesmanship and non-competitive bidding are prohibited and actively monitored by the Federal Energy Regulatory Commission. While transmission constraints can create local market power (Gilbert et al. (2004)), competitive pressures and regulatory limits on bidding behavior make sustained market power difficult to maintain.

I aggregate to the resource and ISO level due to data limitations: hourly generation data are available only at the resource level rather than the generator level. While this aggregation sacrifices granularity that would be valuable in a transmission model, I disaggregate where data permit.

Notation

Let $i \in \{1, \dots, I\}$ index demand nodes, $j \in \{1, \dots, J\}$ index supply (generation) nodes, $k \in \{1, \dots, K\}$ index resources at each supply node, and $t \in \{1, \dots, T\}$ index time periods (hours). Define:

- q_{ijkt}: Quantity of electricity sent from resource k at supply node j to demand node i at time t
- $c_{jkt}(\cdot)$: Cost function for resource k at supply node j at time t
- x_{jkt}: Observable cost shifters for resource jk at time t
- ε_{ijkt}: Unobserved cost shock
- A_{ij}: Transmission capacity between nodes i and j
- O_{jkt}^{MAX}: Maximum generation capacity of resource k at node j at time t
- L_{it}: Load (demand) at node i at time t
- B_{ij}: Transmission loss factor between nodes i and j

The Optimization Problem

The ISO solves a static cost minimization problem each hour, choosing dispatch quantities q_{ijkt} to minimize the total cost of meeting demand:

$$\pi_t = \max_{\{q_{ijkt}\}} \left[-\sum_{j=1}^{J} \sum_{k=1}^{K} c_{jkt} \left(x_{jkt}, \sum_{i=1}^{I} q_{ijkt}, \sum_{i=1}^{I} \varepsilon_{ijkt} \right) \right] \tag{1.15}$$

subject to three constraints.

Transmission Constraint. Flows between any two nodes cannot exceed line capacity:

$$\sum_{k=1}^{K} q_{ijkt} \leq A_{ij} \quad \forall i, j, t. \tag{1.16}$$

Capacity Constraint. Total dispatch from each resource cannot exceed its maximum capacity:

$$0 \leq \sum_{i=1}^{I} q_{ijkt} \leq O_{jkt}^{MAX} \quad \forall j,k,t. \tag{1.17}$$

Load Balance Constraint. Supply must equal demand at each node, accounting for transmission losses:

$$\sum_{j=1}^{J} \sum_{k=1}^{K} (1 - B_{ij}) q_{ijkt} = L_{it} \quad \forall i,t. \tag{1.18}$$

Three maintained assumptions ensure the problem is well-defined: (i) no blackouts, meaning sufficient generation capacity exists to serve all load at every node; (ii) no resource is transmission-constrained to all destinations; and (iii) at least one resource at each generation node has slack capacity at any given time.

When no constraints bind, resources set marginal cost equal to price and serve the highest-priced locations first. When the transmission constraint binds, a resource dispatches up to $A_{ij} - \sum_{k' \neq k} q_{ijk't}$ units to the highest-priced location, then up to A_{ij} units to the next highest-priced location, continuing until either maximum capacity O_{jkt}^{MAX} is reached or marginal revenue equals marginal cost. Prices equal the marginal cost of the last economically dispatched resource at each location.

1.9.2 First-Order Conditions

The Lagrangian for the ISO's problem at time t is:

$$\begin{aligned}\mathscr{L}_t = &- \sum_{k=1}^{K} \sum_{j=1}^{J} c_{jkt} \left(x_{jkt}, \sum_{i=1}^{I} q_{ijkt}, \sum_{i=1}^{I} \varepsilon_{ijkt} \right) \\ &+ \sum_{i,j} \lambda_{ijt} \left(A_{ij} - \sum_{k=1}^{K} q_{ijkt} \right) \\ &+ \sum_{i} \gamma_{it} \left(L_{it} - \sum_{k=1}^{K} \sum_{j=1}^{J} (1 - B_{ij}) q_{ijkt} \right) \\ &+ \sum_{j,k} \theta_{jkt} \left(O_{jkt}^{MAX} - \sum_{i=1}^{I} q_{ijkt} \right), \end{aligned} \tag{1.19}$$

where λ_{ijt} is the multiplier on the transmission constraint, γ_{it} is the multiplier on the load balance constraint, and θ_{jkt} is the multiplier on the capacity constraint.

Optimality Conditions

The first-order condition with respect to q_{ijkt} is:

$$\frac{\partial \mathscr{L}_t}{\partial q_{ijkt}} = -\frac{\partial c_{jkt}}{\partial q_{ijkt}} - \lambda_{ijt} - \gamma_{it}(1 - B_{ij}) - \theta_{jkt} = 0. \tag{1.20}$$

Complementary slackness requires:

$$\lambda_{ijt} \left(A_{ij} - \sum_{k=1}^{K} q_{ijkt} \right) = 0 \quad \forall i, j, t, \tag{1.21}$$

$$\theta_{jkt} \left(O_{jkt}^{MAX} - \sum_{i=1}^{I} q_{ijkt} \right) = 0 \quad \forall j, k, t. \tag{1.22}$$

Rearranging equation (1.46) yields three equivalent expressions:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1-B_{ij}) + \theta_{jkt} = \lambda_{ijt}, \tag{1.23}$$

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1-B_{ij}) + \lambda_{ijt} = \theta_{jkt}, \tag{1.24}$$

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \gamma_{it}(1-B_{ij}). \tag{1.25}$$

Intratemporal Euler Conditions

Three key relationships emerge from comparing first-order conditions across different resource-node combinations.

Condition 1: Same supply node, different resources. For two resources k and k' at the same supply node j, both sending to demand node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \theta_{jkt} = \frac{\partial c_{jk't}}{\partial q_{ijk't}} + \theta_{jk't}. \tag{1.26}$$

Condition 2: Same resource, different demand nodes. For resource k at supply node j sending to two demand nodes i and i':

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1-B_{ij}) + \lambda_{ijt} = \frac{\partial c_{jkt}}{\partial q_{i'jkt}} + \gamma_{i't}(1-B_{i'j}) + \lambda_{i'jt}. \tag{1.27}$$

Condition 3: Different supply nodes, same demand node. For resources at different supply nodes jk and $j'k'$ both sending to demand node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \frac{(1-B_{ij})}{(1-B_{ij'})}\left[\frac{\partial c_{j'k't}}{\partial q_{ij'k't}} + \lambda_{ij't} + \theta_{j'k't}\right]. \tag{1.28}$$

1.9.3 Cost Function Specification and Likelihood

Cost Function

For renewable resources (wind and solar), the cost function is linear in output:

$$c_{jkt}^{renew} = q_{ijkt}\varepsilon_{ijkt}. \tag{1.29}$$

For non-renewable resources, the cost function is quadratic:

$$c_{jkt} = \sum_{i=1}^{I} q_{ijkt}x_{jkt}\beta + \frac{1}{2}\left(\sum_{i=1}^{I} q_{ijkt}\right)^2 \nu_k + \sum_{i=1}^{I} q_{ijkt}\varepsilon_{ijkt}, \tag{1.30}$$

where β and ν_k are parameters to be estimated, with ν_k varying by resource type to capture differences in fuel costs. The marginal cost for non-renewable resources is:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} = x_{jkt}\beta + q_{ijkt}\nu_k + \varepsilon_{ijkt}. \tag{1.31}$$

Identifying the Central Resource

Define a *central resource* at each demand node as a resource for which neither the transmission constraint nor the capacity constraint binds: $\lambda_{ijt} = 0$ and $\theta_{jkt} = 0$. By the no-blackout assumption, every demand node has at least one central resource. For central resources, the observed locational marginal price (LMP) equals marginal cost:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} = LMP_{it}. \tag{1.32}$$

Using equation (1.28) with a central resource $j''k''$ at node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it}. \tag{1.33}$$

Rearranging and substituting the marginal cost specification:

$$x_{jkt}\beta + q_{ijkt}\nu_k + \lambda_{ijt} + \theta_{jkt} - \frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it} = -\varepsilon_{ijkt}. \tag{1.34}$$

Solving for Shadow Prices

From equation (1.25) applied to the central resource at node i:

$$\gamma_{it} = \frac{LMP_{it}}{(1-B_{ij''})}. \tag{1.35}$$

Transmission Shadow Price. Using equation (1.27) and the assumption that each resource can send electricity to at least one unconstrained destination i' (so $\lambda_{i'jt} = 0$):

$$\lambda_{ijt} = q_{i'jkt}\nu_k - q_{ijkt}\nu_k + \frac{LMP_{i't}}{(1-B_{i'j''})}(1-B_{i'j}) - \frac{LMP_{it}}{(1-B_{ij'''})}(1-B_{ij}) + \varepsilon_{i'jkt} - \varepsilon_{ijkt}, \tag{1.36}$$

where j'' and j''' denote the central resources at nodes i' and i, respectively.

Capacity Shadow Price. Using equation (1.26) and the assumption that each generation node has at least one resource k' with slack capacity (so $\theta_{jk't} = 0$):

$$\theta_{jkt} = (x_{jk't}\beta + q_{ijk't}\nu_{k'}) - (x_{jkt}\beta + q_{ijkt}\nu_k) + \varepsilon_{ijk't} - \varepsilon_{ijkt}. \tag{1.37}$$

Constructing the Likelihood Function

Substituting the expressions for λ_{ijt} and θ_{jkt} into equation (1.34) yields the estimating equation. Define the composite error:

$$\tilde{\varepsilon}_{ijkt} = \varepsilon_{ijkt} - \varepsilon_{i'jkt} - \varepsilon_{ijk't}, \tag{1.38}$$

where i' is the unconstrained destination for resource jk and k' is the unconstrained resource at node j.

The deterministic component of the estimating equation is:

$$\begin{aligned}\tilde{V}_{ijkt} = & \underbrace{x_{jkt}\beta + q_{ijkt}\nu_k}_{\text{MC of resource } jk} - \underbrace{\frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it}}_{\text{MC of central resource}} \\ & \underbrace{+ q_{i'jkt}\nu_k - q_{ijkt}\nu_k + \frac{LMP_{i't}}{(1-B_{i'j''})}(1-B_{i'j}) - \frac{LMP_{it}}{(1-B_{ij'''})}(1-B_{ij})}_{\text{cost of transmission constraint}} \\ & + \underbrace{(x_{jk't}\beta + q_{ijk't}\nu_{k'}) - (x_{jkt}\beta + q_{ijkt}\nu_k)}_{\text{cost of capacity constraint}}. \end{aligned} \tag{1.39}$$

For resources with transmission constraints but no capacity constraints:

$$\tilde{\varepsilon}_{ijk't} = \varepsilon_{ijk't} - \varepsilon_{i'jk't}. \tag{1.40}$$

For resources with neither constraint binding:

$$\tilde{\varepsilon}_{i'jk't} = \varepsilon_{i'jk't}. \tag{1.41}$$

By substitution, each individual error ε_{ijkt} can be recovered.

Maximum Likelihood Estimation

Let $P_t(\cdot)$ denote a multivariate normal density with mean zero and variance-covariance matrix Ω. The likelihood function is:

$$\mathscr{L} = \prod_{t=1}^{T} P_t\left(\tilde{V}_{111t}, \ldots, \tilde{V}_{IJKt}\right). \tag{1.42}$$

Maximizing this likelihood function yields estimates of β (the effect of observable cost shifters) and ν_k (resource-type-specific marginal cost parameters), thereby recovering the short-run cost structure.

1.10 Long-run Model Computational Details

1.10.1 Algorithm Overview

This appendix details the computational methodology for estimating the long-run capacity investment model. The estimation combines a nested fixed point algorithm for computing equilibrium with generalized method of moments (GMM) estimation for parameter recovery. The approach follows the dynamic game estimation methodology developed by Pakes and Ericson (1998) and utilizes algorithmic improvements from Gowrisankaran and Schmidt-Dengler (2024) to identify feasible choices and accelerate convergence.

The computational procedure consists of three main components: (i) state space discretization, (ii) equilibrium computation via nested fixed point iteration, and (iii) GMM estimation using simulated moments. Each component is described in detail below.

1.10.2 State Space Discretization

The dynamic investment problem is solved over a discretized approximation to the continuous state space. Discretization renders the Bellman equation computationally tractable while preserving economically relevant variation in prices, quantities, and profits. The grid is constructed to capture empirically observed ranges of capacity, demand, and prices, with bounds chosen to exceed historical maxima to accommodate counterfactual investment paths.

State Variable Bins

The state space is discretized as shown in Table 1.9. Each state variable is discretized using evenly spaced bins over a bounded interval informed by historical data and engineering constraints.

Table 1.9: State Space Discretization

State Variable	Bins	Range
Fixed cost (F_k)	20	$[0, \bar{R}_k]$
Entry cost (E_k)	20	$[0, 10 \times \bar{R}_k]$
Demand state	5	$[0.75 \times \bar{L}, 1.5 \times \bar{L}]$
Own capacity (O_{jkg})	15	$[0, 1.5 \times \bar{G}_{jg}]$
Rival capacity (O^{-}_{jkg})	10	$[0, 1.5 \times \bar{L}]$
Average price (P_{jkg})	10	$[0, 1.1 \times \bar{P}]$

Notes: $\bar{R}_k$ denotes maximum observed revenue per MW for resource type k; $\bar{L}$ denotes the historical maximum system load; $\bar{G}_{jg}$ denotes the maximum observed generation by owner g in region j; and $\bar{P}$ denotes the historical maximum average price.

Demand uncertainty is modeled using a multiplicative demand scaling factor applied to historical load profiles. The five demand states span $\pm 25\%$ around baseline demand and are intended to capture persistent variation in system conditions rather than transitory hourly shocks.

Own capacity and rival capacity are discretized separately. Own capacity reflects firm-specific investment decisions, while rival capacity aggregates the competitive fringe and other strategic firms into a single state variable, consistent with an oblivious equilibrium approximation. Capacity bounds exceed historical maxima to allow for endogenous entry and expansion in counterfactual scenarios.

Short-Run Outcome Mapping

For each point in the discretized state space, the model computes short-run equilibrium outcomes by solving a linear or quadratic dispatch problem subject to generator capacity constraints, zonal load balance, and transmission limits (see Section 1.13). The solution yields locational marginal prices, dispatched quantities, production costs, and operating profits.

Outcomes are computed as weighted averages across representative operating conditions (e.g., seasonal or load-duration blocks), producing a smooth mapping from state variables to expected period profits. These expected profits are stored and treated as primitives in the dynamic investment problem.

Investment Choice Set

The firm's investment choice is restricted to a discrete set of ten capacity adjustment options. These include incremental additions for expanding technologies and retirement options for declining resources. Capacity is constrained to remain weakly non-negative, ruling out infeasible states. This restriction substantially reduces the dimensionality of the choice set while preserving the economically relevant margins of adjustment.

Approximation Accuracy and Computational Feasibility

The number of bins for each state variable reflects a tradeoff between approximation accuracy and computational feasibility. Finer grids are used for own capacity and cost states, which directly affect firm profits, while coarser grids are used for rival capacity and prices. Sensitivity checks using alternative grid densities yield similar qualitative investment patterns, indicating that results are not driven by discretization artifacts.

The full state space is evaluated using batched computation with intermediate checkpointing, enabling parallel execution and recovery from job interruption. This approach allows the model to evaluate tens of thousands of state points per resource type while maintaining feasible memory and runtime requirements. A schematic overview of the dataset construction is provided in Figure 1.4.

1.10.3 Equilibrium Computation

This section describes the numerical procedure used to compute equilibrium investment behavior in the dynamic oligopoly model. The equilibrium is solved using a nested fixed point algorithm, combining backward induction over time with an outer loop over firms' beliefs about rival behavior.

Nested Fixed Point Procedure

The equilibrium is computed through a nested fixed point procedure. The outer loop iterates on beliefs over rival investment behavior until beliefs are consistent with optimal policies, while the inner loop solves each firm's dynamic programming problem conditional on these beliefs.

This approach follows the standard structure of dynamic oligopoly models with forward-looking firms and strategic interactions, and is computationally necessary given the high-dimensional state space.

Algorithm (Outer Loop):

1. *Initialize beliefs:* Each generation company initializes beliefs over the probability distribution of rival investment actions. Beliefs are defined over aggregate rival capacity changes rather than individual firm actions, consistent with the equilibrium concept described below.

2. *Value function iteration:* Conditional on these beliefs, each firm solves its dynamic optimization problem via backward induction. This inner loop yields optimal investment policies and associated value functions for all states.

3. *Update beliefs:* Beliefs are updated using the implied distribution of investment actions generated by firms' optimal policies.

4. *Check convergence:* The procedure iterates until the maximum deviation between successive policy functions satisfies

$$\max |\sigma^{(n+1)} - \sigma^{(n)}| < \varepsilon.$$

If convergence is not achieved, the algorithm returns to Step 2.

Convergence is assessed jointly over value functions and policy functions to ensure stability of both expected payoffs and equilibrium behavior.

Equilibrium Concept

Rather than solving for a full Markov Perfect Equilibrium (MPE), I employ an oblivious equilibrium concept. Under this approach, firms condition their decisions on their own state variables and a low-dimensional summary of competitors' states, namely the distribution of aggregate rival competitive capacity.

Firms do not track the full joint distribution of rival capacities across individual competitors. Instead, beliefs must be correct with respect to the evolution of total rival capacity. This substantially reduces the dimensionality of the state space while preserving the key strategic channel through which rivals' investment decisions affect prices and profits.

This approximation is particularly well suited to electricity markets, where individual generators are small relative to the market but investment decisions influence prices through aggregate capacity.

Value Function Solution

In the final period T, firms solve a stationary problem with no further investment opportunities. The value function is given by:

$$\upsilon_{jgT}(\Omega) = \max_{\Delta O_{jkg}} \left\{ \Pi_{jg}(\Omega) + \sum_{k} \varepsilon_{jkg}(\Delta O_{jkg}) + \frac{1}{1-\beta} \Pi'_{jg}(\Omega') \right\} \tag{1.43}$$

subject to the law of motion for capacity:

$$O'_{jkg} = O_{jkg} + \Delta O_{jkg} \quad \forall k \tag{1.44}$$

The state vector Ω includes firm-specific capacity holdings, demand conditions, and cost states. The term ε_{jkg} represents an idiosyncratic choice shock that smooths the policy function and ensures well-defined choice probabilities in the numerical implementation. The final period is treated as stationary with no further capacity investments, so firms receive the discounted perpetuity of steady-state profits.

The final period is treated as stationary: after period T, firms receive the discounted perpetuity of steady-state profits conditional on post-investment capacity levels.

Period profits are defined as:

$$\begin{aligned}\Pi_{jg} = \sum_k \Big[& D_{jkg}(d_{jkg}, O_{jkg}, O^{-}_{jkg}) P_{jkg} - C_{jkg}(d_{jkg}, O_{jkg}, O^{-}_{jkg}) D_{jkg}(d_{jkg}, O_{jkg}, O^{-}_{jkg}) \\ & - F_k O_{jkg} - E_k \max(\Delta O_{jkg}, 0) \Big]\end{aligned} \tag{1.45}$$

where D_{jkg} is dispatch, P_{jkg} is price, C_{jkg} is variable cost, F_k is per-period fixed cost, and E_k is entry cost for new capacity.

Prices, dispatch, and variable costs are outcomes of a short-run operations model that clears the electricity market given total installed capacity and demand conditions. Fixed costs are incurred on all installed capacity, while entry costs apply only to new investment.

First-Order Conditions

Ignoring the non-differentiable choice shock term, the first-order condition with respect to ΔO_{jkg} is:

$$\frac{\partial \upsilon_{jgT}}{\partial \Delta O_{jkg}} = \frac{\partial \Pi_{jkg}}{\partial \Delta O_{jkg}} + \frac{1}{1-\beta} \frac{\partial \Pi'_{jkg}}{\partial \Delta O_{jkg}} = 0. \tag{1.46}$$

For capacity reductions, $\max(\Delta O_{jkg}, 0) = 0$, yielding:

$$\frac{\partial \left[D'_{jkg} P'_{jkg} - C'_{jkg} D'_{jkg} \right]}{\partial \Delta O_{jkg}} = F_k. \tag{1.47}$$

For capacity expansions, $\max(\Delta O_{jkg}, 0) = \Delta O_{jkg}$:

$$E_k + \frac{\beta}{1-\beta} F_k = \frac{\beta}{1-\beta} \frac{\partial \left[D'_{jkg} P'_{jkg} - C'_{jkg} D'_{jkg} \right]}{\partial \Delta O_{jkg}}. \tag{1.48}$$

The derivatives of dispatch, prices, and costs with respect to capacity are well defined because the short-run operations problem is a linear program. Sensitivity of equilibrium outcomes to capacity constraints can therefore be computed using standard LP dual variables.

Handling Multiple Equilibria

The dynamic investment game may admit multiple equilibria. I impose a sequential move structure within each period, where firms move in descending order of installed capacity. The largest incumbent firm chooses its investment first, followed sequentially by smaller firms.

This equilibrium selection rule both reduces multiplicity and reflects empirical patterns in electricity markets, where large incumbents typically lead capacity expansion decisions.

Backward Induction

For periods $t < T$, the model is solved via backward induction. Relative to the terminal period, two features differ:

1. Firms account for expectations over future profit streams conditional on stochastic evolution of demand and cost states.

2. Capacity adjustment costs enter through the parameter ψ_{1k}, affecting average fixed costs and smoothing intertemporal investment behavior.

Expected continuation values are computed by integrating over discretized transition kernels for exogenous state variables and deterministic capacity evolution.

1.10.4 GMM Estimation

Parameter Space

The long-run model contains 32 parameters to estimate. For each of the four resource types $k \in \{\text{natural gas, coal, solar, wind}\}$, the parameters include:

- **Fixed cost process:** initial value (F_k^0), autoregressive coefficient (ρ_k^F), innovation standard deviation (σ_k^F), and integrated component coefficient
- **Entry cost process:** initial value (E_k^0), autoregressive coefficient (ρ_k^E), innovation standard deviation (σ_k^E), and integrated component coefficient

Moment Conditions

The estimation uses 36 moment conditions to identify the 32 parameters, yielding an overidentified model. Following Gowrisankaran et al. (2024), the moment conditions are constructed to capture key features of the investment distribution.

For natural gas, solar, and wind resources:

- $\mathbb{1}(\Delta O_{jkg} > 0) \times \Delta O_{jkg}$ — positive investment indicator interacted with quantity
- $(\Delta O_{jkg})^2$ — total quantity squared
- $\mathbb{1}(\Delta O_{jkg} > 500)$ — indicator for large investments (over 500 MW)
- Interactions of the above with total capacity $\sum_j O_{jkg}$
- $\mathrm{Var}(\Delta O_{jkg})$ — investment variance
- $|\min(\Delta O_{jkg}, 0)|$ — quantity of capacity retired

For coal resources:

The moment conditions focus on retirement behavior rather than new investment, reflecting the empirical pattern of declining coal capacity. One investment-related moment is included to capture any residual new construction.

Estimation Procedure

The GMM estimator minimizes the weighted distance between simulated moments from the model and their empirical counterparts:

$$\hat{\theta} = \arg\min_{\theta} \left[m(\theta) - m^{\text{data}}\right]' W \left[m(\theta) - m^{\text{data}}\right] \tag{1.49}$$

where $m(\theta)$ is the vector of simulated moments, m^{data} is the vector of data moments, and W is the weighting matrix.

The estimation proceeds in two stages:

1. *First stage:* Estimate parameters using an identity weighting matrix $W = I$ to obtain consistent (but inefficient) estimates $\hat{\theta}^{(1)}$.

2. *Second stage:* Re-estimate using the optimal weighting matrix $W^* = \hat{\Sigma}^{-1}$, where $\hat{\Sigma}$ is the variance-covariance matrix of the moment conditions computed from first-stage residuals.

Standard errors are computed using the standard GMM sandwich formula:

$$\text{Var}(\hat{\theta}) = \frac{1}{n} \left(G'WG\right)^{-1} G'W\Sigma WG \left(G'WG\right)^{-1} \tag{1.50}$$

where $G = \partial m(\theta)/\partial \theta'$ is the Jacobian of moments with respect to parameters.

Integration with Short-Run Model

The long-run investment model is estimated conditional on parameters from the short-run operations model. The short-run model parameters—demand elasticities, cost function parameters, and market clearing conditions—are estimated first and held fixed during long-run estimation. This sequential approach is computationally necessary given the nested structure of the model, where long-run investment decisions depend on expectations of short-run market outcomes.

The derivatives of dispatch, prices, and costs with respect to capacity changes are computed from the short-run model. Since the short-run operations model consists of linear programs, these derivatives are well-defined almost everywhere and can be computed efficiently using sensitivity analysis of the LP solutions.

1.11 Construction of Short-Term Electricity Market Data

This section documents the construction of the short-term electricity market dataset used in the empirical analysis. The goal of this procedure is to produce an hourly, generator-level dataset that links production, fuel costs, prices, weather conditions, and market outcomes across U.S. wholesale electricity markets. The transmission grid and interzonal flow constraints are discussed separately in Appendix 1.13.

Figure 1.4 provides a schematic overview of the full dataset construction pipeline.

1.11.1 Overview

The short-term dataset is constructed in four stages. First, raw ISO-specific fuel mix data are combined with generator characteristics from EIA Forms 860 and 923 to recover hourly generation at the generator level. Second, these ISO-level datasets are standardized and merged into a unified panel. Third, generation is allocated to load-serving zones through an economic dispatch procedure that accounts for interzonal transfers (described in Appendix 1.13). Finally, the resulting data are augmented with weather conditions, fuel prices, and time indicators to produce the final short-term estimation sample.

The pipeline covers six major U.S. Independent System Operators (ISOs): ERCOT, ISO New England, MISO, PJM, SPP, and NYISO.

1.11.2 Construction of Zonal Geographic Maps

A central requirement of the analysis is the ability to match generators to geographic areas in a manner consistent with ISO market structure while preserving as much spatial detail as possible. Because publicly available GIS shapefiles for ISO-defined market zones are incomplete, inconsistent, or entirely unavailable, the construction of zonal maps required substantial manual effort. I view this process as a significant contribution of the paper.

The guiding principle throughout was to aggregate only where necessary to permit consistent geographic matching and econometric analysis, while keeping the data at the lowest feasible spatial resolution. In all cases, zonal boundaries were explicitly restricted to lie within ISO boundaries using GIS shapefiles defining Independent System Operator footprints. This prevents generators, loads, or weather observations from being incorrectly assigned across ISOs.

General Approach. The mapping procedure proceeds in three steps. First, ISO boundaries are imposed as a hard geographic constraint. Second, ISO-specific zones are mapped to counties or county aggregates using a combination of published documentation, utility maps, and GIS overlays. Third, generators are assigned to zones using EIA Form 860 identifiers where available, and county-level geographic matching otherwise. Hand-constructed crosswalk tables are used to reconcile inconsistent geographic schemas across datasets.

Where counties do not align cleanly with ISO zones, counties are split proportionally based on geographic overlap and engineering judgment, applied consistently across datasets.

ERCOT. For ERCOT, zones are based primarily on the system's published weather zones, which in most cases map cleanly to county boundaries. Where weather zones do not align perfectly with counties, I split counties across zones using GIS overlays to preserve contiguity and proportional area. Several ERCOT datasets rely on alternative geographic schemas (e.g., hub- or region-based reporting); in these cases, I construct hand-built mapping tables to reconcile these schemas with the zonal structure used in the analysis.

PJM. PJM required the most extensive manual mapping effort. PJM publishes a large number of zones, many of which do not correspond cleanly to county boundaries. I manually mapped each PJM zone to counties using a combination of PJM documentation, utility service maps, and GIS inspection. In cases where counties span multiple PJM zones, judgment was applied to split counties as cleanly as possible while preserving geographic continuity. Where appropriate for the analysis, zones were subsequently aggregated into regions to support the distribution of generation data.

SPP. SPP zones were mapped using a similar procedure, combining transmission zone definitions with county-level utility maps and GIS shapefiles. In several states, publicly available utility maps with county labels were used to guide assignments. As with PJM, counties were split only where necessary, and always within ISO boundaries.

NYISO. For NYISO, zones were mapped based on reliability assessment zones using a zonal county map. These zones align relatively well with county boundaries, allowing for a clean assignment in most cases.

MISO. MISO presents a comparatively clean case: all zonal boundaries correspond to state-level regions. As a result, counties were assigned directly to MISO zones based on state boundaries, eliminating the need for county splitting.

ISO New England. ISO New England zones are primarily defined at the state level, with the notable exception of Massachusetts, which is divided into three wholesale zones. For Massachusetts, I manually mapped counties to zones using GIS overlays and ISO documentation, splitting counties where necessary to maintain geographic consistency.

Generator Assignment. Generators are assigned to zones using ISO labels from EIA Form 860 wherever possible, which minimizes the risk of misassignment across ISOs. When ISO labels are unavailable or insufficiently granular, generators are matched using county-level geographic information derived from plant location data. This layered approach ensures consistency between generator assignment, zonal geography, and ISO market structure.

Overall, this mapping procedure enables the construction of consistent zonal maps that balance geographic precision with analytical tractability, forming the backbone of the spatial analysis in the paper.

1.11.3 Generator-Level Generation by ISO

Each ISO publishes hourly generation by fuel type rather than by individual generator. To recover generator-level output, I combine these fuel mix data with plant-level characteristics and monthly generation shares from EIA Forms 860 and 923.

For each ISO, hourly generation by fuel type is distributed across generators proportionally to their monthly generation shares within that fuel category. Let g index generators, k fuel types, t hours, and $m(t)$ the month corresponding to hour t. Generator-level hourly output is constructed as

$$q_{gt} = s_{gkm(t)} \cdot Q_{kt},$$

where Q_{kt} is total ISO-level generation for fuel type k at time t, and s_{gkm} is generator g's share of monthly generation within fuel type k.

This procedure is implemented separately for each ISO to accommodate differences in reporting conventions, fuel classifications, and renewable aggregation (e.g., hub-level wind and solar reporting in ERCOT and PJM). Special handling is applied where renewables are reported at regional or zonal levels rather than by unit.

1.11.4 Capacity Constraints

To ensure physical feasibility, generator output is capped at nameplate capacity. In earlier versions of the dataset, excess generation was redistributed across generators of the same fuel type within the same ISO and hour. In the current implementation, generator output is capped at nameplate capacity without redistribution, and a capacity-constrained indicator is recorded. This approach avoids introducing artificial correlations across generators while preserving information on binding capacity constraints.

1.11.5 Combining ISO-Level Datasets

After constructing generator-level output for each ISO, the datasets are combined into a unified panel. Datetime formats, fuel categories, and variable definitions are standardized across ISOs. Fuel types are consolidated into consistent categories (e.g., multiple coal types mapped to a single coal category).

The combined dataset is merged with EPA emissions data where available. When observed EPA generation data exist, these values override the constructed generation estimates to improve measurement accuracy.

1.11.6 Allocation to Load and Short-Term Market Outcomes

Generator-level production is next allocated to load-serving zones using an economic dispatch algorithm that determines how generation serves demand across zones and ISOs. This step produces measures of exports, marginal resources, congestion indicators, and delivered prices. The details of this procedure, including transmission constraints and losses, are described in Appendix 1.13.

1.11.7 Augmentation with Weather, Fuel Prices, and Controls

In the final stage, the dataset is augmented with hourly weather variables matched to generation zones, including temperature and wind conditions. Daily fuel prices for natural gas and coal are merged and interpolated to the hourly frequency.

Time indicators (hour, month, year), zone fixed effects, and interactions between fuel indicators and prices are constructed to support econometric analysis. The resulting dataset contains generator-level observations at the hourly frequency, with detailed information on costs, constraints, market conditions, and environmental factors.

1.11.8 Final Output

The final output is an hourly generator-level panel suitable for short-term market analysis. It forms the basis for the reduced-form and structural estimations in the main text.

1.12 Transmission Capacity and Line Loss Estimation

1.12.1 Overview

This appendix describes the construction of the transmission network dataset used in the empirical analysis, as well as the design of the counterfactual transmission scenarios. The goal is to construct a consistent panel of interregional transmission capacity and losses that can be incorporated into market equilibrium and dispatch models. The methodology combines geographic transmission line data, engineering-based approximations of electrical properties, and historical transmission project information to reconstruct the evolution of the U.S. transmission network over time.

1.12.2 Baseline Transmission Network Construction

Data Sources

The baseline transmission network integrates four primary data sources. First, a national shapefile of high-voltage electric power transmission lines provides geographic geometry, voltage class, and current type (AC or DC). Second, U.S. county boundary shapefiles are used to geolocate transmission line endpoints. Third, a commercial transmission projects database provides information on historical and planned line additions and retirements, including in-service dates, voltage levels, and construction type. Finally, Independent System Operator (ISO) county-to-zone mapping tables are used to associate transmission line endpoints with market pricing zones for ISONE, NYISO, PJM, MISO, SPP, and ERCOT.

Geographic Processing

Transmission line lengths are computed directly from shapefile geometries and converted into physical distance units. For each line, start and end points are extracted from the geometry using the first and last coordinates for `LineString` and `MultiLineString` objects. These endpoints are spatially joined to county boundaries to identify the corresponding county, state, latitude–longitude coordinates, and time zone.

Each endpoint is then mapped to an ISO pricing zone using a hierarchical merge procedure. Counties not directly covered by predefined ISO mapping tables are assigned to the ISO with the largest geographic overlap based on spatial intersection with ISO boundary shapefiles.

Temporal Network Construction (2000–2024)

The transmission network is constructed as a balanced annual panel spanning the years 2000–2024. The reference year for the network is 2022, corresponding to the vintage of the underlying shapefile. For years after 2022, new transmission lines are added based on project commissioning dates. For earlier years, the historical network is reconstructed by removing post-year additions and reinstating lines that were retired after the relevant year.

This procedure assumes that the reference shapefile is comprehensive and that the transmission project database captures the universe of major additions and retirements. While imperfect, this approach allows for a consistent reconstruction of interregional transmission capacity over time.

1.12.3 Electrical Characterization of Transmission Lines

Each transmission line is assigned electrical parameters using standard power-systems engineering approximations.

Resistance

Line resistance is computed as

$$R = \frac{\rho L}{A}, \tag{1.51}$$

where ρ denotes conductor resistivity, L is line length, and A is the conductor cross-sectional area. A uniform conductor type is assumed across lines.

Reactance and Impedance (AC Lines)

For overhead AC transmission lines, inductance per unit length is approximated as

$$\mathscr{L} = 2 \times 10^{-7} \ln\left(\frac{D}{r}\right), \tag{1.52}$$

where D is the distance between phase conductors and r is the conductor radius. Conductor spacing is interpolated based on voltage class using standard utility design practices. Inductive reactance is then computed as

$$X = 2\pi f \mathscr{L}, \tag{1.53}$$

with frequency $f = 60$ Hz. The magnitude of total impedance is given by

$$Z = \sqrt{R^2 + X^2}. \tag{1.54}$$

Flow Limits

For AC lines, maximum transferable power is approximated using a simplified surge-impedance-loading expression:

$$P_{AC}^{\max} = \frac{V^2}{X}, \tag{1.55}$$

where V denotes line voltage. For DC lines, thermal limits are used:

$$P_{DC}^{\max} = 2VI^{\max}, \tag{1.56}$$

assuming a bipole configuration.

Line Losses

Losses are calculated at maximum flow. For AC lines,

$$P_{AC}^{loss} = 3I^2R, \qquad I = \frac{P^{\max}}{\sqrt{3}V}, \tag{1.57}$$

while for DC lines,

$$P_{DC}^{loss} = I^2R. \tag{1.58}$$

1.12.4 Zone-Level Aggregation

Transmission lines are aggregated to the ISO zone-pair level by year. For each ordered pair of zones (z, z'), the dataset reports total transfer capacity, total losses at maximum flow, and the average loss rate as a fraction of transmitted power. These aggregates form the transmission constraint inputs used in the market equilibrium analysis.

1.12.5 Counterfactual Transmission Scenarios

Targeted Capacity Modification Scenarios

The first set of counterfactuals examines the effects of selective changes to interzonal transmission capacity. A "no-limits" scenario sets all interzonal flow limits to a sufficiently large value, approximating a copper-plate network and providing an upper bound on the gains from eliminating congestion. Additional scenarios impose targeted capacity restrictions on specific corridors or introduce fixed-capacity expansions between selected zone pairs, with losses scaled proportionally using observed average loss rates.

HVDC Upgrade Scenarios

A second set of counterfactuals examines the implications of upgrading portions of the AC transmission network to High-Voltage Direct Current (HVDC). Scenarios range from 10% to 100% conversion, with lines selected in descending order of length. Upon conversion, AC reactance is eliminated, impedance collapses to resistance only, flow limits increase, and losses are reduced. While converter station costs and operational control benefits are not explicitly modeled, these scenarios provide a reduced-form characterization of how HVDC adoption alters network capacity and efficiency.

1.12.6 Assumptions and Limitations

The transmission dataset relies on several simplifying assumptions. First, conductor properties are assumed to be uniform across lines. Second, thermal and stability limits are approximated using simplified formulas rather than detailed engineering models. Third, aggregation to ISO pricing zones abstracts from intra-zonal congestion. Fourth, flow limits represent static steady-state constraints and do not capture N–1 reliability requirements. Finally, HVDC scenarios exclude the capital costs of converter stations. These assumptions are discussed when interpreting the empirical results.

1.12.7 Summary

The methodology produces a consistent panel of interregional transmission capacity and losses from 2000 to 2024, along with economically interpretable counterfactual scenarios. These data allow the analysis to isolate the role of transmission constraints and technology choice in shaping electricity market outcomes.

1.13 Calculating the Cost of Line Additions

Estimating the cost of new transmission line additions is central to evaluating the welfare implications of network expansion. Transmission investment costs vary substantially with voltage class, conductor type, terrain, right-of-way acquisition, and regional labor markets, making any single estimation approach subject to considerable uncertainty. To address this, I employ four complementary methods: a hedonic regression using project-level data, a bottom-up engineering cost model, a synthesis of estimates drawn from the literature and announced projects, and an ensemble method that combines the preceding three. This appendix details each approach.

1.13.1 Calculation Method 1: Hedonic Regression

The first method applies a hedonic pricing framework to estimate the cost of transmission line additions as a function of observable project characteristics. Hedonic regression, originally developed in the context of housing markets (Rosen, 1974), decomposes the price of a differentiated good into the implicit prices of its constituent attributes. In this application, the "good" is a transmission line project and the attributes include engineering specifications, geographic characteristics, and temporal factors.

Data

The primary data source is the S&P Capital IQ transmission line database, which contains project-level records of transmission investments across the United States. Each record includes the estimated or reported project cost, voltage rating, line length, line type (overhead versus underground, AC versus DC), energization date, and geographic information.

Specification

I estimate the following hedonic regression:

$$\ln(\text{Cost}_i) = \alpha + \boldsymbol{\beta}_1 \mathbf{X}_i^{\text{eng}} + \boldsymbol{\beta}_2 \mathbf{X}_i^{\text{geo}} + \boldsymbol{\beta}_3 \mathbf{X}_i^{\text{constructed}} + \gamma_t + \varepsilon_i \tag{1.59}$$

where Cost_i is the estimated cost of project i (in $\$_{\text{TODO: base year}}$ per mile, or total—specify), and the regressors are organized into three groups:

Engineering characteristics ($\mathbf{X}^{\text{eng}}$). These include voltage class (kV), line length (miles), and line type indicators (overhead AC, overhead DC, underground AC, underground DC).

Geographic and regulatory characteristics ($\mathbf{X}^{\text{geo}}$).

Constructed features ($\mathbf{X}^{\text{constructed}}$). I construct several derived variables to capture the economic function of each line. These include the thermal flow limit (MW), computed from conductor ratings and voltage, and estimated resistive losses (MW-miles), derived from conductor resistance and expected loading.

Time controls (γ_t). Year fixed effects or a time trend capture secular changes in construction costs, commodity prices, and labor markets.

Estimation and Results

Equation (1.59) is estimated by OLS.

The hedonic regression provides a flexible, data-driven mapping from project characteristics to cost. Its principal advantage is that it captures the joint influence of multiple attributes as revealed by actual project expenditures. However, it is limited by the sample available in the Capital IQ database, which may not be representative of all potential projects, and by the possibility of omitted variable bias if unobserved project features (e.g., permitting difficulty, terrain complexity beyond what is captured by controls) are correlated with included regressors.

1.13.2 Calculation Method 2: Bottom-Up Estimation

The second method constructs a cost estimate from the bottom up by aggregating the costs of individual components and inputs required to build a transmission line. This engineering-economic approach provides a transparent, physically grounded estimate that does not rely on regression assumptions.

Cost Components

The total cost of a transmission line can be decomposed into several major categories:

Conductor and hardware. The cost of conductor wire (e.g., ACSR, ACSS, or HTLS conductors), insulators, and associated hardware depends on the conductor type, number of circuits, and bundling configuration.

Structures. Tower or pole costs vary with structure type (lattice steel, tubular steel, wood H-frame), height, span length, and voltage class.

Right-of-way and land acquisition. Land costs depend on terrain, land use (agricultural, urban, forested), and regional property values.

Labor and construction. Construction labor costs are a substantial share of total project costs and vary by region, terrain difficulty, and prevailing wage requirements.

Permitting, engineering, and contingency. Soft costs including environmental review, engineering design, project management, and contingency allowances.

Assembly

The total estimated cost per mile is computed as:

$$C^{\text{BU}} = C^{\text{conductor}} + C^{\text{structures}} + C^{\text{ROW}} + C^{\text{labor}} + C^{\text{soft}} \tag{1.60}$$

where each component is parameterized by voltage class, line type, and region.

The bottom-up method's strength is its transparency and its ability to generate cost estimates for hypothetical line configurations not observed in historical data. Its principal limitation is that input cost estimates may themselves be uncertain, and interactions between components (e.g., terrain affecting both structure and labor costs simultaneously) may not be fully captured.

1.13.3 Calculation Method 3: Literature-Based and Announced Estimates

The third method draws on three categories of external estimates to benchmark and validate the regression and bottom-up approaches.

Academic and Technical Literature

Several studies have estimated transmission line costs in the context of grid expansion planning, renewable energy integration, and cost-benefit analysis.

Announced Project Costs

Utilities and transmission developers periodically announce cost estimates for planned or completed projects in regulatory filings, investor presentations, and press releases.

Congressional Budget Office and Legislative Cost Estimates

The Congressional Budget Office (CBO) has produced cost estimates associated with transmission-related provisions in proposed legislation.

Synthesis

The literature-based estimates provide an independent check on the data-driven methods. Where estimates from Methods 1 and 2 fall within the range reported in the literature and in announced projects, confidence in the primary estimates is strengthened. Where discrepancies arise, they may point to omitted factors (e.g., permitting delays, opposition-related cost escalation) or differences in project scope.

1.13.4 Calculation Method 4: Ensemble

The fourth method combines the estimates from Methods 1–3 into an ensemble estimate. The motivation for an ensemble approach is standard: individual estimation methods are subject to distinct sources of error, and a weighted combination can reduce overall estimation variance provided the errors are not perfectly correlated.

Combination Procedure

Let $\hat{C}^{(m)}$ denote the cost estimate from method $m \in \{1,2,3\}$ and w_m denote the weight assigned to method m, with $\sum_m w_m = 1$. The ensemble estimate is:

$$\hat{C}^{\text{ens}} = \sum_{m=1}^{3} w_m \hat{C}^{(m)} \tag{1.61}$$

Uncertainty Quantification

The ensemble approach provides the cost estimates used in the main analysis. By drawing on data-driven estimation, engineering fundamentals, and external benchmarks, the ensemble mitigates the weaknesses of any single method while preserving the strengths of each.

1.14 Comparison of Estimates with Literature

1.15 Robustness Checks

This appendix describes a series of robustness checks designed to assess the sensitivity of the main results to key modeling assumptions, estimation choices, and counterfactual specifications. Each subsection describes the check, its motivation, implementation, and the expected diagnostic output.

1.15.1 Reliability Diagnostics

The baseline model assumes that blackouts do not occur (Assumption 1 in Section 4.3.1). While this assumption is necessary for the dispatch problem to have a well-defined solution, it raises the question of whether transmission expansion makes the assumption more or less plausible across counterfactual scenarios. I conduct three complementary reliability diagnostics.

Capacity Shadow Price Analysis

The dispatch linear program produces shadow prices θ_{jkt} on the capacity constraints (Equation 4). These shadow prices represent the marginal value of an additional MW of capacity for resource k at generation node j at time t. High values of θ_{jkt} indicate hours where the system is near its capacity limits.

For each counterfactual scenario s, I compute the distribution of $\theta_{jkt}^{(s)}$ across all zone-hours and report the mean, median, 95th percentile, 99th percentile, and the fraction of zone-hours where $\theta_{jkt}^{(s)}$ exceeds a threshold $\bar{\theta}$. The threshold $\bar{\theta}$ is set equal to the 99th percentile of the baseline distribution, so the diagnostic measures how frequently each counterfactual pushes the system into the tail of baseline capacity stress.

Formally, define the reliability stress indicator for scenario s as:

$$R^{(s)} = \frac{1}{|\mathscr{J}| \cdot |\mathscr{K}| \cdot |\mathscr{T}|} \sum_{j,k,t} \mathbf{1} \left[\theta_{jkt}^{(s)} > \bar{\theta} \right]. \tag{1.62}$$

This diagnostic requires no additional computation beyond what is already performed to solve the dispatch problem.

Effective Reserve Margin

For each zone i and hour t under each counterfactual scenario, I compute the effective reserve margin as:

$$\text{ERM}_{it}^{(s)} = \frac{\sum_j \sum_k \min \left(O_{jkt}^{\text{MAX}}, A_{ij}^{(s)} \right) (1 - B_{ij}) - L_{it}}{L_{it}}, \tag{1.63}$$

where $A_{ij}^{(s)}$ is the transmission capacity in scenario s, O_{jkt}^{MAX} is available generation capacity, B_{ij} is the line loss factor, and L_{it} is load. The effective reserve margin captures the fact that transmission expansion increases available capacity at each node by enabling imports from distant generators, even without new generation investment.

I report the distribution of $\text{ERM}_{it}^{(s)}$ across zones and hours for each counterfactual, focusing on the lower tail. I also report the fraction of zone-hours where $\text{ERM}_{it}^{(s)}$ falls below standard planning reserve margins (typically 15%).

Monte Carlo Generator Outage Simulation

To directly assess how transmission expansion affects reliability under realistic contingencies, I conduct a Monte Carlo exercise in which generators experience forced outages. For a representative sample of high-stress hours—defined as hours in which the aggregate capacity shadow price exceeds its 90th percentile—I randomly derate generator capacity according to technology-specific forced outage rates from NERC's Generating Availability Data System (GADS).

Specifically, for each simulation draw d and each generator g, the available capacity is:

$$\tilde{O}_{jkgt}^{\text{MAX},(d)} = O_{jkgt}^{\text{MAX}} \cdot \mathbf{1}\left[u_{jkgt}^{(d)} > \text{FOR}_k\right], \tag{1.64}$$

where $u_{jkgt}^{(d)} \sim \text{Uniform}(0,1)$ is an independent draw and FOR_k is the forced outage rate for resource type k. I then re-solve the dispatch problem with the derated capacities, checking whether the problem remains feasible. If the problem is infeasible—meaning demand cannot be met at some node—the hour is recorded as a loss-of-load event.

The loss-of-load probability (LOLP) for scenario s is estimated as:

$$\widehat{\text{LOLP}}^{(s)} = \frac{1}{|\mathscr{T}^*| \cdot D} \sum_{t \in \mathscr{T}^*} \sum_{d=1}^{D} \mathbf{1}\left[\text{dispatch infeasible under } \tilde{O}_t^{(d)} \text{ in scenario } s\right], \tag{1.65}$$

where $\mathscr{T}^*$ is the set of high-stress hours and D is the number of simulation draws per hour. I use $D = 500$ draws and report LOLP estimates with bootstrapped confidence intervals.

Comparing $\widehat{\text{LOLP}}^{(s)}$ across counterfactual scenarios provides a direct measure of how transmission expansion affects system reliability under contingency conditions, without requiring modification to the model structure.

1.15.2 Alternative State Space Discretizations

The long-run model relies on discretization of the continuous state space (Table **??**). If results are sensitive to the choice of grid density, this would suggest that the discretization introduces artifacts into the counterfactual predictions.

I re-estimate the long-run model under two alternative discretization schemes. The first uses a coarser grid with approximately half the number of bins for each state variable (F_k: 10 bins, E_k: 10 bins, demand: 3 bins, own capacity: 8 bins, rival capacity: 5 bins, price: 5 bins). The second uses a finer grid with approximately 50% more bins (F_k: 30 bins, E_k: 30 bins, demand: 8 bins, own capacity: 22 bins, rival capacity: 15 bins, price: 15 bins).

I report parameter estimates, simulated capacity paths, and key counterfactual outcomes (average price change, renewable adoption by 2050, and welfare gains) under each discretization and compare these to the baseline results. The check is passed if the qualitative conclusions are unchanged and the quantitative magnitudes are within 20% of the baseline estimates.

1.15.3 Sensitivity to Transmission Capacity Measurement

The baseline transmission network is constructed using engineering approximations applied to the HIFLD shapefile (Appendix D). Several assumptions in this construction may affect the results: uniform conductor properties, simplified thermal and stability limits, and the use of static steady-state flow limits that do not account for N$-$1 reliability requirements.

Scaled Transmission Capacity

To assess sensitivity to the overall level of estimated transmission capacity, I re-run the dispatch model with all inter-zonal capacities A_{ij} scaled by factors $\alpha \in \{0.75, 0.9, 1.1, 1.25\}$. These multiplicative adjustments capture the possibility that the engineering approximations systematically over- or underestimate true transfer capability. I report how the distribution of locational marginal prices, the frequency of binding transmission constraints, and the welfare estimates from the HVDC upgrade counterfactual change under each scaling.

Alternative Loss Specifications

I replace the baseline loss estimates with two alternatives: (i) uniform losses set to the load-weighted average loss rate across all corridors, and (ii) losses calibrated to match aggregate transmission and distribution losses reported in EIA's Annual Energy Outlook. These alternatives test whether spatial heterogeneity in line losses is important for the price and welfare results.

1.15.4 Market Power

The equivalence between the social planner's problem and generators' profit maximization relies on the absence of market power (Section 4.3). Transmission constraints can create local market power by limiting competition from distant generators. To assess the potential impact of market power on the results, I implement two checks.

Pivotal Supplier Index

For each zone i and hour t, I compute the pivotal supplier index (PSI), which identifies whether the removal of any single generation company g from zone j would cause demand at node i to exceed remaining available supply (inclusive of imports):

$$\mathrm{PSI}_{igt} = \mathbf{1}\left[L_{it} > \sum_{j}\sum_{k}\sum_{g' \neq g} \min\left(O^{\mathrm{MAX}}_{jkg't}, A_{ij}\right)(1 - B_{ij})\right]. \tag{1.66}$$

I report the frequency with which generators are pivotal in the baseline and in each counterfactual scenario. Transmission expansion should reduce the frequency of pivotal suppliers by increasing the pool of generators that can serve each node. If the frequency of pivotal suppliers is low in the baseline and decreasing in counterfactuals, this supports the maintained assumption that market power does not materially affect the results.

Markup Bounding Exercise

Following Borenstein et al. (2002), I compute residual demand elasticities at the zone-hour level and use these to bound the Lerner index that a profit-maximizing generator could charge. I compare actual locational marginal prices to the upper bound on prices under monopolistic behavior. If the gap between competitive prices and the monopoly bound is small relative to the price effects found in the counterfactuals, this provides evidence that the competitive pricing assumption does not materially affect the welfare conclusions.

1.15.5 Alternative Demand Specifications

The baseline model treats demand as perfectly inelastic and exogenous. Two aspects of this assumption merit robustness investigation.

Price-Responsive Demand

I introduce a small price elasticity of demand into the dispatch model by replacing the fixed load L_{it} with a linear demand function:

$$L_{it}(\mathrm{LMP}_{it}) = L_{it}^0 - \eta \cdot \mathrm{LMP}_{it}, \tag{1.67}$$

where L_{it}^0 is the baseline load and η is calibrated to match short-run demand elasticity estimates from the literature. I use η values corresponding to elasticities of -0.05 and -0.10 evaluated at mean price and load, consistent with estimates from Ito and Reguant (2016) and Jessoe and Rapson (2014).

Introducing price-responsive demand modifies the dispatch problem from a linear program to a quadratic program, but the problem remains convex and computationally tractable. The key question is whether price-responsive demand amplifies or dampens the welfare effects of transmission expansion. Because transmission expansion reduces prices in high-cost regions (where demand response would increase consumption) and raises prices in low-cost regions (where demand response would reduce consumption), the net effect on welfare depends on the relative magnitudes of these adjustments.

Load Growth Sensitivity

The long-run counterfactuals use load projections from EIA's Annual Energy Outlook. These projections may understate future load growth if electrification of transportation and heating proceeds faster than anticipated, or overstate growth if energy efficiency improvements exceed expectations. I re-run the long-run counterfactuals under alternative load trajectories corresponding to the AEO's high and low economic growth scenarios. This tests whether the complementarity finding between IRA and BIL is robust to uncertainty in demand growth.

1.15.6 Alternative Equilibrium Concepts

The long-run model uses an oblivious equilibrium concept in which firms condition investment decisions on their own state and aggregate rival capacity rather than the full joint distribution of competitor states (Appendix B.3.2). While this approximation substantially reduces computational burden, it may miss strategic interactions that matter for investment dynamics.

I implement two checks. First, I increase the dimensionality of the rival capacity state by splitting it into "same-fuel rival capacity" and "other-fuel rival capacity," allowing firms to condition investment on the composition of competitive supply. This tests whether the aggregation of rivals into a single state variable obscures economically important variation.

Second, I compare the oblivious equilibrium investment paths to those generated by a myopic model in which firms treat current prices and costs as permanent when making investment decisions (i.e., $\beta = 0$ in the value function). If the qualitative results are similar under myopic and forward-looking behavior, this suggests that the specific equilibrium concept is not the primary driver of the counterfactual results. If the results differ substantially, this identifies the role of strategic forward-looking behavior in shaping investment responses to transmission expansion.

1.15.7 Sequential vs. Joint Estimation

The baseline estimation proceeds sequentially: short-run parameters are estimated first and held fixed during long-run estimation (Appendix B.4.4). This approach is computationally necessary but may introduce bias if estimation error in the short-run parameters affects the long-run estimates non-linearly.

To assess the magnitude of this concern, I construct bootstrap confidence intervals that account for first-stage estimation uncertainty. Specifically, I draw from the estimated asymptotic distribution of the short-run parameters, re-compute the short-run profit mapping for each draw, and re-estimate the long-run model conditional on the perturbed short-run inputs. I report the resulting distribution of long-run parameter estimates and counterfactual welfare effects. If the bootstrap confidence intervals are substantially wider than those computed from second-stage GMM standard errors alone, this would indicate that first-stage uncertainty is quantitatively important.

1.15.8 Sensitivity of Counterfactual Policy Parameters

The policy counterfactuals in Section 7.3 require assumptions about the magnitude of IRA tax credits, the timing of BIL transmission expansions, and the corridor-level allocation of new capacity under the BIG WIRES Act. I assess sensitivity to each of these inputs.

IRA Credit Magnitude

The baseline implementation sets the investment tax credit reduction τ_k^{IRA} based on the statutory credit values. I re-run the IRA counterfactual with τ_k^{IRA} scaled by factors of 0.5

and 1.5, corresponding to scenarios where effective credits are lower (due to qualification barriers, domestic content requirements, or phase-outs) or higher (due to bonus credits for energy communities or low-income areas) than the statutory values.

BIL Expansion Magnitude and Allocation

For the BIL counterfactual, I compare the corridor-specific specification (where $\Delta A_{ij}^{\text{BIL}}$ is assigned to individual lines based on announced projects) with the system-wide proportional expansion ($\Delta A_{ij}^{\text{BIL}} = \alpha \cdot A_{ij}^{\text{baseline}}$) under alternative values of $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. This tests whether the welfare and investment results are driven by the specific corridors receiving capacity additions or by the overall increase in system transfer capability.

Ramp Function Sensitivity

Policy wedges are phased in using a ramp function $\rho(t; t_0, R)$ with baseline ramp period $R = 1$ year. I test $R \in \{0, 2, 5\}$ years to assess whether the speed of policy implementation affects the long-run equilibrium. Instantaneous implementation ($R = 0$) provides an upper bound on short-run disruption, while slower phase-in periods test whether gradual adjustment changes the investment dynamics.

1.15.9 Exclusion of Individual ISOs

To verify that the results are not driven by the idiosyncratic characteristics of any single ISO, I re-estimate the short-run model excluding each ISO in turn and report the resulting parameter estimates and counterfactual price effects. ISOs differ substantially in resource mix, congestion patterns, and market size (Table 1), so parameter stability across these leave-one-out samples would provide evidence that the estimates reflect general features of the dispatch technology rather than market-specific artifacts.

For the long-run model, full re-estimation excluding individual ISOs is computationally intensive. Instead, I compute the counterfactual welfare effects and renewable adoption paths excluding each ISO's zones from the simulation while holding parameters fixed. This tests whether the aggregate results are driven disproportionately by investment dynamics in any single market.

1.15.10 Temporal Stability of Short-Run Estimates

The short-run model is estimated on data from 2018–2023, a period that includes both the COVID-19 pandemic (which substantially depressed electricity demand in 2020) and the natural gas price spike associated with the 2022 energy crisis. To test whether these events distort the parameter estimates, I re-estimate the short-run model on two subsamples: 2018–2019 (pre-pandemic) and 2021–2023 (post-pandemic). Parameter stability across subsamples would support the maintained assumption that the dispatch technology is time-invariant, while instability would suggest that structural breaks in fuel markets or demand patterns affect the estimates.

I also estimate the model on rolling two-year windows (2018–2019, 2019–2020, 2020–2021, 2021–2022, 2022–2023) and plot the evolution of key parameters over time. Persistent trends in the fuel cost parameters would indicate time-varying marginal costs not captured by the included fuel price covariates.

1.15.11 Summary of Robustness Architecture

Table 1.10 provides an overview of all robustness checks, the assumption being tested, and the primary diagnostic output.

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Table 1.10: Summary of Robustness Checks

Check	Assumption Tested	Diagnostic Output
Capacity shadow prices	No blackouts	Distribution of θ_{jkt} across scenarios
Effective reserve margin	No blackouts	Fraction of zone-hours below 15% reserve
Monte Carlo outages	No blackouts	LOLP estimates with confidence intervals
Alternative discretizations	State space grid density	Parameter and counterfactual sensitivity
Scaled transmission capacity	Transmission measurement	Price distribution, welfare sensitivity
Alternative losses	Loss heterogeneity	Price and welfare effects
Pivotal supplier index	No market power	Frequency of pivotal generators
Markup bounding	No market power	Gap between competitive and monopoly prices
Price-responsive demand	Inelastic demand	Welfare amplification or dampening
Load growth sensitivity	AEO projections	Robustness of complementarity finding
Rival capacity decomposition	Oblivious equilibrium	Investment path sensitivity
Myopic vs. forward-looking	Dynamic behavior	Role of expectations in investment
Bootstrap first-stage	Sequential estimation	Confidence interval comparison
IRA credit scaling	Statutory credit values	Welfare and investment sensitivity
BIL allocation	Corridor-specific vs. proportional	Welfare decomposition
Ramp function	Phase-in speed	Long-run equilibrium sensitivity
Leave-one-out ISOs	No single-ISO dependence	Parameter and welfare stability
Temporal subsamples	Time-invariant technology	Parameter evolution over time

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Chapter 2

Learning-by-Doing and Industrial Policies: Evidence from China's Solar Panel Industry

Abstract.

Crystalline silicon (c-Si) technology has dominated in the solar panel manufacturing and the production costs have declined by XX% over the past decade. This study investigates the role of learning-by-doing (LBD) in driving this cost reduction and forming the technological pathway of the industry, and its interaction with input price and government subsidy policies.

Keywords:

JEL Classification Codes:

2.1 Introduction

The development of solar energy plays a crucial role in the transition to a sustainable and low-carbon electricity grid and the mitigation of climate change. In recent decades, production costs of solar panel have fallen by XX%. Meanwhile, the share of solar panel manufactured using crystalline silicon (c-Si) technology has increased from 90% ? in 2000 to 98% in 2020. Industry experts have attributed this technological dominance and this substantial cost reduction largely to learning-by-doing (LBD), the maturing of upstream industries, and industrial policies. Nonetheless, during this period of time, polysilicon, the main input of c-Si technology, experienced a price reduction of 95% between 2008 and 2013 while the prices of the main input of competing thin-film technology have remained relatively stable.

While industrial policies are often controversial in many sectors, certain important features of renewable energy, such as solar panel in this study, deserve careful consideration. Subsidies on either demand or supply side can be justified due to various market failures, including environmental externalities and knowledge spillovers. In addition, individual countries may benefit from government support or protection when the industry is still in its infancy. However, how LBD interacts with industrial policies to shape production costs and technological pathways is not well understood. This paper aims to fill this gap by quantifying the extent to which LBD contributes to reducing solar panel cost and driving the growth of the c-Si technology, and by evaluating how industrial policies on both demand and production sides influence solar panel diffusion, the expansion of the c-Si technology, and overall social welfare.

Quantifying LBD is crucial for evaluating the impacts of various industrial policies. Subsidies aimed at promoting solar panel adoption have been widely implemented by governments worldwide. LBD creates a positive “feedback loop", analogous to the mechanism described in Barwick et al. (2025b) on the interaction between the EV subsidies and the LBD in battery manufacturing. In our setting, demand-side subsidies stimulate greater solar panel uptake, which increases manufacturers' production experience. This accumulated experience strengthens LBD, driving down production costs and solar panel prices, which in turn further increases adoption and magnifies the overall impact of demand-side subsidies.

Industrial policies aimed at fostering the development of upstream industries have been adopted by most governments in recent years. China, for example, supported the establishment of a domestic silicon sector, although initially to provide inputs for semiconductors rather than solar panel. This support was proved to be advantageous when silicon prices spiked in late 2007, which was also the first year the solar industry used more silicon worldwide than the chip industry. More recently, the U.S. has offered generous support to the manufacture of rare earth elements which are key inputs of semiconductors. LBD also generates a positive "feedback loop": falling prices of a key input of a particular technology induce firms to adopt that technology, which increases experience in production of that cheaper-input technology. LBD associated with the enhanced production experience of the cheaper-input technology further reduces the cost of the cheaper-input technology and solar panel price. These cost and price reductions of the cheaper-input technology, in turn, further accelerate the adoption of that technology, amplifying the direct effects of demand- and supply-side policies.

[Describe our model!]

Our empirical analysis delivers three key findings. First, the learning rate is estimated to be XX% after controlling for .... That is, doubling solar panel manufactures' production experience would reduce unit production costs by XX%. This implies that LBD alone accounted for XX% of the overall decline in the solar panel costs from 2000 to 2013. Compare with other studies on learning rates. For example, Barwick et al. (2025b) finds a learning rate of 7.5% of EV battery production from 2013 to 2020.

Second, LBD greatly amplifies the impact of the price decline of silicon through positive feedback loops. In the absence of LBD, the price drop of silicon is estimated to increase cumulative solar panel sales by XX% and increase the share of c-Si technology by XX%

during the sample period. When LBD was also in effect, solar panel sales surged by XX% and the share of s-Si technology increased by XX% relative to the baseline with neither silicon price drop nor LBD. This combined effect is XX% greater than the sum of the effects from silicon price drop and LBD individually, suggesting their complementarity.

Third, LBD greatly amplifies the impact of subsidies, including subsidies on installation and subsidies on production costs and entry costs. In the absence of LBD, subsidies across different regions are estimated to increase cumulative solar panel sales by XX% during the sample period. When both subsidies and LBD were in effect, solar panel sales surged by XX% relative to the baseline with neither subsidies nor LBD. These findings highlight the importance of accounting for LBD when evaluating the efficacy and cost-effectiveness of government policies designed to promote solar panel production and adoption. Ignoring LBD would lead to a substantial underestimation of the long-term benefits of such policies.

Literature

Industrial Policies of solar panel Our paper contributes to the growing literature that studies policies in the solar panel industry. Many of these studies, including De Groote and Verboven (2019), focus on the impact of demand-side subsidies. One exception is Gerarden et al. (2025), who examines how import tariffs affect market equilibrium outcomes and compares import tariffs to supply-side subsidies. To the best of our knowledge, our study is the first to study the impact of both demand- and supply-side policies on the production cost and technological pathways, explicitly accounting for LBD. Our results highlight that ignoring LBD would significantly underestimate the impact of government policies on solar panel adoption.

Learning by Doing Our paper contributes to the empirical literature on LBD that has been documented in a variety of industries, for example, Head (1994) and Ohashi (2005) on steel, Thompson (2001) and Thornton and Thompson (2001) on shipbuilding, Benkard

(2000) and Benkard (2004) on aircraft, Su et al. (2025) on space launch and etc. Almost all previous studies relied on data on input requirements or costs associated with producing a product, but these data are often hard to obtain for researchers. Exceptions include Covert and Sweeney (2022) and Barwick et al. (2025b). Our approach is in the same spirit of theirs by exploiting variations in prices and quantities of the final products (solar panel) and main inputs (polysilicon for c-Si technology and Cadmium and Tellurium for thin-film technology).

In particular, our paper contributes to the growing literature studying learning-by-doing in low-carbon technology sectors, such as solar panel, wind generating systems, and electric vehicles. Barwick et al. (2025b) ... Helveston et al. (2022) estimates a two-factor learning model using the annual installed capacities and prices of solar panel modules in Germany, U.S., and China between 2006 to 2020 and finds that the learning rates are 20%, 26%, and 33% in these three countries, respectively.

Technological Path

Estimation of Dynamic Games Finally, our paper is build on the literature on the estimation of dynamic games, including Bajari et al. (2007), Ryan (2012), and Sweeting (2013).

2.2 Solar Panel Industry and Data

2.2.1 Technologies of Solar Panel Manufacturing

There have been two major technologies to manufacture solar panel in the past three decades. The first one is the crystalline silicon technology (c-Si), which was invented by scientists at Bell labs in 1954. Crystalline silicon technology was initially utilized in space installations before being used in the first solar building at the University of Delaware in

1973. Silicon panels quickly improved from below 10% efficiency at inception to 20% efficiency in the field currently. Crystalline silicon benefits greatly from its durability and efficiency. A power plant developed using crystalline silicon will require a significantly smaller area than a power plant developed using thin-film technology. The technology was invented and first commercialized in the United States with the United States still the largest manufacturer through 2010 before becoming a technology dominated by China in the 2010s.

Most c-Si solar panels are produced by vertically integrated manufacturers that use polysilicon to manufacture wafers, then process these wafers into solar cells that generate electricity through the photovoltaic effect, and finally assemble the cells into solar panels (or “modules"). Because this technology is highly commoditized, we treat solar panels as homogeneous, conditional on the underlying production technology. This assumption is further supported by Gerarden et al. (2025) who analyze a dataset of solar panel shipments to the US, constructed from quarterly IHS Markit data, and find very little price dispersion across manufacturers at any given time.

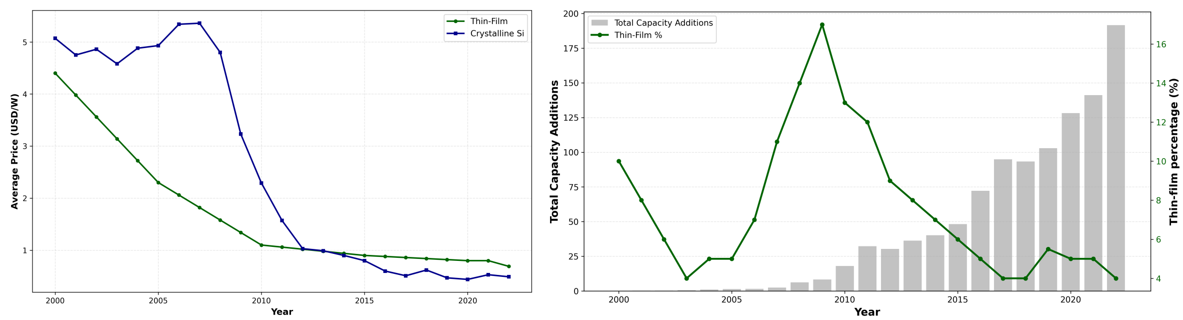
Crystalline silicon competes the thin-film technology, which can be made from a variety of materials including Cadmium, Tellurium, and Silver. Compared with c-Si technology, the thin-film technology is less efficient – that is, it converts a smaller proportion of solar energy into electricity. However, the thin-film technology is more flexible and potentially can be used in a wider variety of configurations. The lower fixed costs and flexibility associated with thin-film technologies made them ideal for traveling outdoors and other adaptable situations. This made them more commonly used in countries like the United States and less commonly used for utility-scale power generation. Moreover, material costs are stable and projected to be cheaper. The major inputs such as Cadmium and Tellurium are the by products of steel production.

Figure 2.1 displays the price and installed capacity of the two types of solar panel since 2000. Over the last two decades, there has been a significant decrease in global solar panel prices, with the most noticeable decline occurring after 2008. This price decline has spurred the installation of solar panels worldwide. Additionally, the rapid expansion of solar panel installations is driven by government subsidies on a global scale.

Figure 2.1: Price and Installed Capacity

(a) Price

(b) Installed Capacity

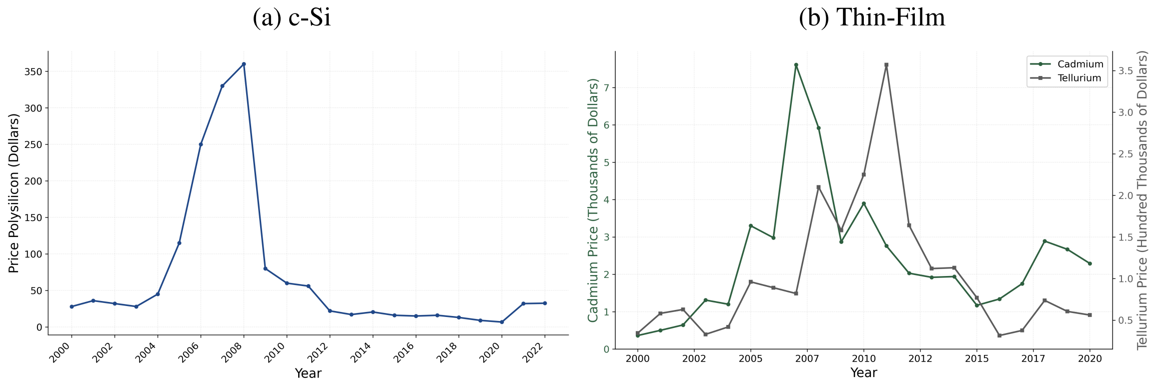


Noticeably, the price of c-Si solar panels was notably high and even rising before 2008, but it began to drop rapidly after 2008 and has remained low since. A major factor in this trend is the price of polysilicon, the main component of c-Si solar panels, which experienced a sharp increase in late 2007, as shown in Figure 2.2(a). This surge was mainly due to the growing demand from the semiconductor sector. This situation led to increased investment, which do not depend on polysilicon, and the prices of the primary inputs for this type of solar panels remained quite stable, as illustrated in Figure 2.2(b). Meanwhile, it also triggered substantial investments in polysilicon production, primarily in the U.S.. As new production came on line, along with the global recession staring in 2008, the price of polysilicon declined rapidly, falling even below pre-surge levels.

2.2.2 Industrial Policies

The rapid expansion of solar panel adoption is partly driven by a variety of industrial policies implemented around the world. These policies fall into two primary categories. The first consists of demand-side policies, designed either to encourage consumers to install solar panel systems or to discourage the use of certain types of solar panels. The second comprises production-side policies, which directly affect manufacturers' costs related to

Figure 2.2: Price of Primary Inputs



producing solar panel, as well as entering or exiting the market.

Demand-Side Policies Demand-side subsidies to encourage the adoption of solar panels have been implemented worldwide. The most widely used form is feed-in tariffs, in which governments provide a fixed payment for each unit of renewable electricity fed into the grid. These tariff rates differ substantially over time and across countries. A second form of demand subsidy is output-based support, such as net metering. A third common type of subsidy is direct payments or tax credits to cover a proportion of the upfront investment costs.

The primary form of subsidy in the EU has been feed-in tariffs, in place since the early 2000s. For example, Germany and Spain offered solar feed-in tariffs of 40–60 cents per kWh, far exceeding the wholesale electricity prices received by other forms of electricity generations. By contrast, the primary federal demand-side subsidy in the US is the Investment Tax Credit, which provided a 30% subsidy on upfront capital costs. In 2009, China launched the "Golden Sun Program" to speed up the deployment of solar panel power plants, covering up to 50% of investment costs through 2013, after which it was replaced by other policy instruments.[1] In addition, China implemented feed-in tariffs in 2010, with rates comparable to those offered in the EU.

[1]Further details on the program can be found in International Energy Agency (2021).

The rapid decline in solar panel prices has not been without controversy. Nemet (2019) uses "gift to the world" to refer to the dramatic production cost reductions achieved by Chinese solar panel manufacturers. However, other nations view China's dominance in solar panel manufacturing as a competitive threat. This perception is partly due to accusations of unfair trade practices and industrial policies by the Chinese government. In response, both the US and the EU have implemented a range of measures aimed at safeguarding their domestic industries since 2010. The first round of anti-dumping and countervailing duties was introduced in 2012. In addition, both the US and the EU start to impose tariffs on c-Si solar panels imported from China. Additional information on US tariffs on solar panels is provided in Gerarden et al. (2025).

The sharp drop in solar panel prices has generated considerable debate. Nemet (2019) characterizes the substantial reductions in production costs achieved by Chinese solar manufacturers as a "gift to the world." Yet many countries regard China's dominant position in solar panel production as a competitive threat. This concern is fueled in part by allegations of unfair trade practices and government-backed industrial policies in China. In reaction, the US and the EU have, since 2010, adopted various measures to protect their domestic industries. The first wave of anti-dumping and countervailing duties was imposed in 2012. Moreover, both the US and the EU began levying tariffs on c-Si solar panels imported from China. Further details on US tariffs on solar panels can be found in Gerarden et al. (2025).

Production-Side Policies Direct subsidies to producers are relatively limited in the EU and the US,[2] but are prevalent in many emerging economies, especially in China. China supported the solar panel manufacturing industry through an extensive range of policy instruments. During the initial growth phase, most of solar panel production subsidies were granted by provincial and local government. These took the form of reduced land prices or

[2] Some manufacturers have benefited from targeted subsidies, such as R&D loans or funding to renewable energy.

rents, low-interest loans, tax exemptions, and decreased electricity rates.[3]

During the later stage of the surge, the central government stepped in. In 2010, the State Council declared solar panel as more than just renewable energy, labeling it a "strategic emerging industry". State-owned banks supported this industry by extending billions of dollars in credits, as mentioned in Lexology (2010) and Nahm (2023). The central government also supported the creation of a domestic silicon industry, although the initial goal was to supply inputs for semiconductors instead of solar panels. According to data from WaferPro (2024), China's polysilicon production surged from 6,000 Metric Tons in 2000 to 410,000 in 2019.

However, systematic data on the quantitative impact of production-side industrial policies is still scarce. Since these policies cannot be directly measured in practice, we follow an indirect strategy: we use detailed firm-level data on production and entry/exit to infer the implied subsidies on production and entry costs by assuming that the observed decisions arise from the equilibrium of a suitably specified model. This methodology has been employed, among others, by Barwick et al. (2025a) in their analysis of the Chinese shipbuilding industry.

2.2.3 Data

Our analysis combines data from multiple sources spanning the years 2000 to 2013.

Chinese Solar Panel Export Volumes The first source is the *Chinese Customs Transactions*, which contains the quantity and trade value of all Chinese solar panel export transactions. Using this dataset, we construct export volumes at the firm–destination–year level. For the empirical analysis, we group export destinations into five regions: the EU, India, Japan, the US, and the Rest of the World (ROW).

[3] For example, Suntech in Jiangsu benefited from \$3.7 billion in low-interest loans from a state-owned bank and an additional \$1.4 billion in tax rebates and other export support subsidies from 2005 to 2012, as noted in Urban et al. (2016).

Chinese Solar Panel Manufacturers The second data source is the *Annual Survey of Manufacturing*, an annual survey administered by the Chinese National Bureau of Statistics that provides detailed information on the manufacturing firms surveyed. This dataset has been used in various academic studies, including Roberts et al. (2018) on the footwear industry and Barwick et al. (2025a) on shipbuilding, among others. [Some description about how we identify solar panel manufacturing firms.] For each manufacturer in each year, the survey reports its output, total fixed assets, and a set of time-varying characteristics, such as location (province), ownership status (whether it is state-owned, privately owned, or foreign firm), and the year in which the firm was established. Using data on total fixed assets, we construct measures of capital stock and investment following the methodology in Brandt et al. (2012) and Barwick et al. (2025a).

Regional-Level Solar Panel Installation and Demand Incentives We obtain quarterly information on the solar panel installation for each technology for each of the six economic regions – China, EU, India, Japan, US, and ROW – based on data from Our World in Data (2025) and percentages drawn from various news and literature sources cite. In addition, we collect information on demand subsidy policies in each economic region, including tax credit or rebates, feed-in tariff rates, and import tariff rates on c-Si solar panels. The data on tax credits and rebates and feed-in tariff rates is sourced from *IEA's database* and also Gerarden (2023).

Prices of Solar Panel and Inputs for each Technology We obtain the quarterly global price per Watt for each of the two technologies, along with the quarterly global price for Polysilicon and the annual prices for Cadmium, and Tellurium. The quarterly Polysilicon price data is sourced from *Bloomberg New Energy Finance* and *Bernreuter Research*, while the annual prices of Cadmium, Tellurium, silver, and aluminum are obtained from the *United States Geospatial Services (USGS)*.

Auxiliary Data We supplement our main data with several auxiliary data. First, we compile aggregate annual production figures for the two solar panel technologies in regions outside China over the sample period. Second, we collect socioeconomic variables (such as GDP and total electricity consumption) for each economic region. Third, we obtain the global prices for key inputs used in the two solar panel manufacturing technologies, including polysilicon, cadmium, and tellurium.

2.2.4 Descriptive Evidence

Figure 2.3(a) presents China's production of the two solar panel technologies and its share in the global market. China's solar panel manufacturing industry was miniscule in the early 2000s. The cell production was 3 megawatts (MW) in 2000, accounting for less than 2 percent of global output. China's share then jumped to 14 percent in 2006 and grew fast over the following five years. By 2011 it reached 60 percent and has remained above that level ever since. The figure also shows that China's solar panel manufacturing is dominated by c-Si technology, whereas the global share of thin-film fluctuates over time, particularly in response to changes in polysilicon prices, as shown in Figure 2.1(b).

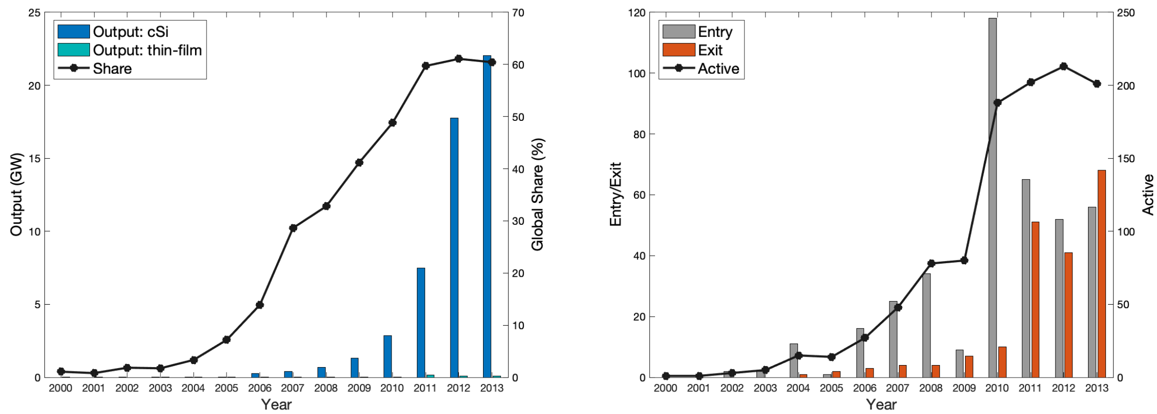
Figure 2.3(b) displays the number of new entries, exits, and active solar panel firms in China during this time period. It indicates that a massive wave of new solar panel firm entries occurred after China declared solar panel as a "strategic emerging industry" in 2010. New entries jumped to 118 in 2010 and remained above 50 in subsequent years. At the same time, a comparable volume of firm exits kept the total number of active firms relatively stable at about 200.

As illustrated in Figure 2.1(a), solar panel prices stayed high in the early 2000s, reached a peak in 2007, started to fall after 2008, and remained low from 2011 onward. In contrast, China's production grew steadily before 2009 and then accelerated after 2010, closely aligning with the timing of major industrial policy rollouts.

Figure 2.3: China's Market Share and Number of Firms

(a) Market Share

(b) Number of Firms



2.3 Model

In this section, we present a dynamic model of firm entry, technology adoption, exit, and production. Time is discrete and a period is a year, denoted as t. There are two types of technologies, with $n = 1$ denotes c-Si and $n = 2$ denotes thin-film solar panels. The global market is divided into $m = 1, ..., m$ economic regions, including China, EU, India, Japan, US, and rest of the world (ROW). In any given period, there exist $j = 1, ..., J_{nt}$ firms in the world that produce solar panels using technology n.

At the beginning of every period, incumbent firms observe their scrap values and decide whether to exit the market. If they choose to stay, these incumbents observe their production cost shocks for each region and determine how many units to produce for each region. Meanwhile, potential entrants observe their entry shocks and decide on whether to enter and, if they do, choose which technology to adopt.

2.3.1 Per-Period Demand

The aggregate inverse demand for technology-n solar panel is expressed as:

$$Q_{nmt} = Q_{nm}(P_{1mt}, P_{2mt}, X_{mt}), \tag{2.1}$$

where Q_{nmt} is the total quantity of technology-n solar panel products demanded by consumers in region m during time period t. The vector X_{mt} includes various demand shifters, such as overall economic activity indicators, regional fixed effects, and year fixed effects. The variable P_{nmt} is the price faced by consumers to install a technology n solar panel system, encompassing both the retail panel price and installation costs. This consumer price may differ from the global price P_{nt} due to region-technology-specific tariffs and installation subsidies. Specifically, the consumer price is given by:

$$P_{nmt} = P_{nt} \times (1 + \text{tariff}_{nmt}) + P^I_{mt} - \text{subsidy}_{mt},$$

where tariff_{nmt} is the tariff rate, which applies only to c-Si technology, P^I_{mt} is the installation costs, and subsidy_{mt} is the subsidy amount provided by region m in period t.

2.3.2 Incumbents: Production and Exit

At the beginning of each period, incumbents observe their private scrap values that are drawn from a technology-specific distribution, and then decide to either exit or continue operations. We denote the scrap value for technology-n incumbent j as ϕ_{jnt} and its distribution as $F_{\phi,n}$. If a firm exits, it receives its scrap value. If it remains active, it observes its marginal cost shocks across all regions, denoted as $\omega_{jnt} = \{\omega_{jnmt}\}_{m=1}^{M}$. Remaining active, firms engage in Cournot competition while considering how their current output decisions will influence future production costs. They simultaneously decide the quantity of solar panels to produce for each region, $\mathbf{q}_{jnt} = \{q_{jnmt}\}_{m=1}^{M}$, to maximize their discounted profit, taking as given the production decisions of rival firms. This new output, $\sum_{m=1}^{M} q_{jnmt}$, is added to the next period's accumulated experience, which in turn affects its future production costs.

Production Cost The production cost for firm j using technology n, when choosing a vector $\mathbf{q}_{jnt}$, consists of two parts:

$$TC_n(\mathbf{q}_{jnt}, s_{jnt}, \omega_{jnt}) = C_0(\sum_{m=1}^{M} q_{jnmt}, s_{jnt}) + \sum_{m=1}^{M} C_m(q_{jnmt}, \omega_{jnmt}), \tag{2.2}$$

where the first component, $C_0(\sum_{m=1}^{M} q_{jnmt}, s_{jnt})$, represents the total production cost of $\sum_{m=1}^{M} q_{jnmt}$, which depends on a vector of state variables s_{jnt}. These state variables include the firm's accumulated experience e_{jnt}, and firm characteristics such as capital, age, location, ownership status, as well as aggregate cost shifters affecting all firms (e.g., input price), and policy dummies that reflect the impact of industrial policies on production costs, including indicators for specific time periods and provinces. The second component, $C_m(q_{jnmt}, \omega_{jnmt})$, accounts for the additional cost of transporting q_{jnmt} to region m in year t.

The experience of firm j, denoted as e_{jnt}, can potentially reduces production costs in the future, incentivizing the firm to increase output to gain more experience in the current period. This experience is defined as the cumulative past production:

$$e_{jnt} = \sum_{\tau < t} \sum_{m} q_{jnm\tau},$$

where $q_{jnm\tau}$ is the units sold by firm j to region m during year τ. The primary parameter of interest is the learning coefficient, which indicates how rapidly the unit cost of solar panel manufacturing declines as the production experience grows.

Incumbent Value Function The value function of a technology-n incumbent j, after observing the scrap value ϕ_{jnt} but before observing the realization of the production cost shock ω_{jnt}, is defined as:

$$V_n(s_{jnt}, \phi_{jnt}) = \max\{\phi_{jnt}, CV_n(s_{jnt})\}, \tag{2.3}$$

where the continuation value $CV_n(s_{jnt})$ is expressed as:

$$\begin{aligned} CV_n(s_{jnt}) \equiv \; & \mathbb{E}_\omega \Big[max_{\mathbf{q}} \{ \sum_m P_{nm}(q_m, \mathbf{q}_{-mt}, X_{mt}) q_m - TC_n(\mathbf{q}, s_{jnt}, \omega_{jnt}) \\ & + \beta \mathbb{E}[V_n(s_{jnt+1}) | s_{jnt}, \mathbf{q}] \} \Big]. \end{aligned} \tag{2.4}$$

Here, $\mathbf{q} = \{q_m\}_{m=1}^M$ is a vector of output choice for each region, while $\mathbf{q}_{-mt}$ denotes the output vectors that are chosen by other competing solar panel firms for region m. The expected value $\mathbb{E}_\omega$ is calculated with respect to the production cost shocks ω_{jnt},

Optimal Exit Decision The optimal decision to exit follows a threshold structure. A firm exits the market if the scrap value exceeds the continuation value; otherwise, it remains active. Therefore, the probability of exiting is:

$$p_n^x(s_{jnt}) \equiv Pr(\phi_{jnt} > CV_n(s_{jnt})) = 1 - F_{\phi,n}(CV_n(s_{jnt})). \tag{2.5}$$

Optimal Production Decision Provided that firm j with technology-n stays active, the optimal output level, $q_{jnmt}^* = q_{nm}^*(s_{jnt}, \omega_{jnmt})$, satisfies the following first-order condition:

$$\begin{aligned} \frac{\partial TC_n(\mathbf{q}_{jnt}^*, s_{jnt}, \omega_{jnt})}{\partial q_{jnmt}^*} = \; & \underbrace{P_{nmt} + q_{jnmt}^* \frac{\partial P_{nm}(q_{jnmt}^*, \mathbf{q}_{-mt}^*, X_{mt})}{\partial q_{jnmt}}}_{\text{Impact on current profit}} \\ & + \underbrace{\beta \frac{\partial \mathbb{E}[V_n(s_{jnt+1}) | s_{jnt}, \mathbf{q}_{jnt}^*]}{\partial q_{jnmt}}}_{\text{Impact on future profits via LBD}}, \end{aligned} \tag{2.6}$$

where $\mathbf{q}_{-mt}^*$ is a vector of equilibrium outputs that are chosen by all other competing solar panel firms for region m during year t.

Equilibrium Global Price In each period, the global price of technology-n solar panels, denoted as P_{nt}, equates the aggregate demand and supply, where the aggregate demand is described by equation (2.1) and the aggregate supply is the sum of q_{jnmt}^* defined in equation

(2.6) across all firms using technology n.

2.3.3 Potential Entrants: Entry and Technology Selection

At the end of each period t, a total of $\bar{N}$ potential entrants observe their profit shocks ζ_{j0t} for not entering, as well as entry costs $\kappa_1(s_{jnt}) - \zeta_{j1t}$ for entering with c-Si technology, and $\kappa_2(s_{jnt}) - \zeta_{j2t}$ for entering with thin film technology. The terms $\kappa_1(s_{jnt})$ and $\kappa_2(s_{jnt})$ denote the expected entry costs associated with the two technologies. The vector s_{jnt} includes policy dummies to capture the impact of industrial policies on the decisions regarding entry and technology selection.

Upon observing their profit shocks ζ's, potential entrants make a decision about entering the market and, if they choose to enter, select a technology. Should they decide to enter, they incur the entry cost during the current period and start earning profit from the product market starting from the following period. Therefore, the potential entrant j's maximization problem can be formulated as:

$$max\{\zeta_{j0t}, max_{n\in\{1,2\}}\{-\kappa_n(s_{jnt}) + \zeta_{jnt} + \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\}\}\} \tag{2.7}$$

A firm stays out of the market if its profit shock from not entering exceeds the expected profit of entering with either technology option. It chooses to enter with technology n if this option provides the highest profit. We use $p^{e,n}(s_{jnt})$ to denote the probability that a potential entrant j enters with technology n, given by

$$\begin{aligned} p^{e,n}(s_{jnt}) \quad \equiv \quad & \Pr\{\zeta_{jnt} - \zeta_{jn't} > \kappa_n(s_{jnt}) - \kappa_{n'}(s_{jn't}) + \beta\mathbb{E}[V_{n'}(s_{jn't+1}) \mid s_{jn't}] - \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}] \\ & \& \ \zeta_{jnt} - \zeta_{j0t} + > \kappa_n(s_{jnt}) - \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\} \end{aligned} \tag{2.8}$$

2.3.4 Markov-Perfect Equilibrium

A Markov-Perfect Equilibrium is characterized by policy function, $q^*_{nm}(s_{jnt}, \omega_{jnmt})$, $p^x_n(s_{jnt})$, $p^e_n(s_{jnt})$, value function $V_n(s_{jnt})$, and global price P_{nt}, such that production policy function $q^*_{nm}(s_{jnt}, \omega_{jnt})$ satisfies (2.6), exit policy function $p^x_n(s_{jnt})$ satisfies (2.5), entry policy function $p^e_n(s_{jnt})$ satisfies (2.8), global price P_{nt} ensures that aggregate demand and aggregate supply are in balance, while the incumbent's value function $V_n(s_{jnt})$ satisfies (2.3).

2.3.5 Discussion of Model Assumptions

Before moving on to the estimation, we discuss two key simplifying assumptions made by our model. First, cost shocks ω_{jnmt} are assumed to be i.i.d., because allowing for a persistent time-varying unobserved state variable in a dynamic model would introduce substantial computational complexity. Nonetheless, our model does incorporate persistent effects through the accumulation of experience. When firms receive positive demand shocks, they increase quantity, which in turn increases their stock of experience. Since experience is a state variable, firms with more experience have lower production costs and thus choose higher output levels, further building experience and reducing long-run costs. This feedback mechanism is the key driver of why firms that are ex-ante identical can exhibit divergent growth paths over time.

Second, following most of the existing literature such as Ryan (2012) and Barwick et al. (2025a), we assume that all firms view government policies as permanent. Relaxing this assumption would require modeling firms' expectations and their adjustment to a changing policy environment, which would greatly increase the complexity of our analysis. This is an important avenue for future research; see Gowrisankaran et al. (2025) and Chen (2024) for further discussion.

2.4 Estimation Strategy

In this section, we outline the strategy for estimating model parameters. In Section 2.4.1, we detail our demand model, specify the production cost function, as well as the distributions related to the scrap value and entry costs. In sections 2.4.2 and 2.4.3, we present the two-stage procedure for estimating dynamic parameters, including those associated with the production cost function and the distributions for the scrap value and entry cost shocks.

We estimate the dynamic parameters using the firms' observed choices of production, exit, and entry, including not entering, entering with c-Si technology, or entering with thin-film technology. Using the BBL approach from Bajari et al. (2007), we estimate these parameters in two stages. In the first stage, we flexibly estimate policy functions, including production, exit, and entry with technology selection, as well as the transition process of state variables directly from the data. After that, we approximate the value function using a set of B-spline basis functions of state variables, as described in (Sweeting (2013)). In the second stage, we estimate the dynamic parameters. We provide additional details in Appendix XX.

2.4.1 Specifications

Logit Demand. We parameterize the solar panel demand, expressed in (2.1), using a simple Logit model. To be specific, the demand of technology-n solar panels in the region m during the year t is given by:

$$Q_{nmt} = \frac{M_{mt} \cdot exp(-\alpha_m P_{nmt} + X_{mt}\eta + \xi_{nmt})}{1 + \sum_{n'} exp(-\alpha_m P_{n'mt} + X_{mt}\eta + \xi_{n'mt})}.$$

Here, M_{mt} is the market size for region m in year t, measured by the total electricity consumption. P_{nmt} is the price of technology-n solar panels that is faced by consumers in region m, calculated by adjusting the global price P_{nt} with local installation cost, tariffs, and subsidies. X_{mt} is a vector of demand shifters, including GPD, feed-in tariff, regional fixed effects, and yearly fixed effects.

For the given demand, the estimation equation is represented as:

$$ln(\frac{Q_{nmt}}{M_{mt}}) - ln(\frac{M_{mt} - Q_{1mt} - Q_{2mt}}{M_{mt}}) \quad = \quad -\alpha_m P_{nmt} + X_{mt}\eta + \xi_{nmt}. \tag{2.9}$$

To address the issue of price endogeneity, we follow Gerarden et al. (2025) and use the prices of silver and aluminum as instruments for the solar panel price. Silver and aluminum are key inputs in both c-Si and thin-film solar technologies. Our identifying assumption is that the prices and production levels of these two inputs are not correlated with the regional demand shocks for solar panels, ξ_{nmt}. This assumption is plausible because the solar industry represents less than 5% of global demand for silver and aluminum throughout the sample period, and the share of demand for solar panels originating from any given region is below 1% of total demand.

Production Costs. As shown in 2.2, the production cost function for firm j of technology-n is decomposed into two components, specified as

$$C_0(\sum_m q_{jnmt}, s_{jnt}) \quad = \quad (\gamma_n^0 + s_{jnt}\gamma_n^s) \cdot (\sum_m q_{jnmt}),$$

and

$$C_m(q_{jnmt}, \omega_{jnmt}) \quad = \quad (\gamma_m^{ex} + \omega_{jnmt}) \cdot q_{jnmt},$$

where γ_n^0 are technology-specific constants, parameters γ_n^s measure the impact of state variables on the unit cost for a specific technology, γ_m^{ex} are parameters particular to each region that capture the additional cost associated with shipping or exporting to that particular region, and ω_{jnmt} are cost shocks, which are assumed to follow an independent and identically distributed normal distribution with a mean of zero and a variance of σ_ω^2.

As a result, the marginal cost that firm j ships q_{jnmt} to region m is:

$$\frac{\partial TC_{jnt}}{\partial q_{jnmt}} = \gamma_n^0 + s_{jnt}\gamma_n^s + \gamma_m^{ex} + \omega_{jnmt}. \tag{2.10}$$

Here, the vector of state variables s_{jnt} includes the firm's cumulative experience e_{jnt}, along with other firm characteristics such as age, location, and ownership type. In addition, it includes technology-specific cost shifters that impact all firms, such as prices of major inputs, and policy variables that reflect the impact of industrial policies on production costs. As mentioned in Section 2.2.4, the central government has been directly supporting this industry since 2009, and several provincial governments also provide generous support. To capture these policy impacts, we incorporate indicators for the year 2009 onward and for the Zhejiang province.

Distributions of Scrap Value and Entry Costs. We assume that the scrap value ϕ_{jnt} follows an exponential distribution characterized by a parameter $1/\sigma_\phi$. (Note: we can allow the parameter to differ across the two technologies, $1/\sigma_{\phi,n}$). Additionally, we assume that the entry cost shocks ζ conform to an independent and identically distributed Type 1 Extreme Value (T1EV) distribution.

2.4.2 First Stage: Policy Functions and State Transition

Exit Policy Function We can estimate the exit policy function using a Probit regression:

$$Pr_n(\chi_{jnt} = 1|s_{jnt}) = \Phi(h_n(s_{jnt})), \tag{2.11}$$

where the variable $\chi_{jnt} = 1$ indicates the exit of firm j with technology n from the market in period t. Here, $h_n(s_{jnt})$ is a polynomial function of the states tailored to the specific technology, and Φ is the cumulative distribution function of a standard normal distribution. We use $\hat{p}_n^x(s)$ to refer to the estimated exit probability given state s.

Production Policy Function Similar to the optimal investment in Bajari et al. (2007), the production policy function $q^*_{nm}(s_{jnt}, \omega_{jnmt})$ decreases monotonically with the cost shock ω_{jnmt} and can be recovered as follows:

$$q^*_{nm}|s_{jnt} = G^{-1}_{nm}(1 - F_\omega(\omega \mid s_{jnt})), \tag{2.12}$$

where $G_{nm}(q \mid s_{jnt})$ is the empirical distribution of production given the state variables, and $F_\omega(\omega \mid s_{jnt})$ is the assumed distribution of ω.

We can express the optimal production as the sum of two components: one is a function of the state variables, and the other is a function of the shock. This is given by

$$q^*_{jnmt} = h_{1nm}(s_{jnt}) + h_{2nm}(\omega_{jnmt}), \tag{2.13}$$

where $h_{1nm}(\cdot)$ and $h_{2nm}(\cdot)$ are two unknown functions that need to be estimated.

We first perform a flexible non-parametric regression of q_{jnmt} on s_{jnt} to get the estimate $\hat{h}_{1nm}(s_{jt})$, which allows us to compute $q^*_{jnmt} - \hat{h}_{1nm}(s_{jnt})$. Then, we estimate $h_{2nm}(\cdot)$ using equation (2.12).

State Transition The vector of state variables s_{jnt} includes the state of all firms in the industry, making it high-dimensional. To reduce the dimension, we assume that firms do not keep track of each competitor's state variables, but instead rely on industry-level prices as a sufficient statistic, akin to the oblivious equilibrium concept introduced by (Weintraub et al. (2008) and Benkard et al. (2015)). This assumption is justified given that the industrial HHI being XXX, indicating that market power is not a significant issue. This approach has often been adopted in the literature, for example, Jeon (2022), Chen and Xu (2023), and Barwick et al. (2025a).

Some of the state variables, such as firm location and ownership status, are fixed over time. The firm age transits deterministically. As for experience stock, it transits as follows:

$$e_{jnt+1} = e_{jnt} + \sum_m q_{jnmt}.$$

where $\sum_m q_{jnmt}$ is the total production of firm j during year t.

The equilibrium price for each type of solar panel clears the market by equalizing aggregate demand and aggregate supply of that type of solar panel, which makes it a complicated object. Following Barwick et al. (2025a) and other studies, we model them as AR(1) processes. We account for different technologies having different price processes. Furthermore, we incorporate changes in constant terms across different periods. We make similar assumptions for input prices.

2.4.3 Second Stage: MLE

Value Function Approximation

The ex-ante value function, prior to the realization of ϕ_{jnt}, is described by

$$\begin{aligned} V_n(s_{jnt}) &= \mathbb{E}_\phi[max\{\phi_{jnt}, CV_n(s_{jnt})\}] \\ &= p_n^x(s_{jnt})\sigma_\phi + CV_n(s_{jnt}), \end{aligned} \tag{2.14}$$

where $p_n^x(s_{jnt})$ denotes the exit probability, and $CV_n(s_{jnt})$ denotes the continuation value as defined in equation (2.4).

Following the existing literature (e.g., Sweeting (2013)), we approximate the ex-ante value function using B-spline basis functions of the state variables. This is expressed as $V_n(s_{jnt}) = \sum_l \rho_{n,l} u_{n,l}(s_{jnt})$ where $u_{n,l}(s_{jnt})$ are basis functions and $\rho_{n,l}$ are coefficients that need to be estimated along with the dynamic parameters.

Specifically, we search for coefficients ρ that minimize the violation of the Bellman equation (2.14) given the dynamic parameters, given by

$$min_\rho \|V_n(s_{jnt};\rho) - \hat{p}^x_n(s_{jnt})\sigma_\phi - CV_n(s_{jnt};\rho)\|_2. \tag{2.15}$$

Here, $\|\cdot\|_2$ is the L^2 norm, $\hat{p}^x_n(s_{jnt})$ is the estimated first-stage exit policy function, and the continuation value is evaluated based on these estimated policy functions, given by

$$\begin{aligned} CV_n(s_{jnt};\rho) &= \mathbb{E}_\omega\Big[\sum_m P_{nmt}\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt}) - TC_{jnt}(\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt}),s_{jnt},\omega_{jnt}) \\ &\quad + \beta\mathbb{E}[V(s_{jnt+1};\rho) \mid s_{jnt},\hat{q}^*_{jnmt}]\Big], \end{aligned}$$

where $\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt})$ is the estimated first-stage production policy function.

Estimation of Production Costs and Scrap Value

Log-Likelihood of Output The optimal output $q^*_{jnm}(s_{jnt},\omega_{jnmt})$ is determined by the first-order conditions in equation (2.6). By construction, $q^*_{jnm}(s_{jnt},\omega_{jnmt})$ is strictly monotonic in ω_{jnmt}. Assuming it is also differentiable, the likelihood of output can be expressed as

$$g_{nm}(q_{jnmt};\theta) = \frac{f_\omega(\omega_{jnmt})}{|\partial q^*_{nm}(s_{jnt},\omega_{jnmt})/\partial\omega_{jnmt}|}, \tag{2.16}$$

where $|\partial q^*_{nm}(s_{jnt},\omega_{jnmt})/\partial\omega_{jnmt}|$ is the absolute value of the derivative of $q^*_{nm}(s_{jnt},\omega_{jnmt})$ with respect to ω_{jnmt}.

Log-Likelihood of Exiting Let χ_{jnt} indicate that firm j of technology n exits in year t in the data. The maximum likelihood estimate of the standard deviation scrap value, σ_ϕ, maximizes the following log-likelihood for exit decisions:

$$log(f_n(s_{jnt};\theta)) = \Big[(1-\chi_{jnt})log(1-exp(-\frac{CV_n(s_{jnt};\rho)}{\sigma_\phi})) - \chi_{jnt}\frac{CV_n(s_{jnt};\rho)}{\sigma_\phi}\Big] \tag{2.17}$$

MLE Let θ summarize the parameters associated with the distribution of scrap values and the production cost functions, where $\theta = (\sigma_\phi, \gamma)$. The maximum likelihood estimator for θ is determined by maximizing the joint log-likelihood of exiting and production:

$$max_\theta L(\theta) \quad = \quad \sum_{j,n,t} log(f_n(\chi_{jnt};\theta)) + \sum_{j,n,m,t} log(g_{nm}(q_{jnmt};\theta)), \tag{2.18}$$

where $f_n(\chi_{jnt};\theta)$ is the exit probability as described in equation (2.17), and $g_{nm}(q_{jnmt};\theta)$ is the output density as described in equation (2.16).

Equation (2.15) is imposed as a constraint when estimating dynamic parameters. Specifically, for each guess of the dynamic parameters, we solve for coefficients ρ that satisfy equation (2.15) and use the estimated $\hat{\rho}$ to construct the sample log-likelihood described in equation (2.18).

Estimation of Entry Costs

The probability that potential entrant j enters with technology n is expressed as

$$p^{e,n}(s_{jnt}) \quad \equiv \quad \frac{exp\Big(-\kappa_n(s_{jnt}) + \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\Big)}{1 + \sum_{n'} exp\Big(-\kappa_{n'}(s_{jn't}) + \beta\mathbb{E}[V_{n'}(s_{jn't+1}) \mid s_{jn't}]\Big)}. \tag{2.19}$$

We estimate the mean entry costs using the observed entry decisions via MLE.

2.5 Estimation Results

2.5.1 Estimates of Demand Equation

Table 2.1 reports the parameter estimates for the demand equation (2.9). Column (I) assumes a homogeneous preference for c-Si solar panels, whereas column (II) allows this preference to vary across economic regions. In both specifications, regional solar panel prices for a given year are instrumented using that year's prices of the major inputs. We use the specification in column (II) as our preferred estimates for the subsequent supply estimation and counterfactual analysis.

All parameter estimates have the expected signs. Among the six economic regions, Japan exhibits the smallest price coefficient in absolute value, while China dislikes price the most, except the rest of the world (ROW). Consumers generally prefer c-Si solar panels over thin-film solar panels, with EU consumers demonstrating the strongest preference. The utility derived from solar panels increases with both the GDP and the feed-in tariff of a region.

Table 2.1: Demand Estimates

	(I)		(II)	
	Est.	SE	Est.	SE
Solar panel price (α_m)				
× China	-1.168***	(0.111)	-1.428***	(0.122)
× EU	-0.997***	(0.105)	-1.116***	(0.107)
× India	-1.317***	(0.116)	-1.243***	(0.130)
× Japan	-0.720***	(0.090)	-0.779***	(0.093)
× US	-1.142***	(0.113)	-1.206***	(0.125)
× ROW	-1.204***	(0.101)	-1.469***	(0.095)
Demand shifters (η_m)				
1(c-Si solar panel)	3.214***	(0.360)		
× China			3.654***	(0.383)
× EU			3.983***	(0.338)
× India			2.641***	(0.402)
× Japan			3.526***	(0.331)
× US			2.388***	(0.359)
× ROW			3.599***	(0.356)
GDP	9.490***	(1.720)	2.410*	(1.340)
Feed-in tariff	1.954***	(0.531)	1.898***	(0.573)
Constant	-30.422***	(0.366)	-29.411***	(0.250)
R^2	0.731		0.809	

Notes: Standard errors in parentheses. * p<0.10,** p<0.05,*** p<0.01.

2.5.2 Estimates of Policy Functions and State Transitions

Exit Policy Function The panel (I) of Table 2.2 reports the estimated exit policy function obtained from a Probit model. Firms with more experience are less likely to exit the market. The estimates also show that their exit decisions also respond to industrial policies. Stronger demand subsidies reduce the likelihood of exit. Moreover, the probability of exit is lower in the period 2006–2010, but rises after 2011. Firms located in Zhejiang or with government ownership are also less likely to exit. Lastly, the exit probability declines as the solar panel price increases and rises as the input prices go up.

Table 2.2: Estimates of Exit and Production Policy Functions

	(I) Exit		(II) Production	
	Est.	SE	Est.	SE
Common Variables:				
Experience	-0.378	(0.389)	3.800***	(0.366)
Feed-In Tariff	1.363***	(0.521)	1.323***	(0.276)
2006–2010	-1.920**	(0.815)	0.335	(0.344)
2011+	0.077	(0.218)	0.157**	(0.092)
Zhejiang	-0.083	(0.162)	-0.096	(0.099)
SOE	-0.193	(0.226)	5.018	(3.200)
Constant	-0.211	(0.320)	0.040	(0.127)
c-Si Firms:				
c-Si price	-2.380**	(0.970)	0.026	(0.025)
Polysilicon price	-1.037**	(0.519)	-0.293	(0.217)
Thin-Film Firms:				
Thin-film price	0.244	(0.347)	-1.010	(1.980)
Cadmium price	-1.008*	(0.555)	-0.044	(0.132)
Tellurium price	7.060	(4.470)	-0.287	(1.620)
Log-Likelihood	-234.52			

Note: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$

Production Policy Function The panel (II) of Table 2.2 reports OLS estimates for the production policy function. In Appendix XX, We also report Tobit estimates, which assume that $h_{2nm}(\omega)$ follows a normal distribution, as well as Censored Least Absolute Deviation

(CLAD) estimates, which are distribution-free with respect to the cost shock and estimate $h_{2nm}(\omega)$ non-parametrically. These alternative approaches yield similar coefficient estimates but exhibit a lower overall model fit. We therefore adopt OLS as our preferred specification.

Firms with greater experience produce more. The output also increases when more generous demand subsidy (feed-in tariffs) are provided. The coefficients on the 2006–2010 and 2011+ policy dummies are both positive, with the latter being statistically significant. This results align with the policy rollout, especially the center government's declaration of solar panels as a "strategic emerging industry" in 2010. Lastly, c-Si firms expand production when solar panel prices rise and scale back production when input prices increase.

2.5.3 Estimates of Dynamic Parameters

Production Cost Parameters. Table 2.3 reports the MLE results for the production cost function as outlined in Equation (2.10). We categorize the parameters into three groups: (i) those specific to c-Si solar panel firms, (ii) those specific to thin-film solar panel firms, and (iii) those shared by both technologies.

We begin with discussing the cost parameters of c-Si solar panel firms. The estimated learning parameter $\gamma_n^{s,e}$ is -0.451, indicating that doubling manufacturing experience reduces marginal costs by $ln(2) \times 0.451 = 0.313$ per MW. At this rate, c-Si solar panel manufacturers' cumulative experience over the sample period lowered prices by $XX (XX%) per MW. compare with the findings of other studies on the learning rate of solar panels. The coefficient γ_n^0 represents the baseline costs, which is the solar panel production cost when a firm begins production and sells to the domestic market in 2000. The baseline cost is estimated to be 0.742 RMB per MW. The production cost is higher when the input price is higher. Specifically, the pass-through rate from the polysilicon price the to c-Si solar panel price is approximately 13%.

Compared with the pre-2006 baseline period, production costs are lower during 2006-2008, and substantially lower after 2010. Our estimates imply that during 2006-2008, Chinese solar panel manufacturers received subsidies of XX RMB per MW of production, corresponding to XX% of the unit production cost being subsidized. These figures help explain the surge of Chinese manufacturers' output after 2005, despite the surge in polysilicon price and sharp drop in solar panel prices.

Although thin-film solar panel firms exhibit cost estimates with the same signs as c-Si firms, their magnitudes differ. The baseline cost is 0.239 RMB per MW higher compared to that of c-Si firms. The estimated learning parameter is minimal in magnitude and not statistically significant. Again, production costs rise with increases in input prices. why input price coefficients are negative?

Finally, we turn to the estimates of these common parameters. Our estimates indicate that exporting to the EU and Japan is more costly than selling to the domestic market, while exporting to the US and India is actually less costly. This might be attributed to China's export tax rebate policy, which effectively lowers exporters' costs for shipments to the US (Chandra and Long (2013)). We also find that state-owned firms incur higher costs.

Note: back of envelop calculation of the subsidy during 2006-2008. According to industrial reports, one unit (kT) of polysilicon can generate 115 units (MW) of solar panel on average during 2006-2008. The price of polysilicon increased by around $250 per unit during this time. It suggests the MC of solar panel per unit is $250/115 higher. The subsidy that makes the MC unchanged should be around 250/115 * 7.5 = 16 RMB per MW, which is 0.016 RMB per watt.

Scrap Value and Entry Cost Parameters. Table 2.4 presents the estimated standard deviation of scrap value and the mean entry costs for the two technologies. As discussed in Section 2.2, the number of new entrants increases sharply after 2010, prompting us to differentiate c-Si firms' entry costs before and after 2010. Our estimates indicate that the entry cost for c-Si firms was roughly 6.362 million RMB lower prior to 2010, representing an estimate of the overall entry subsidy before that year. Additionally, entrants in Zhejiang

Table 2.3: Estimates of Production Cost Parameters

	Est.	SE
(i) c-Si Firms:		
Constant (γ_n^0)	2*0.429	(0.042)
Experience ($\gamma_n^{s,e}$)	-0.077	(0.005)
2006-2008 ($\gamma_n^{s,s}$)	-0.016	()
(ii) Thin-Film Firms:		
Constant (γ_n^0)	0.732	(0.107)
Experience ($\gamma_n^{s,e}$)	-0.097	(0.0747)
(iii) Common Variables:		
Exporting (γ_m^{ex})		
EU	0.0095	(0.005)
India	0.085	(0.008)
Japan	0.012	(0.009)
US	0.010	(0.012)
ROW	0.003	(0.007)
Other state variables ($\gamma^{s,o}$)		
SOE	0.247	(0.012)
SD of shocks (σ^{ω})	0.5*2.943	(0.045)

Notes: Standard errors from reported output.

constitute XX% of all new solar panel firms over the sample period. To account for this locational variation, we allow for differing entry costs in Zhejiang province. Our analysis reveals that the entry cost in Zhejiang is 4.874 million RMB less than in other provinces, implying that the local government may offer substantial subsidies for the entry of solar panel firms.

2.5.4 Model Fit

To assess how well our estimated model matches the data, we simulate the model's equilibrium and plot both the observed and predicted output and the number of active firms. Overall, our model fits the data well. Specially, it captures the rapid expansion in output during the sample period, and the rapid growth in the number of firms before 2011 as well as the subsequent slowdown.

- output: (mean, SD) or draw a graph

Table 2.4: Estimates of Scrap Value and Entry Cost

	Est.	SE
Scrap Value (σ_ϕ):		
2000–2009	1.2*2.953	()
2010+	0.8*2.953	()
Entry Cost:		
c-Si technology (κ_1)		
2000–2009	1.2*8.218	(1.784)
2010+	0.8*8.218	(0.850)
Thin-film technology (κ_2)	9.144	(2.218)
Zhejiang	-0.5*4.855	(1.879)

Notes: Entry costs are in millions of RMB.

- number of exits/exit
- number of active firms

2.6 LBD and Industrial Policies

In this section we quantify the welfare effects of both demand-side and supply-side policies with and without LBD incorporated.

Industrial Policies

3. Demand policies
   (a) feed-in tariff. The coefficient before this variable = 1.898. We set it to be zero to remove the impact of feed-in tariff.
   (b) tax credits. We adjust the local price in each region by removing tax credits.
   (c) import tariff on c-Si technology. We adjust the local price in each region by removing import tariffs.
4. Production cost subsidies

(d) polysilicon price drop in 2008. We assume that the polysilicon price after 2008 follows the same trend as before 2008. In particular, we set the constant term of the polysilicon price after 2008 to be the same as that before 2008.

(e) exporting subsidies. We set the exporting cost to US to be zero. (some positive number?)

(f) other subsidies on production cost. To remove the subsidies either at the national level or at the provincial level, we set the coefficient before 2009+ to be zero, and the coefficient before Zhejiang to be zero.

5. Entry cost subsidies

   (g) subsidy on entry cost of c-Si tech before 2011. Set the coefficient of c-Si entry cost before 2011 to be the same as that after 2011.

   (h) local subsidy on entry. Set the coefficient of Zhejiang to be zero.

LBD creates a positive “feedback loop". To remove the impact of LBD, we set the coefficient before experience in the production cost function γ_n^e to be zero.

Since both production and entry have dynamic impacts, we simulate the industry over a long horizon to assess the impacts of industrial policies. In each counterfactual scenario, we turn on the policy of interest while keeping other policies off, and compute the averages across 50 simulations of the industry, starting in 2000 and ending in 2050. In each simulation, Chinese firms strategically choose their production quantities and whether to exit or enter. The production quantities of firms in other countries are fixed at the levels observed in the data. Equilibrium prices are then pinned down by the interaction of global demand and supply of solar panels.

Section 2.6.3 examines the impact of demand-side policies, including subsidies (feed-in tariffs and tax credits) that are applied across all regions over the sample period, as well as the tariffs on c-Si panels introduced by the EU and the US since 2012. Section 2.6.4 investigates the impact of policies aimed at lowering production cost, including development of the upstream segment that reduces the polysilicon prices, and other forms of subsidies. Section 2.6.5 focuses on policies that intended to lower firms' entry costs.

2.6.1 Outcomes of Interest

The key market outcomes of interest include: (i) global solar panel output and prices, (ii) the market share of c-Si technology, (iii) China's market share and producer surplus, (iv) government subsidy spending, reported separately for China and for other regions, (v) consumer welfare, (vi) environmental benefits, and (vii) employment in the solar panel sector. The first five sets of measures are relatively straightforward and can be directly computed based on the simulated market outcomes. In what follows, we describe how we measure the effects on environmental benefits and employment.

To quantify the impacts on environmental gains from solar panel adoption, we draw on prior estimates of the value of avoided air pollution damages from Sexton et al. (2021), which are also used in Gerarden (2023). For local air quality benefits, we rely on their estimates of damages avoided from emissions of nitrogen oxides, fine particulate matter, and sulfur dioxide. For global climate benefits arising from lower carbon dioxide emissions, we use the U.S. Environmental Protection Agency (2023) estimate of the social cost of carbon, set at \$190 per metric ton of CO_2 for emissions in 2020 (in 2020 dollars).

[Do we need to quantify the pollution from solar panel manufacturing??]

To quantify employment effects, we distinguish between jobs in solar panel manufacturing and those related to deployment. We estimate impacts by multiplying the simulated changes in Chinese manufacturing by the average labor intensity of solar panel production, and by multiplying the simulated changes in solar panel adoption in each region by the average labor intensity of solar panel installation (including sales and distribution) in that region.[4]

[4]The labor intensity of Chinese solar panel manufacturing is calculated by dividing the number of solar panel manufacturing jobs in China by the volume of output produced in China. The labor intensity of solar panel adoption in a given region is calculated by dividing the number of solar panel installation jobs by the

2.6.2 LBD

We quantify the role of LBD when evaluating all policies in place.

- CF0 (baseline): no policies + no LBD
- CF1: policies + no LBD
- CF2: no policies + LBD
- CF3 (model fit or data): policies + LBD

The impact of policies without LBD = CF1-CF0. The impact of policies with LBD = CF3-CF2. We expect to see that the policies in the presence of LBD can lead to much larger impact than that when LBD is not present.

2.6.3 Demand Policies

In this section, we analyze two counterfactual policy configurations: (i) a case where only demand-side subsidies (feed-in tariffs and tax credits) are in place, and (ii) a case where only c-Si tariffs are imposed. For each configuration, we simulate market outcomes and welfare, considering scenarios both with and without LBD. The subsequent figures and tables present the results, comparing simulated outcomes under these counterfactuals to those under the baseline scenario with no industrial policies, again both with and without LBD.

- CF1-0: feed-in tariff + no LBD
- CF1-1: feed-in tariff + LBD
- CF2-0: tax credits + no LBD
- CF2-1: tax credits + LBD

installed capacity in that region.

- CF3-0: import tariffs + no LBD
- CF3-1: import tariffs + LBD

Table 2.5: Welfare Impacts of Demand-Side Policies

	(I) Demand Subsidies		(II) Import Tariffs on c-Si	
	w/ LBD	w/o LBD	w/ LBD	w/o LBD
Δ in Solar Panel Output in 2013				
c-Si				
Thin-Film				
Δ in Solar Panel Price in 2013				
c-Si				
Thin-Film				
Δ in China's Market Share in 2013				
Δ in Chinese Producer Surplus				
Δ in Government Subsidies				
China				
US				
EU				
Δ in Consumer Surplus				
China				
US				
EU				
Δ in Employment				
Chinese Manufacturing				
Worldwide Installation				
Δ in Environmental Benefits				
Local Pollution				
Global Pollution				

Note:

2.6.4 Production Cost Policies

We focus on subsidies on the production cost.

- CF1-0: polysilicon price drop in 2008 + no LBD
- CF1-1: polysilicon price drop in 2008 + LBD

- CF2-0: exporting subsidies + no LBD
- CF2-1: exporting subsidies + LBD
- CF3-0: other subsidies + no LBD
- CF3-1: other subsidies + LBD

Table 2.6: Welfare Impacts of Production Cost Policies

	(I) Input Price		(II) Other Subsidies	
	w/ LBD	w/o LBD	w/ LBD	w/o LBD
Δ in Solar Panel Output in 2013				
c-Si				
Thin-Film				
Δ in Solar Panel Price in 2013				
c-Si				
Thin-Film				
Δ in China's Market Share in 2013				
Δ in Chinese Producer Surplus				
Δ in Government Subsidies				
China				
US				
EU				
Δ in Consumer Surplus				
China				
US				
EU				
Δ in Employment				
Chinese Manufacturing				
Worldwide Installation				
Δ in Environmental Benefits				
Local Pollution				
Global Pollution				

Note:

2.6.5 Entry Cost Policies

We focus on subsidies on entry cost.

- CF1-0: subsidy on entry cost of c-Si tech before 2011 + no LBD
- CF1-1: subsidy on entry cost of c-Si tech before 2011 + LBD
- CF2-0: provincial subsidy on entry + no LBD
- CF2-1: provincial subsidy on entry + LBD

Table 2.7: Welfare Impacts of Entry Policies

	w/ LBD	w/o LBD
Δ in Solar Panel Output in 2013		
c-Si		
Thin-Film		
Δ in Solar Panel Price in 2013		
c-Si		
Thin-Film		
Δ in China's Market Share in 2013		
Δ in Chinese Producer Surplus		
Δ in Government Subsidies		
China		
US		
EU		
Δ in Consumer Surplus		
China		
US		
EU		
Δ in Employment		
Chinese Manufacturing		
Worldwide Installation		
Δ in Environmental Benefits		
Local Pollution		
Global Pollution		

Note:

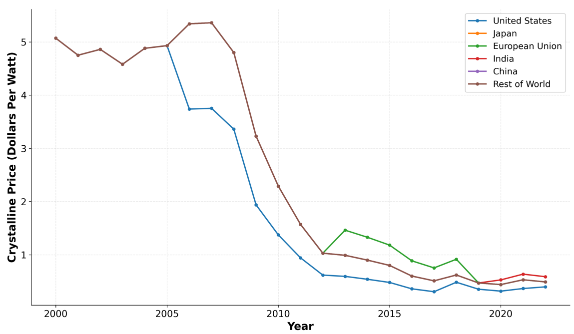
2.7 Conclusion

H Estimation Details

I Data Details

Figure I.4: Solar Panel Price by Region

(a) c-Si



(b) Thin-Film

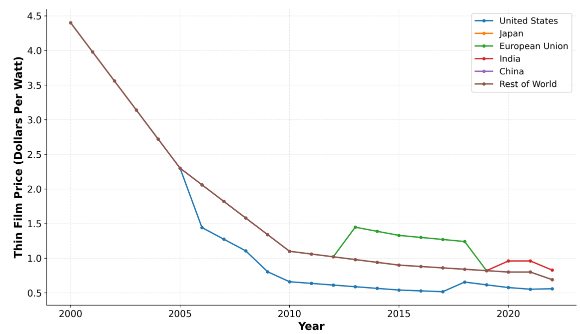
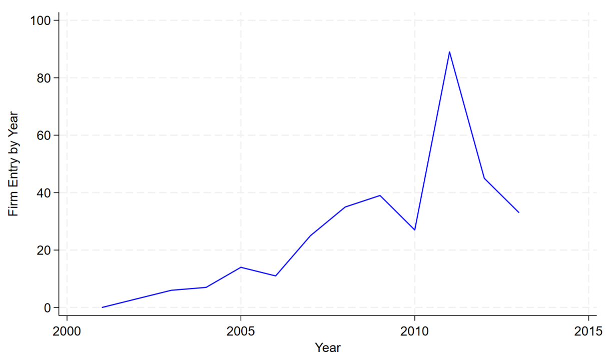


Figure I.5: Entry and Exit of Chinese Solar Panel Manufacturers

(a) Entry



(b) Exit

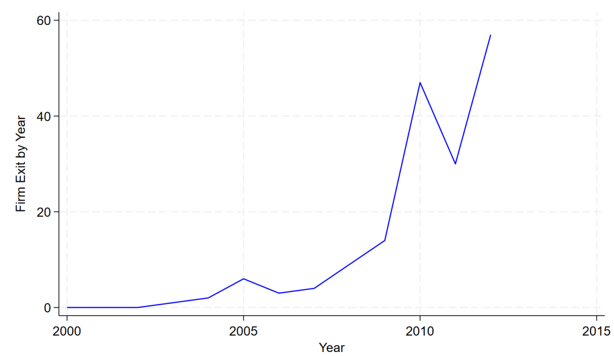


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Table I.8: AR(1) Estimates of State Transition Process

	Output Price		Demand Subsidies	Input Price		
	Poly.	Thin-film	Common	Poly.	Cadmium	Tell.
λ_0^{pre}	1.596	-1.570	0.021	36.799	2.272	3.238
	(0.022)	(0.099)	(0.033)	(9.1)	(0.584)	(0.662)
λ_0^{post}	-1.761	7.223	0.131	6.07		
	(0.048)	(0.439)	(0.229)	(9.72)		
λ_1	0.804		-0.053	0.195	7.099	7.164
	(0.054)		(1.634)	(0.064)	(0.082)	(0.092)
N	22	22	13	228	31	30
R^2	0.988	0.998	0.660	0.760	0.633	0.615
AIC	-29.671	-97.774		2362.69	44.499	32.491

Notes: The dependent variable is the price in year t. Standard errors in parenthesis. We use different year as the cutoff to define Pre" and Post". The cutoff year is 2008 for the c-Si solar panel and polysilicon price, 2010 for the thin-film solar panel price, 2007 for the Cadmium price, 2011 and for the Tellurium price. The demand subsidies are measured in two ways. The first one is weighted by a region's share of demand among the all demand, i.e., $\sum_m sub_{mt} * \frac{Q_{mt}}{\sum_{m'} Q_{m't}}$, where sub_{mt} is the demand subsidy in region m and Q_{mt} is the total demand in region m. The second one is weighted by the share of a firm's output to a region among this firm's total output, i.e., $\sum_m sub_{mt} * \frac{q_{jmt}}{\sum_{m'} q_{jm't}}$, where sub_{mt} is the demand subsidy in region m and q_{jmt} is firm j's sales to region m. Structural break years: Poly solar panel (2009), Thin-film (2010), Input (2019), Cadmium (2007), Tellurium (2001).

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Chapter 3

An Econometric Model for Measuring Impacts of AI on United States Power Grids

Abstract.

Data centers now consume 4.4% of U.S. electricity—more than double their 1.9% share six years ago—yet AI's effects on grid stability remain poorly quantified. We exploit staggered Large Language Model releases as a natural experiment, using difference-in-differences to estimate how model training and inference affect local power grids. AI activity significantly degrades power quality and increases fossil fuel generation near data centers, with high-intensity periods **adding 0.5–1 outages per year.** We find an increase in fossil fuel power demand from AI **equivalent to $\sim$ 47,000 household-years** during inference. We construct counterfactuals examining alternative AI development trajectories. **A 1% increase in model parameters raises local power demand by 0.15% post-release.** Fully implementing on-site generation would improve power quality by **0.17 reliability events per month—potentially making AI a net**

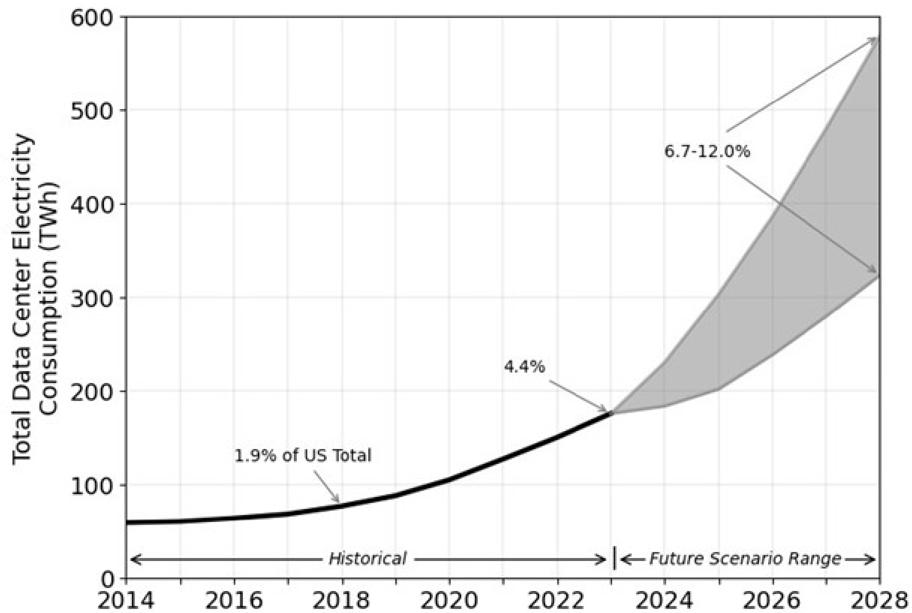

Figure 3.1: Projected data center growth over time. Source: DOE.

positive for grid reliability.

3.1 Introduction

Over the past six years, data centers have expanded from consuming 1.9% to 4.4% of total U.S. electricity demand (Figure 3.1, DOE). This rapid growth—driven largely by artificial intelligence—follows two decades of flat electricity consumption and coincides with nationwide efforts to electrify transport and heating. The combination poses a challenge: grid infrastructure was planned for stable demand, not simultaneous load growth from multiple sectors. Recent anecdotal evidence points to data centers reducing local power quality and raising prices Nicoletti et al. (2024a), yet systematic empirical evidence on these grid-level impacts remains scarce.

Existing work does not provide causal estimates of how AI deployments affect power systems at the *grid level*. Techno-economic assessments from international organizations broadly characterize rising AI workloads in the data center market IEA (2024); EPRI

(2024). Micro-level studies examine energy performance of AI accelerators and identify workload management opportunities Shankar and Reuther (2022); Patel et al. (2024). A large computer science literature quantifies the compute, energy, and environmental costs of model development and deployment Strubell et al. (2019); Schwartz et al. (2020); Wu et al. (2022); Berthelot et al. (2024); Morrison et al. (2025). These provide a foundation for understanding datacenter-level dynamics, but do not address the externalities experienced by other grid users—impacts on power quality, prices, and reliability that matter for energy security.

Quantifying and understanding these power grid level impacts is critical from the perspectives of energy reliability and security. In this paper, we ask seek to answer four specific questions: 1) What are the effects of AI model releases on local power quality and power frequency? 2) How do AI training and inference affect electricity consumption around data centers owned by model developers? 3) How do these impacts translate into price effects in the wholesale and retail markets? Additional demand will likely increase prices, but understanding the magnitude of these price increases is important to determining the severity of the issue. 4) How will these impacts be different based on counterfactual scenarios articulating different technical evolutions of AI computing at scale? These scenarios include a shift away from cloud-based inference to edge-based computing, a geographic realignment of data centers away from population clusters, an evolution towards primarily inference-based or primarily training-based workloads, and a significant improvement in overall energy efficiency.

One way to answer the above questions is to extrapolate from energy usage measurements (micro measures) within a datacenter to system-level (macro measures) impacts on the grid. This is however, is near infeasible because it requires a great deal of specific internal information that isolates what is run within the datacenter and when. While this data with level of detail is not available (for proprietary and other practical reasons), there are data sources that provide high-level observational data about various aspects of power consumption and quality at the grid-level over the time periods when models were trained and released. *How can we answer specific questions about the impact of AI models with this high-level observational data alone?*

We can draw upon econometric approaches for addressing this problem. In particular, we focus on the Difference-in-Differences (DiD) models which allows us to estimate the causal effects of AI model deployments on power systems, even when some contributing factors are unobservable. By comparing areas around data centers running AI models against control areas, and partitioning observations into pre- and post-release periods, we isolate AI-specific impacts while controlling for temporal and geographic factors.

There are multiple technical challenges in applying this standard DiD approach in our setting: (i) There is a lack of specific information about when and where models are trained and released. Further model releases can be staggered, affecting the validity of the standard model which only uses one timepoint. (ii) The DiD method at its core relies on the parallel trends assumption i.e. we would know what would happen if the AI model were not run; in other words the estimates are valid only if we have a fairly accurate estimate of the trend of the variable of interest over time. (iii) There could be other confounding factors contributing to change in the variable of interest (e.g. extreme weather events coinciding with model release times could also cause increased power demand). (iv) No one clean dataset exists including all data useful for analysis. Often data are at different scales, lack linking tables, or require significant processing prior to use for analysis. In some cases, this requires significant efforts for manual mapping, combination, and labeling.

Our solution methodology addresses the above challenges:

1) Addressing Inexact Information and Staggered Releases We use a *Staggered DiD* approach with multiple horizons, which has been used to overcome similar challenges in other applications Cengiz et al. (2019); Deshpande and Li (2019). Staggered DiD with multiple horizons extends the standard two-period (pre/post), two-group setup to settings where treatment activates for different units at different times, leveraging this variation in rollout timing to assess different potential points of impact—analogous to how a staged feature rollout lets you compare early adopters against users still in the control group at each point in time. For the question on fossil fuel demand effects, we address confounds such as alternative fuel prices or generator efficiency issues using a two stage regression, where we first factor out the impact of these confounds, and explain the remaining demand using a second stage staggered DiD. Similarly, for estimating price effects, we need to isolate effects from other sources of power demand that can cause price changes. We handle confounds such as weather-driven and industry production driven shifts in demand (and thus price) via a two-stage regression.

2) Addressing validity of the DiD method and other confounds We conduct a wide-array of statistical robustness checks supported by integration of large amounts of data from a diverse array of sources. We run statistical tests to show that we cannot reject the null hypothesis of parallel trends. We also run placebo tests with random treatment dates and randomized locations to show that our treatment effects are not the result of randomness, and we vary treatment definitions geographically and temporally. We use time series testing for anomaly detection and structural breaks to show that the differences we find in our regressions show up in model free analysis. Together, these robustness checks provide evidence of an experiment that is valid across all relevant dimensions.

3) Addressing data disparity and relevance issues Training dates and facilities are generally unobserved, and for inference—which occurs continuously—these dates are not well-defined. We therefore rely primarily on model release dates. To strengthen identification, we supplement this with training locations, training dates, and API release dates gathered from press releases and news articles. We obtain verified training information for four models and confirmed API release dates for over half of the foundational models in our sample. We construct estimates using this most robust subset, then expand our model definitions to exploit exogenous variation for counterfactual analysis. Throughout, we use the most conservative feasible treated subset for each analysis. To construct suitable control groups, we employ propensity score matching as detailed in Appendix Subsection 3.10.9, excluding observations without comparable pre-treatment trends.

We apply this methodology using publicly available data and proprietary on datacenter locations, power grid conditions, AI model release dates, and controls including weather and market prices. Our data creation process is described in detail in Section 3.5. While all data pre-exist our analysis, the combination of the data creates a novel dataset that represents a significant contribution. Because the data are from government agencies, ISOs, or proprietary datasets, the data are highly trustworthy. We perform data validation to verify this. We estimate DiD models for outcomes measuring both power quality and electricity demand.

Our findings presented in Section 3.6 speak to the large magnitude of the impacts of AI data centers and the increasing effects of training ever-larger models over time. We find that large model releases (including the likes of GPT-3, Claude 2.1, Llama 2 documented in subsection 3.10.5) cause significant power quality deterioration in nearby areas – **well above half the standard deviation of U.S. power-quality distributions**– and increase fossil fuel consumption on the order of terawatt-hours— **equivalent to around 45,000 households annual consumption**– in both training and inference phases. We also find **wholesale electricity prices increase in treated PJM zones with increases of 25%** during treatment in the American Electric Power (AEP) PJM Zone. Using these estimates, we show in Subsection 3.8 how altering the trajectory of AI along disparate dimensions impacts the grid. We are able to provide evidence for the separate impacts of training and inference, the effects of a shift to edge computing or more concentrated training loads, and how differential efficiency improvements impact AI's effects on power usage. We also show compelling evidence that significantly increasing on-site power generation provides a promising avenue for future reductions in grid impact.

This work makes three contributions. First, we develop a methodology for assessing macro-level grid impacts of AI models using publicly available data, even when fine-grained operational information is unavailable. Second, we provide the first causal estimates of these impacts for major frontier models, complementing the emerging literature on environmental and system-level effects of AI data centers Murino et al. (2023); Guidi et al. (2024); Thangam et al. (2024). Finally, we provide policy-relevant counterfactual estimates for the impacts of technical evolution of AI on the grid.

3.2 Related Work

This paper contributes to four interrelated literatures. First, techno-economic assessments document the scale of data center electricity demand: Lawrence Berkeley estimates U.S. data center consumption could reach 6.7–12% of national electricity by 2028 Shehabi et al. (2024), while utility five-year peak demand forecasts jumped from 38 GW to 128 GW between 2023 and 2024, with approximately 90 GW attributable to data centers Goggin and Gramlich (2025). Recent simulation studies examine AI-driven load growth implications for grid planning Lin et al. (2024b); Chien et al. (2023), but these assessments characterize aggregate trends rather than isolating causal mechanisms.

Second, computer science research quantifies AI's computational and environmental costs, from training emissions Strubell et al. (2019); Schwartz et al. (2020) to debates over efficiency improvements Patterson et al. (2021, 2022) and inference-time scaling Kim et al. (2025), including data center carbon emissions Maji et al. (2025). Related work has developed carbon intensity forecasting methods Maji et al. (2022a); Li et al. (2023) and examined power forecasting for grids Lechowicz et al. (2025); Maji et al. (2022b), but not in the context of AI workload impacts on market outcomes. Recent empirical projections analyze the tension between AI energy growth and grid decarbonization Maji et al. (2024), while others consider grid planning for AI demand Lin et al. (2024a), yet the causal effect of AI deployment on wholesale markets remains unstudied.

Third, emerging evidence points to grid-level externalities from data center concentration.

Sensor data reveals spatial correlation between data center proximity and power quality degradation Nicoletti et al. (2024b), PJM capacity prices increased from $30 to $270/MW-day between 2023–2024 amid data center load growth Luby et al. (2025), and transmission costs are increasingly socialized across ratepayers Jacobs (2025). The systems literature has examined carbon-aware geographical load shifting using locational marginal prices Lindberg et al. (2021) and data center demand response participation Chen et al. (2019); Liu et al. (2014); Klingert et al. (2018). These studies establish correlational patterns or propose optimization frameworks but lack causal identification.

Fourth, the econometric literature on electricity markets provides methodological foundations, including difference-in-differences for demand responses Roth et al. (2023) and event studies for policy interventions Pacific Northwest National Laboratory (2022), but has not applied quasi-experimental methods to AI deployment impacts.

We address this gap by treating AI model releases as plausibly exogenous demand shocks, using difference-in-differences to estimate causal effects on locational marginal prices and power quality. Unlike prior work building from computational requirements or facility audits, we analyze grid-level impacts using wholesale market data, capturing realized demand effects without requiring proprietary operational information.

3.3 Problem and Challenges

Our goal is to investigate three primary effects of AI data centers on power markets. We also conduct counterfactual analyses that could inform policy: assessing different scenarios of parameter size, training versus inference proportions, edge compute preference, relocation to low-population areas etc.

3.3.1 Problem introduced by AI workloads

First, we quantify the effect of frontier AI models on power quality. AI workloads affect power quality through three main mechanisms. First, data centers and AI computing facilities add substantial baseload demand that can strain grid capacity and compromise voltage regulation, especially during peak periods. Second, AI workloads fluctuate rapidly with computational needs, creating unpredictable load variations that challenge stability and frequency regulation. Third, the switching power supplies and electronics essential to AI hardware generate harmonic distortion, introducing waveform distortions that spread through distribution networks and degrade power quality. See Figure 3.6 in Appendix for an illustration of this harmonic distortion.

Second, we analyze how training and inference workloads differentially affect local power demand given that training produces sharp, episodic spikes while inference generates steadier ongoing consumption.

Third, we estimate how wholesale electricity prices have responded to LLM development. AI workloads affect wholesale electricity prices through the demand channel, with potentially indeterminate long-run retail market impacts. Wholesale prices are determined hourly in day-ahead markets via sealed-bid, uniform-price auctions subject to network constraints. Locational price variation reflects transmission limitations within the network. Increased demand raises wholesale prices by necessitating dispatch of higher-cost marginal generators. The wholesale price increases are passed through to retail consumers with substantial lag. Understanding wholesale demand and price shifts thus provides useful intuition regarding the direction and magnitude of retail price changes.

3.3.2 An Econometric Model

We want to quantify the effects of running AI models on the power grid but we only have overall observational data about power demand, power quality, prices, exogenous controls, locations of data centers, and dates of model activity. We do not have access to data that

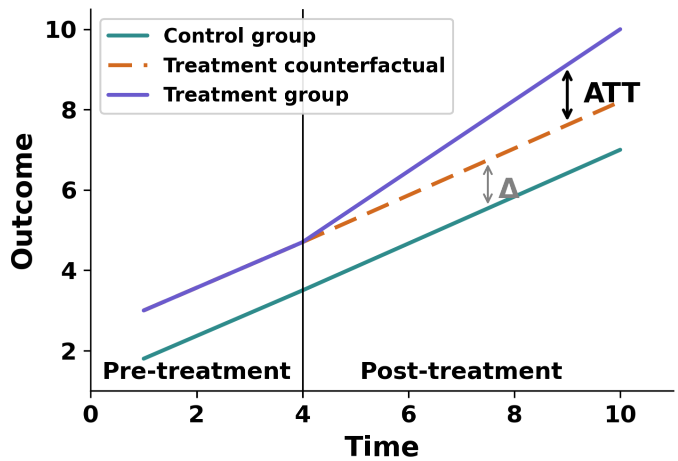

Figure 3.2: Difference-in-differences estimation. Under the parallel trends assumption, the counterfactual (dashed) projects treatment group outcomes absent intervention. The ATT is identified as the divergence between observed and counterfactual outcomes post-treatment.
specifically isolates the effects with respect to running the AI models because these require proprietary knowledge.

To address this challenge, we can turn to Difference-in-differences (DiD), a widely-used statistical estimation technique in econometrics for causal inference with observational data. This approach has been used to answer a wide range of empirical economics questions from observational data such as telecom demand from price and quantity variation, Fowlie et al. (2012) evaluate environmental regulation in electricity using emissions and price data, and Card and Krueger (1994) study minimum wage effects via cross-state employment variation.

The key idea behind this technique is to estimate treatment effects by comparing the change in outcomes over time for a treated group to the change for a control group, as illustrated in Figure 3.2.

For power quality, as an example, this translates to estimating the change in utility-level power quality within the geographically-defined utility area before and after the release of a major AI model. The treated group in this case would be utility areas containing data centers engaged in model training or inference during the time period of activity, and the control group would be utilities without AI activity in the same time period. The outcome variable, power quality in our example, is modeled as a function of the treatment (i.e. the running of the AI model), and other contributing factors. Formally, we can specify the power quality effect as:

$$Y_{it} = \alpha + \beta\,(\text{Treatment}_i \times \text{Post}_t) + \gamma_i + \delta_t + \varepsilon_{it}, \tag{3.1}$$

where Y_{it} is power quality for utility i at time t, Treatment_i indicates proximity to a data center, and Post_t captures periods after the release of a major AI model. The coefficient of interest, β, measures the causal impact of model releases on local power quality. γ_i captures unobserved factors common across all utilities, δ_t captures unobserved factors that remain fixed across time, and ε_{it} is an error term that captures random disturbances or shocks.

We can learn the co-efficients, given appropriate data about the outcome variables over time — in this case with data about the power-quality values over time across a range of utilities containing AI data centers, and the knowledge of when and where the AI models of interest are run.

3.3.3 Technical Challenges and Validating Model Assumptions

The validity of this basic DiD approach for our problem is affected by the following issues:

1. *Lack of specific information about when and where models are trained and released* The model as described requires knowing specifics of the training windows, e.g., when training began, and when it was released. It is not always possible to obtain such specific information.

2. *Staggered Events* The model also assumes a single temporal event of interest to capture pre and post treatment effects. However, models are not always released at a single time point and are often staggered.

3. *Parallel Trends Assumption* The effects of running a model on an AI data center is estimated by the difference from what would have been in the model were not run on that data center. This assumes that we have a fair estimate of the trend that the

variable of interest would have over time but for the training and inference of the AI models.

4. *External Confounds* The effects we observe and attribute to AI models could be due to other confounds such as random weather events, pre-existing trends in load, or variation in fuel price.

5. *Data Engineering Challenge* No single clean dataset contains all variables needed for analysis. Data often exist at different scales, lack linking tables, or require substantial preprocessing—sometimes necessitating manual mapping, combination, and labeling. An example is the zonal mapping effort used to assign generators to zones with the zonal map shown in Figure 3.8 in the Appendix.

We address the first two challenges with a stacked modification of the DiD model (subsection 3.4.1). We address the next two through appropriate statistical tests and robustness checks (subsection 3.4.5) and the final issue through our data creation (section 3.5).

3.4 Methodology

To address the inexact information about training windows and release times, we estimate dynamic treatment effects at multiple horizons relative to model release dates. This approach, referred to as an event study or **stacked difference-in-differences** in econometrics, offers three advantages over the standard pre/post comparison discussed earlier: (1) it does not require observing the true training start date, instead using release timing as the anchor; (2) it also provides a direct test of the parallel trends assumption through examination of pre-release coefficients; and (3) it allows the data to reveal the temporal profile of effects rather than imposing an arbitrary discontinuous treatment structure.

3.4.1 Stacked DiD with Multiple Horizons

Formally, we define treatment exposure relative to each model's release date r_m for both training and release: **1) Training windows:** The period $[r_m - \bar{\tau}_{\text{train}}, r_m)$ preceding release, during which computational resources are devoted to model training. We remain agnostic about the exact training start date; our identifying assumption is that training activity is elevated in the months before release and that any pre-existing differential trends would manifest in the earlier leads. **2) Inference window:** The period $[r_m, r_m + \bar{\tau}_{\text{infer}}]$ following release, during which the model is deployed for inference via API access or public availability.

Stacked Difference-in-Difference

Model releases occur at different calendar dates, creating a staggered adoption setting. Recent econometric literature has demonstrated that conventional two-way fixed effects (TWFE) estimators as is the case in the estimates from Equation 3.1. can produce biased estimates under treatment effect heterogeneity Goodman-Bacon (2021); Sun et al. (2021); Callaway and Sant'Anna (2021); Borusyak et al. (2024). Treatment effect heterogeneity means the policy works differently for different places or periods, which can distort standard difference-in-differences estimates. The bias arises because TWFE implicitly uses already-treated units as controls for later-treated units, and differences in treatment effects across cohorts or over time contaminate the estimated average.

In our setting, consider regions as units and AI model releases as treatments inducing increased data center power demand. A "cohort" comprises regions whose data centers began significant AI workloads at the same time. Suppose Region A adopted AI workloads in 2022 while Region B adopted in 2023. When estimating treatment effects for Region B, TWFE implicitly uses Region A as a control—but Region A in 2023 is not untreated; its data centers have been running AI workloads for a year. If treatment effects evolve over time (e.g., power demand grows as AI infrastructure scales), comparing Region B's outcomes against Region A's already-treated outcomes introduces bias. Staggered design provides heterogeneity-robust estimators that solve for this exact issue.

To address this, we use a stacked DiD model Cengiz et al. (2019); Deshpande and Li (2019), where we construct a stacked dataset where each model release defines a separate "sub-experiment." The idea here is to construct, for each treatment cohort, a separate sub-experiment using only units that are genuinely untreated at the time of that cohort's adoption. By stacking these sub-experiments—each with its own clean control group of not-yet-treated regions—we ensure that no already-treated unit ever serves as a control, thereby eliminating the bias that arises when treatment effects are heterogeneous across cohorts or time.

Formally, for each release event m, we create a dataset containing: (i) Treated units – Regions linked to the releasing company's AI data centers. (ii) Control units – Regions with no AI data center exposure during the event window, or regions whose treatment occurs sufficiently far from event m that they serve as clean controls.

- Time window: A symmetric window around r_m, e.g., $[r_m - K, r_m + K]$ for some horizon K.

Following we then stack these sub-experiments and estimate:

$$Y_{i,t,m} = \sum_{k=-K}^{K} \beta_k \cdot D_{ut}^{k} + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.2}$$

where γ_u are unit-by-event fixed effects, δ_u are time-by-event fixed effects, $\mathbf{X}_{ut}'\theta$ represents controls, and the coefficients $\{\beta_k\}$ trace out the dynamic treatment effect at each event-time horizon k. Standard errors are clustered at the unit level to account for correlation across events within the same region.

3.4.2 Estimating Power Quality

Our power quality analysis estimates the following stacked DiD specification at the utility-month level:

$$\text{PowerQuality}_{ut} = \sum_{k=-K}^{K} \beta_k^{PQ} \cdot D_{ut}^k + \gamma_u + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.3}$$

where:

- PowerQuality_{ut} is the Whisker Labs Consumer Power Quality Index for utility u in month t;
- $D_{ut}^k = \mathbf{1}[\text{event-time} = k] \times \text{Treated}_u$ indicates that utility u is k periods from the release date of a model linked to an AI data center in its service territory;
- γ_u are utility fixed effects absorbing time-invariant characteristics;
- δ_t are calendar month fixed effects absorbing common temporal shocks;
- $\mathbf{X}_{ut}$ includes time-varying controls (weather, regional load).

We estimate an analogous specification for total harmonic distortion (THD):

$$\text{THD}_{ut} = \sum_{k=-K}^{K} \beta_k^{THD} \cdot D_{ut}^k + \gamma_u + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.4}$$

3.4.3 Estimating Fossil Fuel Demand

Power demand analysis proceeds at the generator-month level, exploiting finer geographic resolution. We maintain the stacked DiD structure while incorporating instrumental variables to address the price endogeneity concern. Price endogeneity arises because electricity prices and demand are simultaneously determined—high demand pushes prices up, while high prices suppress demand—making it impossible to identify the causal effect of price on consumption from their correlation alone.

Two-stage least squares resolves this by isolating variation in price that is plausibly exogenous: in the first stage, we predict prices using an instrument that affects prices but has no direct effect on demand; in the second stage, we regress demand on these predicted prices rather than observed prices, recovering an unbiased estimate of the price elasticity.

First Stage. We instrument for electricity prices using supply-side cost shifters:

$$p_{it} = \pi_0 + \pi_1 \text{HeatRate}_{it} + \pi_2 \text{GasPrice}_t + \gamma_i + \delta_t + \mathbf{X}'_{it}\pi_3 + \nu_{it} \tag{3.5}$$

where HeatRate_{it} captures the thermal efficiency of generator i and GasPrice_t is the contemporaneous natural gas spot price. These instruments satisfy relevance (they are primary determinants of marginal generation costs) and exclusion (they affect demand only through their impact on prices, not through direct effects on AI workload scheduling or other consumption decisions).

Second Stage. We estimate the stacked DiD specification using predicted prices:

$$\text{FossilDemand}_{it} = \sum_{k=-K}^{K} \beta_k^{FD} \cdot D_{it}^k + \eta \hat{p}_{it} + \gamma_i + \delta_t + \mathbf{X}'_{it}\theta + \varepsilon_{it} \tag{3.6}$$

where D_{it}^k indicates event-time relative to model releases linked to AI data centers in generator i's service area.

The coefficients $\{\beta_k^{FD}\}$ capture the dynamic effect of AI activity on fossil fuel demand. Negative values of k (pre-release) correspond to the training phase; positive values correspond to inference. The price coefficient η controls for supply-driven price movements and provides a scaling factor for welfare calculations.

3.4.4 Estimating Price Effects

Beyond quantity effects, we estimate how AI-driven demand increases affect electricity prices. The key economic insight is that price impacts depend on supply flexibility. In other words, when generators can easily expand output, increases in demand (e.g. AI-driven demand) can be easily absorbed with minimal increases in price. When the demand suddenly spikes (i.e. a demand shock), the prices will increase sharply as well. This is characterized using *supply elasticity* ε^s—the percentage change in quantity supplied per one percent price change—quantifies this responsiveness.

Our estimation proceeds in two steps. First, we use our IV estimates to quantify how AI activity shifts demand. Simply put, the average change in total fossil demand from AI demand is the DiD coefficient times the treatment indicator:

$$\Delta\widehat{\text{FossilDemand}}it = \hat{\beta} \cdot \Delta\text{AI}it \tag{3.7}$$

Second, we estimate a linear supply relationship using demand controls as instruments and supply shifters as controls. While power markets involve optimal dispatch, the fossil-fuel portion of PJM's marginal cost curve is approximately linear over the operating range we study PJM Interconnection, L.L.C. (2025). Let p_i denote the market-clearing price in market i, and let Q^s denote the supplied quantity. We model supply locally as linear in price, $Q^s = a + b \cdot p_i$, where the slope parameter b captures the responsiveness of supplied quantity to price changes (i.e., how flexibly supply can adjust to incentives). Rearranging, the implied price response to a demand shift ΔAI_{it} is given by $\Delta p = \hat{\beta} \cdot \frac{\Delta AI_{it}}{b}$. Expressed in percentage terms, the price impact depends on the flexibility of supply with respect to price:

$$\%\Delta p = \frac{\%\Delta Q^d}{\varepsilon^s}, \tag{3.8}$$

where ε^s denotes the percentage change in supplied quantity per percentage change in price.

We estimate ε^s via IV regression of quantity on price in log-log form, using demand-side instruments (weather-driven load variation and industrial production indices).

We focus on the PJM operator at the zonal level for this analysis. We restrict to a single ISO (operator) because differing market structures make prices incomparable across ISOs, and choose PJM because it is the largest (65 million people, 13 states), includes key data center states (Virginia, Pennsylvania), and provides 20 diverse pricing zones. Because observed training locations are outside PJM, this analysis uses the broader model set only.

3.4.5 Validating Assumptions and Robustness

We implement several diagnostic and robustness tests to assess the credibility of our estimates.

Sample Selection

Our empirical strategy balances internal validity with external relevance. Our main sample comprises four models for which we observe exact training and inference dates as well as training location, and we employ propensity score matching (described in the appendix) to support the parallel trends assumption. We then report results for progressively larger samples of models and data centers, accepting weaker identification in exchange for broader coverage. This approach enables precise, well-identified estimates from our core sample while exploring heterogeneity and counterfactuals in extended samples, with appropriate caveats regarding the latter.

Pre-Trends and Parallel Trends Testing

Our primary test of the identifying assumption examines whether pre-release coefficients $\{\beta_k : k < 0\}$ are jointly indistinguishable from zero. We report:

- Point estimates and confidence intervals for each lead coefficient.

- An F-test of the joint null hypothesis $H_0 : \beta_{-K} = \beta_{-K+1} = \cdots = \beta_{-1} = 0$.

Significant pre-trends would undermine the causal interpretation of post-release estimates.

Placebo Tests with Randomized Treatment Assignment

To verify that our estimates reflect genuine treatment effects rather than spurious correlations or specification artifacts, we conduct placebo tests with randomly assigned treatment:

1. **Random treatment timing:** We hold the set of treated units fixed but randomly reassign release dates drawn from the empirical distribution of actual release dates. Under the null of no effect, placebo estimates should center on zero.

2. **Random treatment assignment:** We hold release timing fixed but randomly reassign treatment status across units. This tests whether results are driven by the specific geographic pattern of AI data center locations.

Sensitivity to Treatment Definition

Given uncertainty in linking models to specific training locations, we assess robustness to alternative treatment definitions in our less conservative analysis:

1. **Intensive margin:** Weight treatment by the number or capacity of AI data centers in the region, rather than using a binary indicator.

2. **Dropping ambiguous cases:** Exclude model releases where the company operates data centers in multiple balancing authorities, restricting to cases with cleaner geographic attribution.

3. **Verified timing subset:** Estimate separately on the subset of models with externally verified training windows.

4. **Geographic Resolution and Robustness.** The generator-level data permit more granular treatment definitions than the utility-level power quality data. We exploit this

flexibility by estimating specifications under alternative treatment definitions in terms of radius.

Heterogeneity by Model Characteristics

We examine whether treatment effects vary with observable model characteristics:

- Model scale (parameter count, reported compute).
- Company (to assess whether effects are driven by specific firms).
- Release year (to assess whether effects change over time).

Heterogeneity analysis both informs mechanisms and serves as a specification check: if effects are concentrated among larger models or models from companies with substantial AI infrastructure, this lends credibility to the AI-driven interpretation.

3.4.6 Counterfactual Analysis

We conduct three primary sets of counterfactual exercises to investigate how alternative evolutionary paths for AI compute might affect grid-level outcomes.

Model Scaling. First, we examine the grid implications of continued growth in AI model size. Using the Epoch database to characterize trends in model parameters, we provide descriptive analysis of how increasing model scale translates into grid-level impacts. We then relate our estimated power demand coefficients to power quality coefficients, enabling analysis of how changes in training and inference energy efficiency propagate to power quality outcomes.

Geographic Redistribution. Second, we leverage the geographic locations of data centers in conjunction with our econometric estimates to conduct counterfactuals focused on spatial reallocation of AI compute. We consider two scenarios. In the first, we simulate a shift from cloud-based to edge-based inference—reallocating inference demand from concentrated hyperscaler facilities to smaller data centers distributed closer to population centers.

We implement this by redistributing our estimated inference demand impacts across data centers not currently classified as AI-focused. In the second scenario, we simulate relocating training workloads from population-dense regions on the East Coast to energy-abundant regions in the center of the country. For both geographic counterfactuals, we provide sensitivity analyses in the appendix to account for potential increases in communication costs and efficiency losses.

Colocated on-site Power Generation. Finally, we exploit a new feature in the Aterio data, namely a flag on data centers with on-site power generation. We re-estimate both our main models separately for generators with and generators without onsite power generation. We then report the impacts to power demand and power quality of either removing all onsite generation or having onsite generation for all data centers.

Together, these counterfactuals illuminate both how the grid might be affected by different trajectories for AI compute and which levers—geographic placement, workload composition, efficiency improvements—may be available to computer scientists and policymakers seeking to mitigate grid impacts.

3.5 Data

Table 3.1 summarizes the datasets we use to address each of our three research questions, distinguishing between variables of interest and the control variables described above. For variables of interest, we rely on **Aterio**, which provides comprehensive information on the location and history of data centers. The Aterio dataset documents when and where centers have been established, expanded, or retired, along with details on ownership and operational capacity (in MW). This allows us to identify data centers owned by specific hyperscalers and to quantify their scale over time. We then combine these data with external sources that provide the necessary controls. The data centers we use include ones that we could verifiably trace as having been the source of training of well-known large scale models such

Table 3.1: Data sources for each analysis question.

Source	Content	Use in Analysis	# Instances
Q1: Difference-in-Differences (Power Quality)			
Aterio	Data center locations, history, ownership, capacity (MW)	Define treatment regions; link AI model releases	3862 data centers
Whisker Labs	CPQI (consumer reliability); THD (waveform distortion)	Outcome variables for DiD regressions	2684 utility-months
EIA	Retail service territory maps	Define treatment/control boundaries	72 utilities
Q2: Instrumental Variables (Fossil Demand)			
EPA CAMPD	Hourly generator demand; heat rates	Fossil demand outcome; heat rate as instrument	22 million generator-hours
S&P Capital IQ	Natural gas prices	Instrument and fuel-cost control	12 Million location-day prices
Aterio	Data center proximity to generators	Treatment near releasing-company centers	3862 data centers
Meteostat	Temperature, precipitation	Demand and renewable controls	22 million location-hours
ISOs	Zonal wholesale prices	Market controls	54 zones
Q3: Counterfactual Analyses (Technical Evolution of AI)			
Whisker Labs	CPQI; THD	Baseline power quality impacts for scaling projections	
Epoch	Model parameters, FLOPs, release dates	Scaling regressions; efficiency counterfactuals	75 models

as GPT-3, PaLM, and Llama 2, among others.

3.5.1 Data and Empirical Specification

We analyze four primary outcomes. **The Consumer Power Quality Index (CPQI)** from Whisker Labs (2024) provides a composite measure of consumer-facing reliability events (surges, sags, brownouts, interruptions), summarizing frequency and severity of power deviations at the household level. **Total Harmonic Distortion (THD)** data from the same source offer a technical measure of waveform distortion, quantifying voltage deviation from a clean 60-Hz sine wave; elevated THD indicates grid stress, reduces motor efficiency, and shortens equipment lifespan (Figure 3.6). Since THD data are more recent than CPQI, fewer model releases are available for THD analysis. We incorporate **generator demand** from EPA's CAMPD database, providing hourly plant-level demand spanning 2021–2023. Finally, we combine **wholesale zonal price data** from Independent System Operators (ISOs) to assess AI impacts on prices.

Our power quality regressions use a parsimonious specification with time and geographic fixed effects. Generator demand regressions employ a richer specification: we instrument for endogeneity using generator heat rates and natural gas prices (S&P Capital IQ Global) in a two-stage least squares framework, and control for weather conditions (Meteostat) to capture demand and renewable variability. EIA retail service territory files reconcile generator-, zonal-, and retail-level observations.

3.5.2 Combining Data Sources

To integrate datasets, we harmonize spatial and temporal units across sources. Figure 3.11 demonstrates the integration at each step for our modeling purposes. Data center locations from Aterio are mapped to ISO zones, EIA territories, and counties, enabling linkage with Whisker Labs reliability indexes (CPQI and THD) and CAMPD generator demand. We are the first study to integrate Aterio's new field indicating which data centers are utilized in processing AI workloads. CPQI has 2,683 observations (mean 0.52, s.d. 0.42), while THD is more dispersed (mean 1.81, s.d. 6.77). CAMPD generator demand averages 218 MW (s.d. 158), and wholesale prices average $51/MWh (s.d. 148). Weather data capture local conditions (mean temperature 17°C, precipitation 0.12 mm), and time series are aligned at hourly or monthly resolution. External controls—natural gas prices, weather, and ISO market data—are merged by geography and time, yielding a unified panel suitable for difference-in-differences and IV regressions.

Our demand model focuses on deregulated markets in the Eastern Interconnection and ERCOT, where data on zones and prices are readily available, while our power quality model draws on Whisker Labs data from 72 retail utilities nationwide (monthly, 2022–2025). The demand regressions use hourly data for several thousand generators (2021–2023). We concentrate on hyperscaler AI companies—including Meta, Microsoft, Amazon, and Google—with Anthropic linked to Google due to its cloud partnership. Appendix materials include maps, sample descriptions, and data tables. In order to determine which data centers should be considered AI data centers and should be included in our sample, we perform a linkage procedure described in appendix Subsection 3.10.7.

3.6 Results

We reports results using our stacked DiD estimator (subsection 3.4.1) which is designed to handle staggered AI model releases. For all power quality and fossil demand effects, we first construct cohort-specific datasets following Callaway and Sant'Anna (2021), for each of the treatment events corresponding to major AI model releases through 2023 (subsection 3.10.5). Table 3.2 provides the full set of results from our main regressions including analysis of the impacts of on-site colocated generation for our counterfactuals.

3.6.1 Power Quality Effects: Utility-Level Strict DiD

We report the power quality effects of AI data centers in seventy one retail electric utility service territories estimated from power quality observations over 1136 utility-months. We use Consumer Power Quality Index (CPQI) and Total Harmonic Distortion (THD) as outcome variables in our stacked DiD models (Equation 3.3).

Panel A in Table 3.2 reports results from the utility-level strict DiD specification. We use strict here to distinguish that due to the sample size, we prefer the traditional DiD as our main specification rather than the stacked alternative. The estimated CPQI coefficient speaks to the reliability of the power supply to the consumers. The CPQI coefficient of 0.3795 indicates a power quality deterioration level that corresponds to moving from a typical U.S. power area toward the bottom quartile of power quality—roughly **the difference between $\sim$ 1 outage/year and $\sim$ 1.5–2 outages/year for the average customer**. Other reliability events such as brownouts and surges are predicted to increase by 1-2 per year. Importantly, power surges are a known ignition source for electrical fires, contributing to the approximately 51,000 annual residential electrical fires in the U.S., with electrical arcing accounting for 74% of ignition sources U.S. Fire Administration (2019).

Table 3.2: Summary of Estimation Results Across Specifications

Analysis	Outcome	Coefficient	Coefficient (Effect) Interpretation
Panel A: Utility-Level Power Quality (Two-Way FE)			
Strict DiD	CPQI	0.3795***	Corresponds to moving from typical U.S. power q the bottom quartile; increases annual outages for the customer from 1 to ∼1.5–2.
Strict DiD	THD (>0.8)	0.0625***	Causes more outages and equipment failure.
Panel B: Generator-Level Demand (Stacked DiD)			
IV 2SLS	Pre-treatment avg	+6.33***	
IV 2SLS	Post-treatment avg	+5.20***	> 500 additional GWh; ∼47,000 household-years
Panel C: PJM AI Price Analysis			
ComEd	Supply Elasticity	31.64	Higher supply elasticity in ComEd zone
ComEd	AI Price Impact	25.39	corresponds to larger AI-induced price
AEP	Supply Elasticity	8.13	impacts; AEP's lower elasticity yields
AEP	AI Price Impact	1.89	substantially smaller price effects
Panel D: Power Quality by On-Site Generation Status			
With on-site	CPQI	−0.122***	On-site generation mitigates grid stress from
Without on-site	CPQI	+0.228***	AI load; utilities without on-site experience
Difference	CPQI	−0.350***	reliability deterioration while those with on-site see improvements

Notes: *** $p < 0.001$, ** $p < 0.05$. Panel A: Utility and time fixed effects with cluster-robust standard errors; treatment defined as presence of AI-affiliated data center in utility service territory. Panel B: 35.6 million observations from 19 treatment events; standard errors clustered by unit-event; IV instruments: natural gas price × zone-average heat rate. OLS = Ordinary Least Squares; IV 2SLS= Instrumental Variables Two Stage Least Squares; CPQI = Consumer Power Quality Index; THD = Total Harmonic Distortion.

The post-treatment coefficient of 0.0625 for THD indicates that utility territories containing AI data centers experienced a 0.0625 percentage point increase in readings above the 0.8 distortion threshold following AI model releases. While a 0.0625 percentage point increase may appear small in isolation, it is still problematic as sustained exposure to power distortions compounds to over the multi-year lifespans of household appliances, accelerating degradation of refrigerators, air conditioners, and other motor-driven equipment National Electrical Manufacturers Association (2016); Laughner et al. (2024).

3.6.2 Fossil Fuel Demand Effects: Stacked DiD

Fossil fuel demand effects are estimated using demand data from a total of 35.6 million generator-hour observations[1], which includes 900,956 treated observations from generators located within 20 miles of AI-affiliated data centers and the other 34.7 million from other generators as control observations.

Average fossil fuel demand effects are shown in the IV 2SLS rows in Panel B of Table 3.2.[2] Figure 3.3 shows the coefficients spread two months before (training) and after (inference) model release.

The training phase fossil fuel power demand corresponds to the $t = -2$ pre-treatment coefficient of 6.33 MWh, which we find to also be statistically significant result. Note that this is the hourly demand at a single generator. When the aggregate impact is calculated, they represent over 500 additional GWh added nearby to a data center over the inference period in some cases. **The additional power is roughly equivalent to that of 47,000 household-years in the United States**. For inference, the $t = +2$ coefficient shows a significant demand increase of 5.20 MWh, representing the demand shift at generators near data centers two months after AI model release, isolated from supply-side price variation through our instrumentation strategy.

3.6.3 Price Impacts

We estimate wholesale electricity price effects by combining our demand estimates with supply curve parameters. This approach, described in Section 3.4.4, recognizes that our IV strategy identifies demand shifts but does not directly estimate equilibrium price

[1]These data are from he Clean Air Markets Program (CAMPD) dataset and include hourly generation by all US utility-scale fossil generators Agency (2022)

[2]Recall that we use two stage instrumental variables (IV) regression here, where the second stage isolates the AI model caused demand-side effects from supply shocks (i.e., generator heat rate and alternate fuel source prices) modeled in the first stage. Details on the specifics of the instrumental variables specification can be found in Appendix Subsection 3.10.7 and results from the first stage regression are presented in Appendix.

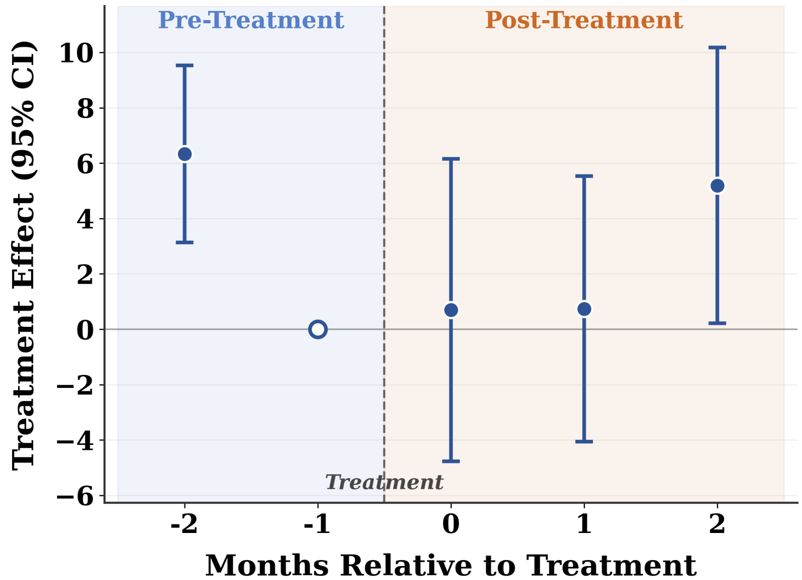

Figure 3.3: Stacked DiD coefficients for power demand (IV specification). Event Time 0 corresponds to AI model publication date. The pre-treatment coefficient at $t = -2$ demonstrates the training demand. Reference period is $t = -1$.

effects.[3] The price impact of AI-induced demand depends on the slope of the supply curve[4]: inelastic supply, where quantity supplied responds weakly to price changes, implies large price effects, while elastic supply implies small effects. We use supply chain elasticities for PJM obtained from our supply-side instrumental variables regression (see Figure 3.12 in Appendix 3.10.6 for a map).

Figure 3.4 showing price increase estimates by zone. On average AI training leads to **substantial increases in wholesale prices in PJM with the highest increase surpassing 25% in the American Electric Power (AEP) Zone**.

[3]Our instrumental variables approach isolates exogenous variation in electricity demand driven by AI model releases. However, observed market prices reflect the intersection of supply and demand in equilibrium. To translate our demand estimates into price effects, we must incorporate information about supply-side responses, which requires additional assumptions about the shape of the supply curve.

[4]Recall, the supply curve describes the relationship between price and quantity supplied. Its slope reflects how much generators must be compensated to produce additional electricity—steeper slopes indicate that marginal generation costs rise quickly as output expands, typically because higher-cost units must be dispatched.

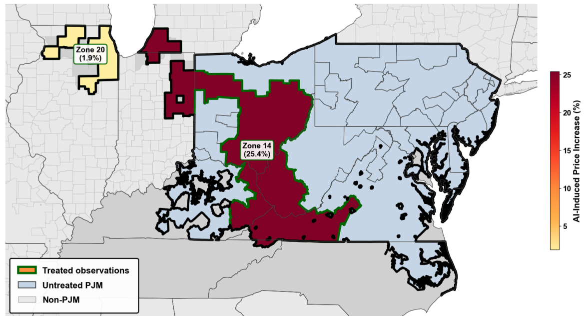

Figure 3.4: The map shows estimated AI-related wholesale prices increases in the PJM Interconnection in treated zones.

We also estimate the expected price rise for retail customers, using estimates from the literature as documented in Appendix Subsection 3.10.8. While we are not able to provide definite evidence of the impacts on retail price pass through, it should be noted that retail electricity prices have increased overall at a rate surpassing inflation since 2020 Nicoletti et al. (2024a); Horsley (2025); Southwest Power Pool, Inc. Market Monitoring Unit (2025). Also, while increased investment in electric generation through power purchase agreements by tech firms may in the long run lead to falling retail prices, it appears that at present data center power usage is increasing wholesale prices by raising total demand, which requires dispatching higher-cost generators and thus increases the market clearing price—a price paid by all retail consumers.

Using the most relevant examples for PJM of retail elasticity of between .53 and .56 De Feuillley and Fontaine (2023), **the expected consumer price rise in AEP, a sub-zone in PJM, would be** $\sim 13\%$. However, using recent long-run econometric estimates of full passthrough from MacKay and Mercadal (2024), the long-run expected rise for consumers may be closer to the full 25%.

3.7 Validity and Robustness

We present two sets of analyses to address issues highlighted in Subsection 3.3.3: formal testing to provide evidence for the parallel trends assumption, and placebo tests address potential issues related to external confounds or lack of specific knowledge about training.

Additional tests are found in Appendix 3.10.10.

1) Formal Testing for Parallel Trends: For power demand specifically, the formal joint test of pre-treatment coefficients using a Wald test of joint significance yields a chi-squared statistic of 0.99 (df = 1) with p-value = 0.3196. **We cannot reject the null hypothesis of parallel pre-trends at conventional significance levels.** This provides support for our assumption that treated and control generators followed similar demand trajectories prior to AI model releases i.e., our identification assumption is valid. Some of our samples demonstrate significant evidence against the the parallel trends test prior to propensity score matching (PSM) that we use for sample selection. However, across all matched samples for both power quality and power demand after PSM, we find *no evidence of violations of the parallel trends assumption.* This implies that the technique correctly handles the issue where it exists.

2) Placebo Tests We conduct two placebo exercises to demonstrate that the treatments are meaningful and not the result of random chance, we implement random date and random location placebo tests. Both yield a p-value of 1.00 for the coefficient in our fossil fuel demand, CPQI, and THD analyses, confirming that our results do not arise from spuriously chosen locations or dates.

3.8 Counterfactual Analyses

We present three main counterfactual exercises to inform stakeholders and policymakers about how alternative evolutionary paths for AI compute might affect grid outcomes. These combine our econometric estimates with data on model characteristics and infrastructure configurations to explore three dimensions: model scaling, geographic redistribution, and workload composition.

1) Model Scaling. We combine data on model size (parameter counts) with our econometric estimates from our least conservative specification to extrapolate grid impacts at different scales.

(a) Power Quality. Fitting a trend line to power quality impacts for Llama-3.1 (405B), PaLM (540B), and Llama 4 Behemoth (2T), we find that scaling from a 2 trillion parameter model to a 4 trillion parameter model would increase power quality impact from **0.321 to 0.434**—a deterioration of roughly 0.113 units, or just over 35% relative to the 2 trillion baseline. The relationship is exponential over log parameter counts, implying that larger models require proportionally greater parameter reductions to achieve the same percentage improvement in power quality. **(b) Power Demand.** Using an analogous extrapolation, we find that **each 1% parameter increase leads to approximately 0.15% higher power demand** within three months of model release. Scaling from 540 billion to one trillion parameters would **increase total power demand by 10.5%**. For DALL-E, halving parameters would **reduce inference demand by approximately 300 GWh**—equivalent to the annual consumption of 27,000–30,000 homes.

2) Training Compute. Training compute requirements also have measurable effects: a 1% increase in training FLOPs is associated with a **0.1% increase in power demand**. For GPT-3.5, halving training compute would reduce energy use by enough to power 8,000–10,000 U.S. homes for a year.

3) Colocated On-site Generation. Data centers increasingly deploy colocated on-site generation to reduce grid dependence. Our inventory identifies 87 facilities with on-site capability and 12,401 without. We test whether on-site capacity moderates estimated effects by interacting treatment with on-site status in the utility-level strict DiD specification. Results indicate that on-site generation substantially mitigates measured grid impacts. The sign reversal for CPQI is notable: data centers *with* on-site generation are associated with improved power quality (negative coefficient), while those *without* are associated with deterioration (positive coefficient). This heterogeneity suggests that **on-site generation absorbs demand spikes that would otherwise propagate to the grid**, and that aggregate grid impacts may understate total AI energy consumption.

Figure 3.5 extends this to policy-relevant scenarios. Remarkably, universal adoption of on-site power would shift impact of data centers from a net detriment to power quality to a net improvement.

4) Geographic Redistribution

(a) Distributed Inference (Edge Computing). In our sample where we know AI data centers but not specific training and inference sites, we simulate a shift from cloud-based to edge-based inference, redistributing inference demand from concentrated hyperscaler facilities to smaller data centers located closer to population centers. Under this scenario, total demand added in a particular zone falls from a baseline of 162 MWh per hour over the entire treatment period to 4 MWh per hour. Figure 3.5 illustrates this effect of shifting the composition of inference.

(b) Concentration in Low-Population, High-Supply Regions. Alternatively, we consider relocating both training and inference to regions with abundant electricity supply and lower baseline load, such as areas in the central United States with substantial renewable or fossil capacity. We simulate this by shifting demand from the baseline case where training and inference are assumed distributed to a case where the only data center for training and inference is the largest data center owned by a hyperscaler. This leads to an extreme concentration of added demand with nearly 100 GWh added near data centers in training locations. Figure 3.5 shows this scenario.

This analysis has the important caveat that shifting to one data center for all training activity is extreme, with many possible configurations between full decentralization and full centralization. Centralization may also yield other benefits, such as concentration near renewable generation and reduced energy consumption from lower communication costs.

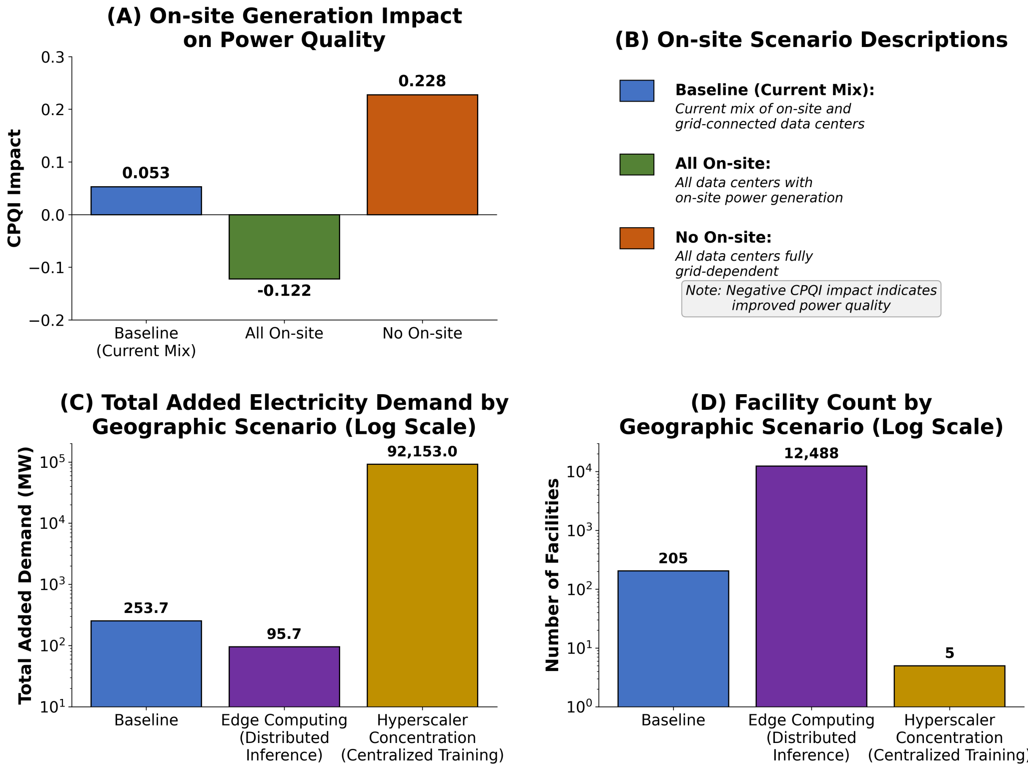

Figure 3.5: Effects of Geographic Concentration and On-site Generation on AI Impacts on the grid.

3.9 Conclusion

Understanding how AI models impact U.S. power grids is critical for ensuring energy reliability and security. This work highlights the challenges in quantifying this impact using existing data sources. It introduces a rigorous econometric methodology to overcome the lack of fine-grained data about contributing factors. In particular, we use difference-in-differences regressions to provide econometric estimates of the amount of power being used by specific AI models as well as the power distortions being caused by AI and the impacts of AI on price. We find evidence that AI is making power quality significantly worse and increasing fossil fuel power demand. Using counterfactual analyses, we provide estimates of the potential impacts of increasing efficiency of AI on the power draw of data centers. We also show how changes in the geographic composition of AI, a potential shift to edge computing, and a shift in the ratio of training to inference conducted impact the power market.

3.10 Appendix

3.10.1 Total Harmonic Distortion

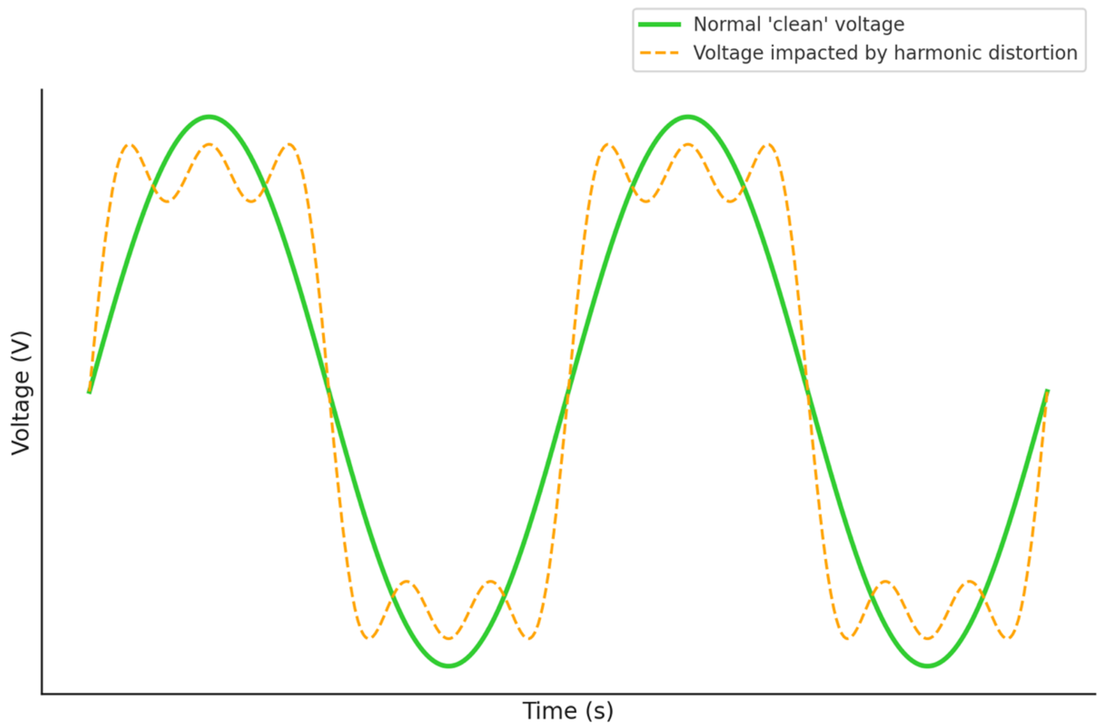

Figure 3.6: Total harmonic distortion representation.

Figure 3.6 illustrates the distortions in the power frequencies. The green lie shows the *clear* voltage curve, which is cleanly sinusoidal, whereas with huge loads can produce harmonic distortions shown as the dotted yellow curve. This *dirtier* voltage can adversely affect the operation and lifetime of household appliances and other products that use electricity.

3.10.2 Expanded Dataset Description

Below, maps can be found providing more detail on the data center dataset alongside tables on the datasets and the dataset sample selection criteria.

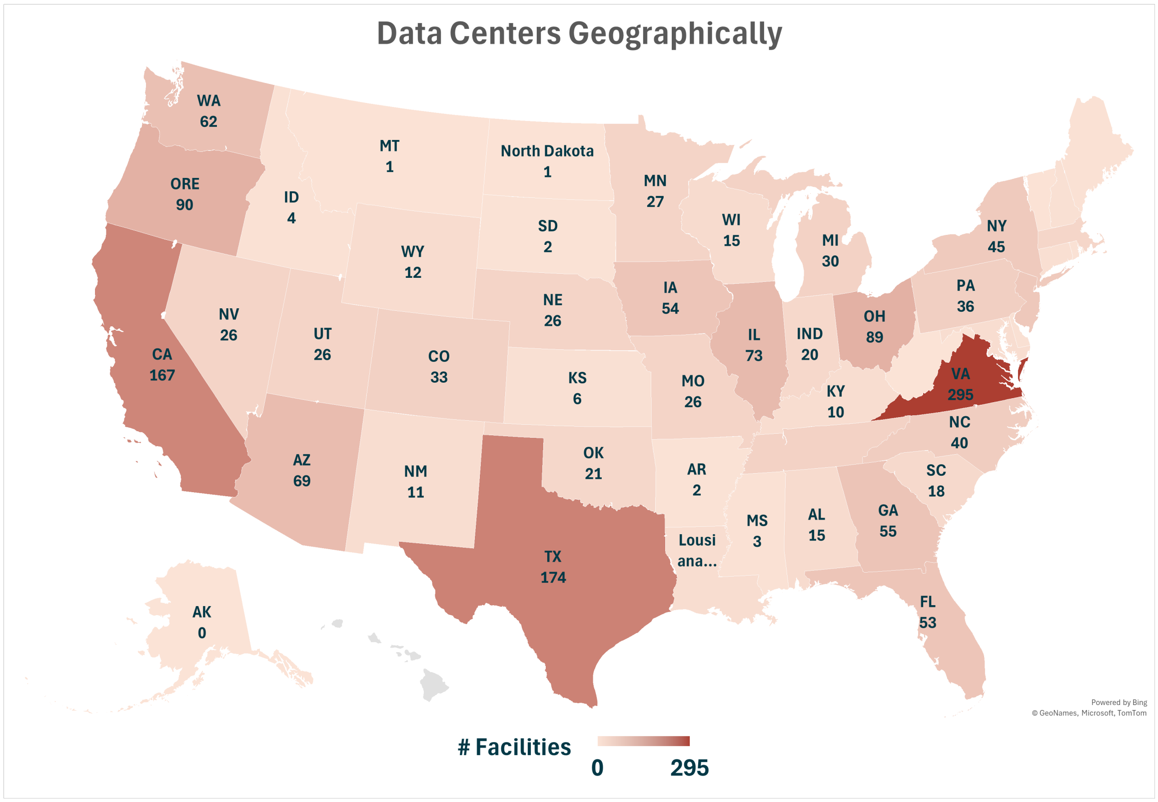

Figure 3.7: Data centers by state.

Choropleth Start_ZoneCode

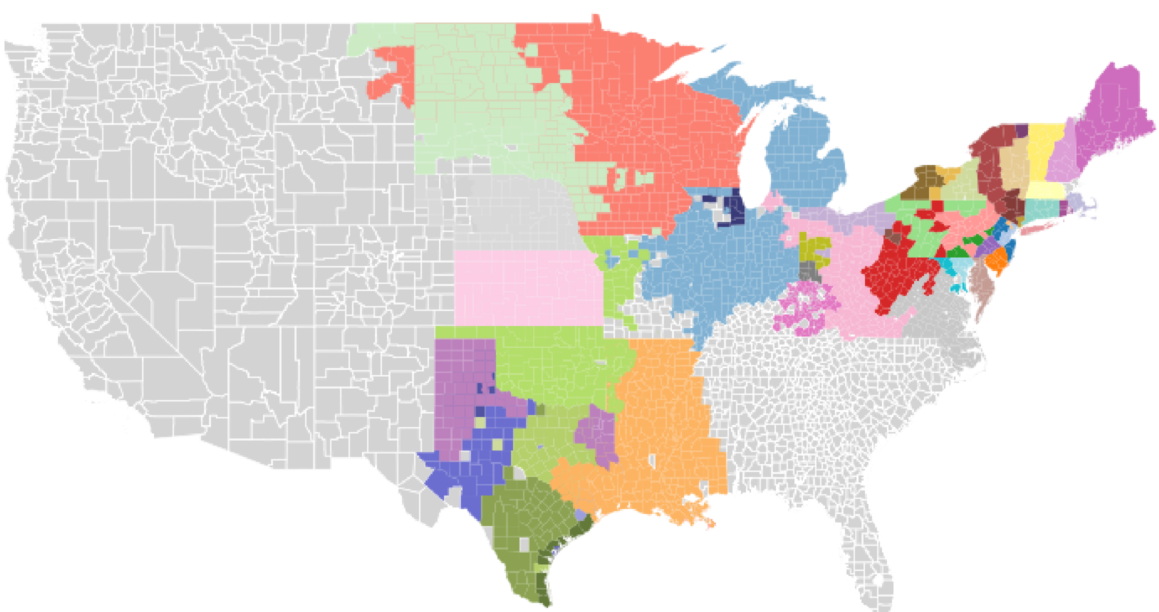

Figure 3.8: Map of zones for demand model.

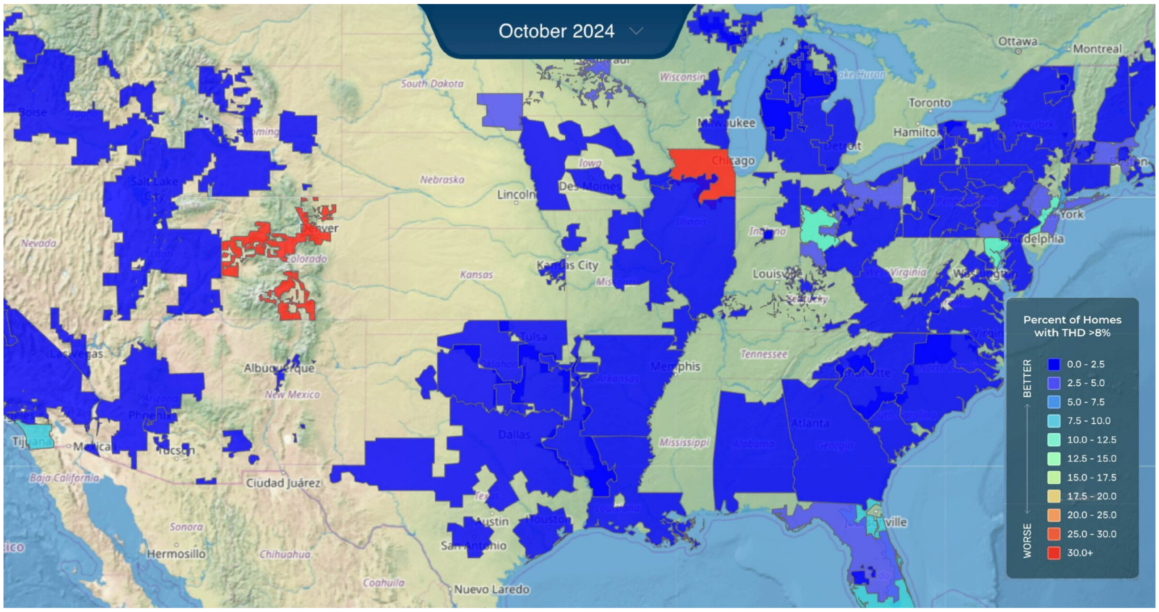

Figure 3.9: Map of Retail Electric Utilities for Whisker Labs Data.

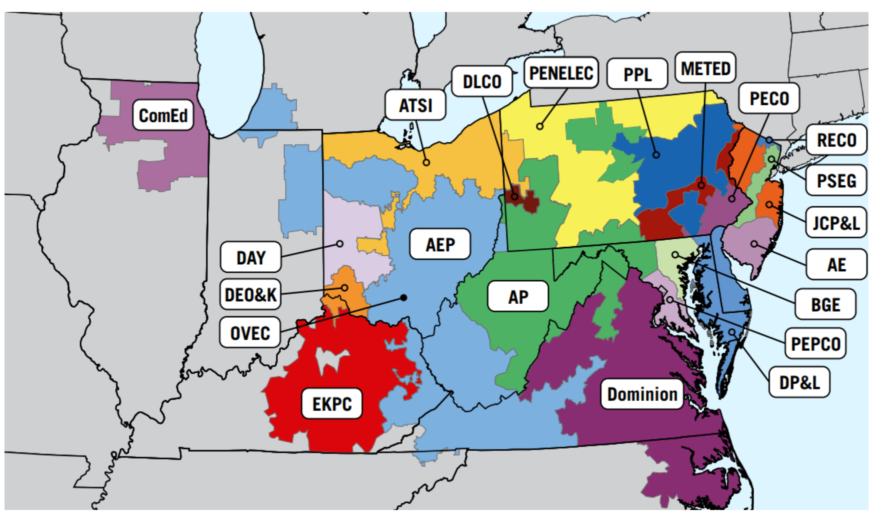

Figure 3.10: Map of PJM by Zones.

Table 3.3: Expanded description of data sources (for appendix).

Source	Content and Granularity	Use in Analysis
	Q1: Difference-in-Differences (Power Quality)	
Aterio	Data center locations, establishment/expansion/retirement dates, ownership, and operational capacity (MW). Data are available at the facility level and updated annually.	Defines treatment regions; links AI model releases to hyperscaler-owned centers.
Whisker Labs	Consumer Power Quality Index (CPQI, composite measure of surges, sags, brownouts, interruptions) and Total Harmonic Distortion (THD, waveform distortion). Provided at county level, hourly frequency.	Outcome variables for DiD regressions.
Meteostat	Hourly temperature and precipitation, aggregated at the county level. Coverage includes all U.S. counties with weather stations.	Controls for demand fluctuations and renewable generation variability.
ISOs	Geographic boundary shapefiles and wholesale price data at zonal level. Boundaries digitized from ISO-provided maps; prices available hourly.	Market alignment and regional controls.
EIA	Retail service territory shapefiles, at utility level, updated periodically.	Used to define treatment/control boundaries and reconcile geographies.
	Q2: Instrumental Variables (Fossil Demand)	
EPA CAMPD	Hourly generator-level demand (MWh) and unit-specific heat rates (Btu/kWh). Coverage 2021–2023.	Generator demand is outcome; heat rates serve as instruments for prices.
S&P Capital IQ	Daily natural gas price series, Henry Hub benchmark. National coverage.	Instrument for price and fuel-cost control.
Aterio	Generator proximity to data centers owned by releasing companies, matched at 20-mile radius.	Defines treatment generators in IV design.
Meteostat	Hourly temperature and precipitation, as above.	Controls for demand and renewable variability.
ISOs	Wholesale price series, zonal level, hourly frequency.	Market-level controls.
	Q3: Counterfactual Analyses (Scaling, Efficiency, & Tech Development)	
Whisker Labs	CPQI and THD (as above). Used to establish baseline deterioration in power quality.	Benchmark outcomes for scaling projections.
Epoch	AI model metadata: release dates, FLOPs, parameter counts. Public dataset with model-level detail.	Used for scaling regressions and efficiency counterfactuals.
GPU Experiments	Energy use measured on NVIDIA A6000 GPUs. Experiments run with batch size 4, sequence length 1024, and 200 inferences. Each bar in results represents total energy; lighter portion indicates communication overhead. Data collected in-house on 8xA6000 cluster.	Provides empirical GPU energy baselines for counterfactual scaling exercises.

Table 3.4: Dataset sample selection and summary statistics

Dataset	Content / Summary Statistics	Sample Selection
Aterio	Data center locations, ownership, and capacity; mapped to ISO zones, EIA service territories, and counties	Hyperscaler data centers (Meta, Microsoft, Amazon, Google; Anthropic via Google)
Whisker Labs	Reliability measures: CPQI ($N = 2{,}683$, mean=0.52, s.d.=0.42); THD (mean=1.81, s.d.=6.77)	72 retail electric utilities, monthly data 2022–2025
CAMPD	Generator-level demand: mean 218 MW (s.d.=158); operating times ≈ 1	Several thousand fossil generators, hourly data 2021–2023
ISOs	Wholesale electricity prices: mean $51/MWh (s.d.=148); zonal boundaries	Deregulated markets in Eastern Interconnection and ERCOT
Weather (Meteostat)	Temperature (mean=17°C, s.d.=11.4); precipitation (mean=0.12 mm, s.d.=0.91)	Matched by county/zone, hourly or daily resolution
External Controls	Natural gas prices, ISO market data, geography/time harmonization	Used for both DiD and IV regressions

Notes: Time series are harmonized at hourly or monthly resolution depending on source. Unified panel supports both difference-in-differences (power quality) and instrumental variable (demand) regressions. Maps of sample regions provided in the appendix.

3.10.3 Dataset Creation Diagram

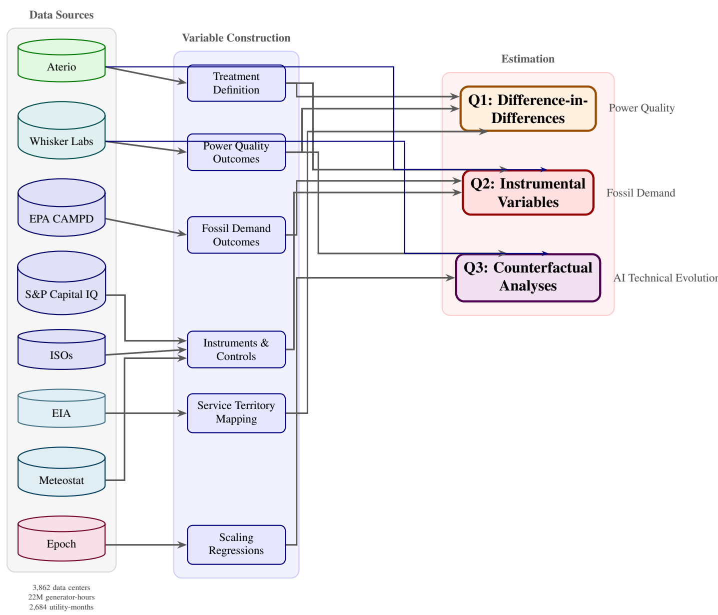

Figure 3.11: Data sources and analysis pipeline. Data sources flow through variable construction into three estimation strategies addressing power quality impacts (Q1), fossil fuel demand (Q2), and counterfactual scaling scenarios (Q3).

3.10.4 Calculation of Comparisons

We calculate the household average energy usage based on data provided by EIA (2024). We simply divide our estimates by the provided numbers to get the number of households for our comparisons.

3.10.5 Specifically Verified Models

Below is a subset of our dataset for specifically verified models to provide an example of the data:

Table 3.5: Documented Training Locations (All Confirmed)

Model	Location	Source
GPT-4	W. Des Moines, IA	KCCI 8 News (2023)
PaLM, LaMDA, MUM	Mayes County, OK	Google Cloud (2023)
Megatron-Turing NLG	Santa Clara, CA	NVIDIA and Microsoft (2021)

Table 3.6: Verified Model Dates

Model	Type	Date	Source
GPT-3	API	2021-11-18	OpenAI (2021)
Claude 2.1	API	2023-11-21	Anthropic (2023)
PaLM API	API	2023-03-14	Google Developers Blog (2023)
Llama 2	API	2023-07-18	Meta Newsroom (2023)
Code Llama	Train	2023-01–07*	Meta AI (2023)

*Date range (month precision)

3.10.6 PJM price Elasticities

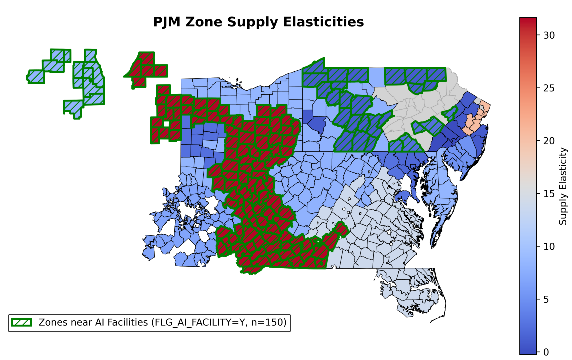

Figure 3.12: PJM Price Elasticity by Zone

3.10.7 Treatment Definition and Data Linkage

Our treatment relies on linking AI model releases to geographic regions through a chain of observable connections: models to companies, companies to data center locations, and data center locations to electricity service territories. We describe each link and its limitations.

Identifying AI Data Centers

We leverage a recently released variable from Aterio that directly classifies data centers by their primary use case, including a designation for AI-focused facilities. This classification improves upon prior approaches that treated all data centers homogeneously, as AI workloads—particularly large-scale model training—exhibit distinct load profiles characterized by sustained high utilization and substantial power draws that differ markedly from traditional cloud computing or storage operations.

We link data centers to electricity service territories at two geographic resolutions:

- **Utility service territory:** For power quality analysis, we match AI data centers to retail electric utility boundaries. A utility is designated as treated if at least one AI data center operates within its service territory.
- **Generator proximity:** For fossil fuel demand analysis, we match data centers to generators by picking generators within a 20-mile radius of an in-sample data center. This finer resolution exploits the greater spatial granularity of EPA generator-level data.

Linking Models to Data Centers

We construct a crosswalk from AI model releases to data center locations through company ownership. For each major model release in our sample, we identify the developing company and link to all AI-classified data centers operated by or contracted to that company.

Identifying Assumption and Limitations. Our approach assumes that training and inference activity for a given model occurs at data centers associated with the releasing company. This assumption is imperfect for several reasons:

1. Training may occur at third-party cloud facilities not captured in our data center inventory.

2. Inference workloads may be distributed across facilities differently than training workloads.

We interpret our estimates as capturing the *average* effect across a company's AI data center footprint, which represents a lower bound on the localized effect at the actual training site if activity is concentrated, or an upper bound if we misattribute activity to facilities that were not involved. We probe robustness to this measurement error through several approaches described in Section 3.4.5.

For a subset of models where public information permits more precise dating, we supplement our treatment timing with verified training and inference windows:

Instrumental Variables Specification

The learned DiD coefficients capture the relationship between AI model releases and generator output, but this relationship may be confounded by supply-side shocks. For example, if AI model releases coincide with changes in natural gas prices or generator availability that independently affect electricity production, our estimates would conflate demand-side and supply-side effects. To isolate the demand-side effects, we instrument for electricity prices using the interaction of generator-level heat rates and regional natural gas prices. This instrumentation strategy exploits cross-sectional variation in generators' fuel efficiency: conditional on natural gas price movements, generators with higher heat rates (lower efficiency) face larger marginal cost shocks, allowing us to trace out the demand curve.

First-Stage and Structural Estimates. The first stage regresses electricity prices on the heat rate–gas price interaction. The structural equation then uses predicted prices to estimate price responsiveness. The estimated price elasticity from the structural equation is 0.135 (SE = 0.013, $p < 0.0001$), indicating economically meaningful and precisely estimated demand responsiveness. As a specification check, we estimate the first stage separately for each of 5 model-specific regressions; the mean price coefficient is -0.331 (SD = 0.005, range: -0.336 to -0.323). The tight clustering of estimates across specifications suggests stable demand elasticity across AI model events and supports the validity of our instrumentation strategy.

3.10.8 Wholesale-to-retail Passthrough Estimates

Table 3.7: Wholesale-to-Retail Price Pass-Through in Deregulated Markets

Study	ISO	PT	Class
Panel A: Wholesale → Retail Pass-Through			
De Feuillley and Fontaine (2023)	PJM (PA)	0.53	Res. (EDC)
De Feuillley and Fontaine (2023)	PJM (PA)	0.56	Res. (EGS)
Brown et al. (2020)	ERCOT	0.43–0.47	Res.
MacKay and Mercadal (2024)	Multi	≈1.0[b]	All
Panel B: Demand Elasticity (Residential)			
Deryugina et al. (2020)	PJM (IL)	−0.09	SR (1 mo.)
Deryugina et al. (2020)	PJM (IL)	−0.27	MR (2 yr.)
Deryugina et al. (2020)	PJM (IL)	−0.35	LR (10 yr.)

Notes: PT = pass-through coefficient. EDC = Electric Distribution Company; EGS = Electric Generation Supplier. SR/MR/LR = short/medium/long-run.

[b]Long-run; \$9.59/MWh cost → \$8.81/MWh retail.

3.10.9 Propensity Score Matching

Here, we describe the propensity score matching (PSM) and inverse probability weighting (IPW) methodology used to enhance the robustness of our difference-in-differences (DiD) estimates. These methods address potential selection bias arising from non-random placement of AI data centers.

Standard DiD estimation relies on the parallel trends assumption: absent treatment, treated and control units would have followed similar trajectories. However, AI data center locations are chosen strategically based on factors such as electricity prices, grid reliability, and climate conditions. If power plants near AI data centers systematically differ from control plants in observable characteristics correlated with electricity demand trends, the parallel trends assumption may be violated.

Propensity score methods address this concern through three mechanisms: (i) balancing covariates to create comparison groups similar on observable characteristics; (ii) enforcing common support to ensure we only compare units that could plausibly receive either treatment status; and (iii) reducing model dependence by making estimates less sensitive to functional form assumptions (Rosenbaum and Rubin, 1983; Hirano et al., 2003).

The propensity score is the conditional probability of receiving treatment given observed covariates:

$$e(\mathbf{X}_i) = \Pr(T_i = 1 \mid \mathbf{X}_i) \tag{3.9}$$

where T_i is a treatment indicator equal to one for observations near an AI data center in the post-treatment period, and $\mathbf{X}_i$ is a vector of pre-treatment covariates.

We estimate propensity scores using logistic regression:

$$\log\left(\frac{e(\mathbf{X}_i)}{1 - e(\mathbf{X}_i)}\right) = \beta_0 + \boldsymbol{\beta}'\mathbf{X}_i \tag{3.10}$$

The covariates included in the propensity score model are: pre-trend load growth, ambient temperature (°F), dew point (°F), precipitation (inches), plant heat rate (BTU/kWh), and natural gas price (\$/MMBtu). Temperature and dew point capture weather-driven demand variation and cooling load requirements. Precipitation provides additional weather controls. Heat rate proxies for plant efficiency and technology type. Natural gas price captures fuel cost variation affecting dispatch decisions.

We restrict the sample to observations with propensity scores in the common support region $[0.1, 0.9]$:

$$\mathscr{S} = \{i : 0.1 \leq e(\mathbf{X}_i) \leq 0.9\} \tag{3.11}$$

This trimming rule excludes observations with extreme propensity scores—units almost certain to be treated or untreated—for which comparable counterfactuals do not exist (Crump et al., 2009). Observations with $e(\mathbf{X}_i) < 0.1$ have no comparable treated units, while those with $e(\mathbf{X}_i) > 0.9$ have no comparable control units. The trimmed sample ensures that treatment effect estimates rely on regions of the covariate space where both treatment statuses are observed.

IPW creates a pseudo-population where treatment assignment is independent of observed covariates by weighting observations inversely to their probability of receiving their actual treatment status (Hirano et al., 2003). We employ stabilized weights to reduce variance:

$$w_i^{\text{stab}} = \begin{cases} \dfrac{\Pr(T=1)}{e(\mathbf{X}_i)} & \text{if } T_i = 1 \\[2ex] \dfrac{1-\Pr(T=1)}{1-e(\mathbf{X}_i)} & \text{if } T_i = 0 \end{cases} \tag{3.12}$$

where $\Pr(T = 1)$ is the unconditional (marginal) treatment probability in the sample.

To prevent extreme weights from inducing instability, we cap weights at the interval $[0.01, 100]$. Within each treatment group, weights are then normalized to sum to the group size:

$$\tilde{w}_i = w_i^{\text{capped}} \times \frac{n_g}{\sum_{j \in g} w_j^{\text{capped}}} \tag{3.13}$$

where $g \in \{\text{treated}, \text{control}\}$ and n_g is the number of observations in group g.

Our baseline DiD specification takes the form:

$$Y_{it} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{Treat}_i + \delta(\text{Post}_t \times \text{Treat}_i) + \mathbf{X}_{it}' \boldsymbol{\gamma} + \varepsilon_{it} \tag{3.14}$$

where Y_{it} is the outcome of interest (electricity demand), Post_t indicates the post-treatment period, Treat_i indicates treatment group membership, and δ is the treatment effect of interest.

With IPW, we estimate a weighted least squares version:

$$\hat{\boldsymbol{\theta}}^{\text{IPW}} = \arg\min_{\boldsymbol{\theta}} \sum_{i,t} \tilde{w}_{it} \left(Y_{it} - \mathbf{Z}_{it}' \boldsymbol{\theta}\right)^2 \tag{3.15}$$

where $\mathbf{Z}_{it}$ includes all regressors and $\tilde{w}_{it}$ are the normalized IPW weights.

Our primary specification combines IPW with instrumental variables to address both selection bias (via IPW) and price endogeneity (via IV). The instruments for electricity price are natural gas price and zone-average heat rate. The resulting IV-IPW-DiD estimator is implemented via weighted two-stage least squares, following the doubly robust approach of Sant'Anna and Zhao (2020).

We assess covariate balance using the standardized mean difference (SMD):

$$\text{SMD}_k = \frac{\bar{X}_{k,\text{treated}} - \bar{X}_{k,\text{control}}}{\sqrt{(s^2_{k,\text{treated}} + s^2_{k,\text{control}})/2}} \tag{3.16}$$

where $\bar{X}_{k,g}$ and $s^2_{k,g}$ denote the sample mean and variance of covariate k in group g.

Following standard practice, we consider $|\text{SMD}| < 0.1$ as indicating good balance, $0.1 \leq |\text{SMD}| < 0.25$ as moderate imbalance warranting caution, and $|\text{SMD}| \geq 0.25$ as substantial imbalance (Austin, 2011). After weighting, we compute weighted SMDs using IPW weights to verify that the reweighting procedure successfully balances covariates across treatment groups.

We implement several robustness checks to validate our PSM-IPW approach.

First, we compare treatment effect estimates from standard (unweighted) DiD against IPW-DiD. Similar estimates across specifications suggest results are robust to selection on observables.

Second, we test sensitivity to trimming thresholds by varying the propensity score bounds (e.g., $[0.05, 0.95]$, $[0.15, 0.85]$, $[0.20, 0.80]$). Stable estimates across thresholds indicate robustness to the choice of common support region.

Third, we conduct a pre-trends test by estimating a specification that includes a placebo pre-treatment indicator. An insignificant coefficient on this term ($p > 0.05$) supports the parallel trends assumption.

Fourth, we replicate the analysis using only confirmed AI training facility locations from verified sources, reducing potential measurement error in treatment assignment.

Fifth, we compare covariate balance statistics before and after weighting. IPW should substantially reduce SMDs across all covariates.

The IPW-DiD estimator provides a causal estimate of the average treatment effect under the following identifying assumptions: (i) conditional independence—treatment assignment is independent of potential outcomes conditional on $\mathbf{X}$; (ii) parallel trends—absent treatment, treated and control units would follow similar outcome trajectories; (iii) common support—there exists positive probability of treatment for all covariate values; (iv) stable unit treatment value assumption (SUTVA)—no spillovers between units; and (v) correct propensity score specification—the logistic regression model is correctly specified (Abadie, 2005; Callaway and Sant'Anna, 2021).

An important limitation is that propensity score methods only control for selection on *observable* characteristics. If unobserved factors influence both data center location decisions and electricity demand trends, our estimates may still be biased. We partially address this concern through instrumental variables estimation, but acknowledge that unobserved confounding cannot be definitively ruled out in observational settings.

3.10.10 Additional Robustness Checks

Robustness Checks

We test result robustness by varying treated population definitions, estimating specifications with alternative geographic boundaries to confirm consistency. This demonstrates findings reflect genuine causal impacts rather than arbitrary boundary choices. While we lack data on which models run at specific centers, our DiD methodology accounts for this by using publication dates as exogenous treatment, utilizing minimal geographic and temporal treatment windows. We plan robustness checks restricting treatment to larger data centers most likely involved in AI training. To verify treatment effects aren't noise-related, we include appendix robustness checks with randomized treatment dates as additional evidence that effects relate to model releases.

Treatment Intensity

Rather than binary treatment indicators, we estimate a continuous treatment intensity specification using total electricity demand (MWh) as the treatment variable. The significant coefficient ($p < 0.001$; THD: $+0.00018$, $p < 0.001$) indicate dose-response relationships: larger electricity loads are associated with larger power quality changes.

Confirmed Locations and Verified Dates

We re-estimate our main specifications using stricter treatment definitions. The confirmed-location analysis restricts treatment to facilities with precisely geocoded training locations (mean effect: 74.19 MWh). The verified-date analysis uses API release timing rather than publication dates from the Epoch database (mean effect: 63.20 MWh). Both analyses corroborate our main findings with expected attenuation from the more conservative treatment definitions.

Callaway-Sant'Anna Estimator

As a robustness check and to provide transparent decomposition of treatment effect heterogeneity, we also implement the Callaway and Sant'Anna (2021) estimator. Like the stacked approach, CS avoids using already-treated units as controls; however, rather than achieving this through sample construction, it does so by estimating separate ATT(g,t) parameters for each cohort g and time period t, then aggregating these to form summary measures. This approach explicitly avoids problematic comparisons and provides a framework for testing parallel trends through examination of pre-treatment ATT(g,t) estimates.

Time-Series Anomaly Detection

As an internal consistency check, we apply time-series anomaly detection methods to the outcome variables in treated units. If AI model releases causally affect grid outcomes, we should observe detected anomalies clustering around treatment dates. Specifically:

1. For each treated unit, we fit a baseline time-series seasonal ARIMA model to the pre-treatment period.

2. We generate one-step-ahead forecasts and prediction intervals for the post-treatment period.

3. We flag observations where realized values fall outside the 20% prediction interval as anomalies.

4. We test whether anomaly frequency is elevated in the post-treatment window relative to the pre-treatment window and relative to control units.

This approach provides model-free evidence of structural breaks coinciding with treatment timing.

3.10.11 Discussions

Limitations

While informative, this approach inherits the limitations of all Difference-in-Differences experimental designs. It relies on the parallel trends assumption, which may be violated if treated and control regions were already diverging or if generators anticipated model releases.

Why our Estimates are Higher than Previous Estimates

Our estimates tend to find larger impacts of AI than previous estimates from the computer science literature Strubell et al. (2019); Schwartz et al. (2020). We attribute this discrepancy to our choice to utilize empirical data on the realization of outcome variables following AI activity. Compared to traditional approaches like PUE, we assert that our approach is better able to consider the full costs of training and inference including both indirect and direct costs and may be more suited to precise and accurate estimation of AI energy usage.

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